ISC Class XII Mathematics

Chapter 7

Integrals

139 Types of Sums Solved

Step-by-Step ISC Board Solutions – ML Aggarwal

Types Covered

1. Standard integrals & algebraic manipulation before integrating
2. Integration by substitution (algebraic, exponential, log, inverse-trig, trig)
3. Trigonometric integrals: product-to-sum, power reduction, rationalizing tricks
4. Standard forms $1/(x^2 \pm a^2)$, $1/\sqrt{a^2 - x^2}$, and reducible variants
5. Completing the square for quadratic denominators
6. Weierstrass substitution for rational functions of $\sin x, \cos x$
7. Integration by partial fractions (all four standard cases)
8. Integration by parts, incl. the cyclic e^x trick and reduction patterns
9. The $e^x[f(x) + f'(x)]$ special form
10. Standard forms $\sqrt{a^2 \pm x^2}$, $\sqrt{x^2 - a^2}$
11. Definite integrals: direct evaluation and substitution
12. Properties of definite integrals, incl. King's Rule

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Type	Method / Pattern	★	Source(s)
1	Direct application of the power rule $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C$ $(n \neq -1)$, including cases where a radical or reciprocal must first be rewritten as x raised to a fractional or negative power.	–	Exercise 7.1, Q1(iii); Exercise 7.1, Q2(i)
2	The integrand looks trig-heavy, but simplifies algebraically (via reciprocal or Pythagorean identities) to a single standard derivative you already know	–	Exercise 7.1, Q2(iii); Exercise 7.1, Q2(ii)
3	Integrate an exponential of the form a^x directly, including cases where a product or quotient of two different exponential bases must first be combined into a single base $(a^x \cdot b^x = (ab)^x)$.	–	Exercise 7.1, Q3(iii); Exercise 7.1, Q3(ii)

Type	Method / Pattern	★	Source(s)
4	Use a double-angle or half-angle identity to collapse a trigonometric numerator (or an entire integrand under a square root) to a constant or a single standard form before integrating	–	Exercise 7.1, Q4(ii); Exercise 7.1, Q4(iii)
5	Simplify an expression of the form $a^{k \log_b x}$ using the identity $a^{\log_a f(x)} = f(x)$	–	Exercise 7.1, Q5(i); Exercise 7.1, Q5(iii)
6	Integrate an algebraic expression term by term using the power rule, where the expression is already a sum of separate power/radical terms (possibly with literal constants a, b, c)	–	Exercise 7.2, Q2(ii)
7	Expand a product, square, or cube of a binomial *before* integrating, since the expanded form is a simple sum of power terms even though the original expression is not.	Yes	Exercise 7.2, Q8(ii); Exercise 7.2, Q6(ii)
8	Split a single algebraic fraction into several simpler fractions by dividing every term of the numerator by the denominator separately, turning one hard-looking integral into several easy power-rule integrals.	–	Exercise 7.2, Q7(i)
9	Recognise a fraction that is already a standard trig derivative in disguise	–	Exercise 7.2, Q3(i); Exercise 7.2, Q11(ii)
10	Use a double-angle identity to rewrite a trig numerator so the whole fraction collapses into one or more standard secant/cosecant-family integrals	–	Exercise 7.2, Q15(ii); Exercise 7.2, Q12(ii)
11	Integrate a fraction built directly from the secant/cosecant family by expanding a product first	–	Exercise 7.2, Q11(i); Exercise 7.2, Q10(i)
12	Rationalize a trig denominator by multiplying top and bottom by a conjugate expression (built from a Pythagorean identity), turning an otherwise unintegrable fraction into a standard secant/cosecant-family sum.	Yes	Exercise 7.2, Q13(ii); Exercise 7.2, Q16(i)
13	Use $\sin^2 x = 1 - \cos^2 x = (1 - \cos x)(1 + \cos x)$ (difference of squares) to factor a numerator so it cancels directly with a matching factor in the denominator	–	Exercise 7.2, Q14(ii)

Type	Method / Pattern	★	Source(s)
14	Simplify the argument of an inverse trig function using double-angle or half-angle identities until it collapses to the function's own inverse acting on a simple angle, i.e. reduces to a form like $\cot^{-1}(\cot x) = x$.	–	Exercise 7.2, Q19(i); Exercise 7.2, Q20(ii)
15	Find a function $f(x)$ given its derivative $f'(x)$ and an initial condition $f(x_0) = y_0$	–	Exercise 7.2, Q21
16	Integrate a "linear-inside" expression directly using the reverse chain rule	–	Exercise 7.3, Q1(i); Exercise 7.3, Q3(ii)
17	Simplify an expression like $\sqrt{e^x}$ or $e^{k \log x}$ into a single power or exponential term before integrating	–	Exercise 7.3, Q4(i)
18	Simplify $\sqrt{1 + \cos x}$ or $\sqrt{1 + \sin x}$ using half-angle identities (writing $1+\cos x$ or $1 + \sin x$ as a perfect square in terms of $x/2$) before integrating.	–	Exercise 7.3, Q6(i); Exercise 7.3, Q6(ii)
19	Simplify a fraction by dividing the numerator by the denominator when the denominator is a monomial or when a linear divisor exactly divides the numerator (found by inspection or trial factoring), reducing to a simpler linear-plus-remainder or fully cancelling form.	–	Exercise 7.3, Q9(i); Exercise 7.3, Q9(ii)
20	Substitute $t =$(the linear expression inside a square root or a squared bracket) to rewrite the whole numerator in terms of t, then integrate the resulting simpler expression in t.	–	Exercise 7.3, Q10(i)
21	Perform polynomial long division when the numerator's degree is at least as high as the denominator's and the two do not divide out cleanly by simple factoring	–	Exercise 7.3, Q11(i)
22	Rationalize a denominator built from a sum or difference of two square roots by multiplying top and bottom by the conjugate surd, which uses the difference-of-squares identity to eliminate the roots from the denominator.	–	Exercise 7.3, Q12(i)
23	Use the product-to-sum formulas to convert a product of two trig functions of different (linear) arguments into a sum of two simple sine or cosine terms, each of which integrates directly.	Yes	Exercise 7.3, Q17(i); Exercise 7.3, Q18(ii)

Type	Method / Pattern	★	Source(s)
24	Use repeated power-reduction (half-angle) identities to rewrite an even power of sine or cosine ($\sin^4 x$, $\cos^4 x$, $\sin^6 x$, etc.) as a sum of constant and $\cos(kx)$ terms, which then integrate directly.	–	Exercise 7.3, Q14(ii)
25	The integrand is a power of some inner function $g(x)$ multiplied by $g'(x)$ (the exact derivative of that inner function)	Yes	Exercise 7.4, Q1(ii); Exercise 7.4, Q3(i)
26	The integrand is exactly $\dfrac{g'(x)}{g(x)}$ (the derivative of an inner function divided by the function itself)	Yes	Exercise 7.4, Q4(iv); Exercise 7.4, Q9(ii)
27	Substitute t for an exponential expression (such as $a+be^x$) so the whole integral becomes a simple power-rule problem in t	–	Exercise 7.4, Q4(i)
28	Substitute $t = \log x$ when the integrand contains a power of $\log x$ together with a factor of $\dfrac{1}{x}$ (which is exactly the derivative of $\log x$)	–	Exercise 7.4, Q4(iii)
29	Substitute t for an inverse trig function (such as $\tan^{-1} x$ or $\sin^{-1} 2x$) when its exact derivative appears as an accompanying factor	–	Exercise 7.4, Q6(i); Exercise 7.4, Q6(ii)
30	Integrate a product of an odd power of one trig function and any power of another (e.g. $\sin x \cos^5 x$) by substituting t for the trig function with the *even* power, peeling off one factor of the odd-power function to supply dt.	–	Exercise 7.4, Q7(i); Exercise 7.4, Q8(i)
31	Substitute t for a trig expression sitting under a square root (or otherwise raised to a power) in the denominator, when the numerator supplies exactly the derivative needed	–	Exercise 7.4, Q11(i)
32	Reduce an integral like $\tan^4 x$ or $\cot^4 x$ by repeatedly applying the Pythagorean identity to peel off a factor of $\sec^2 x$ (or $\text{cosec}^2 x$), splitting the integral into simpler pieces that combine substitution with direct standard forms.	–	Exercise 7.4, Q14(i)

Type	Method / Pattern	★	Source(s)
33	Recognise a mixed algebraic-trigonometric expression of the exact form $\dfrac{g'(x)}{g(x)}$ where $g(x)$ itself is a sum of an algebraic and a trig term (e.g. $x + \log \sec x$, or $x + \sin^2 x$)	–	Exercise 7.4, Q15(i); Exercise 7.4, Q17(i)
34	Manipulate an expression algebraically first	–	Exercise 7.4, Q19(ii); Exercise 7.4, Q21(i)
35	The integrand is a trig function wrapped around a power of x (e.g. $\sin(x^2+1)$, $\cos(x^4)$), multiplied by exactly the derivative of that power	–	Exercise 7.5, Q1(i)
36	Substitute $t = \sqrt{x}$ when the integrand is a trig function of $\sqrt{x}$ divided by $\sqrt{x}$	–	Exercise 7.5, Q2(i)
37	Substitute $t = \dfrac{1}{x}$ when the integrand is a trig function of $\dfrac{1}{x}$ divided by x^2	–	Exercise 7.5, Q5(ii)
38	Substitute $t = \log x$ when the integrand is a trig function of $\log x$ divided by x	–	Exercise 7.5, Q6(ii)
39	Substitute $t = \tan^{-1} x$ when the integrand is an exponential or trig function of $\tan^{-1} x$ divided by $1 + x^2$	–	Exercise 7.5, Q7(i); Exercise 7.5, Q7(ii)
40	Substitute t for a compound inner expression (such as xe^x) whose derivative requires the product rule	–	Exercise 7.5, Q9(ii)
41	Integrate an odd power of one trig function multiplied by another trig factor with a compound (non-x) argument, or paired with a square root	–	Exercise 7.5, Q12(i); Exercise 7.5, Q14(ii)
42	Integrate a higher odd power of sine or cosine (such as $\sin^5 x$, $\sin^7 x$) by peeling off one factor and rewriting the remaining even power fully via the Pythagorean identity, then expanding before integrating term by term	–	Exercise 7.5, Q13(i)
43	Recognise the special standard derivatives $\dfrac{d}{dx} \log\left(\tan \dfrac{x}{2}\right) = \operatorname{cosec} x$ and $\dfrac{d}{dx} \log(\operatorname{cosec} x - \cot x) = \operatorname{cosec} x$	Yes	Exercise 7.5, Q16(ii); Exercise 7.5, Q15(i)
44	Integrate $\sec^4 x$ (or $\tan^2 x \sec^4 x$) by splitting off a factor of $\sec^2 x$ and converting the remaining $\sec^2 x$ into $1 + \tan^2 x$, then substituting $t = \tan x$	–	Exercise 7.5, Q16(i); Exercise 7.5, Q11(ii)

Type	Method / Pattern	★	Source(s)
45	Integrate $\dfrac{N(x)}{D(x)}$ where $D(x)$ is a single shifted trig function like $\sin(x-\alpha)$ (or a linear combination of $\sin x, \cos x$) by writing the numerator as $A \cdot D(x) + B \cdot D'(x)$ for constants A, B	Yes	Exercise 7.6, Q4(i); Exercise 7.6, Q9
46	Integrate $\dfrac{1}{\sin(x-\alpha)\sin(x-\beta)}$ by writing the constant numerator as $\dfrac{1}{\sin(\alpha-\beta)} \cdot \sin\big((x-\beta)-(x-\alpha)\big)$, splitting the fraction into a difference of two simple cot/log-style integrals.	–	Exercise 7.6, Q5(i)
47	Convert a product of two shifted sines in the denominator into a sum using product-to-sum, when the numerator is itself a double-angle sine	–	Exercise 7.6, Q7(i)
48	Integrate $\dfrac{1}{\sin x \cos^2 x}$ (or similar products of trig powers in the denominator) by writing $1 = \sin^2 x + \cos^2 x$ in the numerator, splitting the fraction into a sum of two simpler standard forms.	–	Exercise 7.6, Q7(ii)
49	Simplify an algebraic-trig ratio using basic reciprocal/quotient identities until it collapses to a recognisable standard form	–	Exercise 7.6, Q2(i); Exercise 7.6, Q3(i)
50	Rationalize a square-root trig denominator like $\sqrt{1-\sin x}$ by first substituting $x = 2t$ so the expression under the root becomes a perfect square in t, $(\sin t - \cos t)^2$, then rewriting the result as a single shifted cosine to reach a standard sec integral.	–	Exercise 7.6, Q8(i)
51	Recognise $(\sin x + \cos x)^2 = 1 + \sin 2x$ to simplify a denominator built from $1 + \sin 2x$, cancelling directly with a matching numerator.	–	Exercise 7.6, Q8(ii)
52	Direct application of the two standard log-form integrals $\displaystyle\int \frac{dx}{x^2-a^2}$ and $\displaystyle\int \frac{dx}{a^2-x^2}$	–	Exercise 7.7, Q1(i); Exercise 7.7, Q1(ii)

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53	Direct application of the two standard inverse-trig-form integrals $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1}\frac{x}{a} + C$ and $\int \frac{dx}{x^2+a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$.	–	Exercise 7.7, Q1(iii); Exercise 7.7, Q1(iv)
54	The same four standard forms as Types 52–53, but with a coefficient in front of x^2	–	Exercise 7.7, Q2(i); Exercise 7.7, Q2(ii)
55	Direct application of the standard $\sec^{-1}$-form integral $\int \frac{dx}{x\sqrt{x^2a^2-b^2}}$, after factoring so the expression matches the exact pattern $x\sqrt{(cx)^2-b^2}$.	–	Exercise 7.7, Q3(i)
56	Direct application of the two standard log-form integrals $\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left\lvert x+\sqrt{x^2+a^2}\right\rvert + C$ and $\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left\lvert x+\sqrt{x^2-a^2}\right\rvert + C$.	–	Exercise 7.7, Q3(ii)
57	When the numerator's degree is at least the denominator's degree (an "improper" rational function), divide first to get a polynomial plus a proper-fraction remainder, then apply a standard arctan or log form to what remains.	–	Exercise 7.7, Q6(ii)
58	Substitute $t = x^2$ (or x^3, x^4, depending on the powers present) when the integrand is an odd power of x times a function of an even power of x	–	Exercise 7.7, Q7(i); Exercise 7.7, Q7(ii)
59	Split an algebraic fraction into two pieces using partial-fraction-style numerator manipulation before applying two different standard forms	–	Exercise 7.7, Q9(ii)
60	Substitute t for a trig function (usually $\sin x$ or $\tan x$) to convert a trig-algebraic mixed integrand into one of the standard quadratic-denominator forms in t.	–	Exercise 7.7, Q11(i)
61	Complete the square in a quadratic denominator (rewriting x^2+bx+c as $(x+p)^2+q$ or $(x+p)^2-q$) so it matches one of the standard forms from Types 52–53, then apply the arctan or log formula directly.	Yes	Exercise 7.8, Q1(i); Exercise 7.8, Q2(i)

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62	For $\frac{\text{linear}}{\text{quadratic}}$, first decompose the numerator as $A \cdot$ (derivative of the quadratic) + B, splitting into a log-form piece (from the A part) and a completed-square arctan piece (from the B part).	Yes	Exercise 7.8, Q2(ii)
63	Substitute for an exponential or trig expression first (recognising it as the natural inner variable), then complete the square in the resulting algebraic quadratic before applying a standard form.	–	Exercise 7.8, Q4(i)
64	Complete the square under a square root (rewriting $bx - x^2$ or $c + bx - x^2$ as $a^2 - (x-p)^2$) to match the standard arcsin form.	–	Exercise 7.8, Q5(i)
65	Complete the square under a square root where x^2 has a coefficient other than ± 1	–	Exercise 7.8, Q7(i)
66	For $\frac{\text{linear}}{\sqrt{\text{quadratic}}}$, decompose the numerator as $A \cdot$ (derivative of the quadratic) $+ B$ exactly as in Type 62, but here the A-piece becomes a square-root standard form and the B-piece becomes an arcsin (or log) form after completing the square.	–	Exercise 7.8, Q8(ii)
67	Substitute for an expression like $\log x$ or x^3 first, then complete the square in the resulting quadratic before applying a standard form	–	Exercise 7.8, Q11
68	Integrate $\frac{1}{a + b\cos x}$ (or the analogous form with $\sin x$) using the Weierstrass substitution $t = \tan\frac{x}{2}$, which converts *every* rational function of $\sin x$ and $\cos x$ into an ordinary rational function of t	Yes	Exercise 7.9, Q1(ii)
69	Integrate $\frac{1}{a + b\sin x + c\cos x}$ (mixing both sine and cosine, or sine alone) using the same Weierstrass substitution	–	Exercise 7.9, Q3(ii)

Type	Method / Pattern	★	Source(s)
70	Integrate $\dfrac{1}{1 \pm \tan x}$ or $\dfrac{1}{1 + \cot x}$ by multiplying numerator and denominator by $\cos x$ (or $\sin x$), converting the denominator into a $\sin x \pm \cos x$ combination	–	Exercise 7.9, Q4(ii)
71	Given $\displaystyle\int \frac{k \tan x}{\tan x - c}\,dx = ax + b \log\lvert \sin x - c\cos x\rvert + C$, find the constants a, b by converting to $\sin x, \cos x$ and applying the numerator-decomposition technique from Type 45 to a tan-based denominator.	–	Exercise 7.9, Q5
72	Decompose a proper rational function with a denominator that factors into distinct (non-repeated) linear factors into a sum of simple fractions, one per factor, then integrate each piece as a standard log form.	Yes	Exercise 7.10, Q2(i)
73	Decompose a proper rational function with a *repeated* linear factor in the denominator	–	Exercise 7.10, Q5(ii)
74	Decompose a proper rational function where the denominator includes an irreducible quadratic factor (one that doesn't factor over the reals)	–	Exercise 7.10, Q18(i)
75	When the numerator's degree is at least the denominator's degree, divide first to get a polynomial plus a proper-fraction remainder	–	Exercise 7.10, Q2(ii)
76	Substitute $t = x^2$ (or another even power) first when the rational function is genuinely a function of x^2 alone (odd powers of x appear only as a single multiplying factor), reducing to an ordinary partial-fraction problem in t.	–	Exercise 7.10, Q9(i); Exercise 7.10, Q11(ii)
77	Substitute for a trig or exponential expression first (recognising the "true" variable), then apply partial fractions to the resulting algebraic rational function in the new variable.	–	Exercise 7.10, Q13(ii); Exercise 7.10, Q14(ii)
78	For a denominator with a symmetric or reciprocal structure (like $x^4 - x^2 - 12$, secretly quadratic in x^2), substitute $t = x^2$ directly since every power of x present is already even	–	Exercise 7.10, Q19(ii)

Type	Method / Pattern	★	Source(s)
79	The most basic integration-by-parts pattern: a single polynomial factor x multiplied by a standard exponential or trig function, chosen as the "first function" using the ILATE priority order (Inverse trig, Log, Algebraic, Trig, Exponential	–	Exercise 7.11, Q1(i)
80	The same basic by-parts pattern as Type 79, but the trig/exponential factor has a linear (compound) argument like $\sin 3x$ or $\cos 2x$	–	Exercise 7.11, Q3(ii)
81	The "second function" dv is itself not a standard integral, but a recognisable derivative-of-a-standard-form pattern (like $\sec^2 x \tan x$)	–	Exercise 7.11, Q4(i)
82	A polynomial multiplied by $\log x$	Yes	Exercise 7.11, Q6(i)
83	Integrate $\log x$ on its own by treating it as $(\log x) \cdot 1$	–	Exercise 7.11, Q6(ii)
84	A polynomial multiplied by log of a *compound* expression (like $\log(1 + x)$, not just $\log x$)	–	Exercise 7.11, Q7(ii)
85	A trig function multiplied by log of a *related* trig function (like $\sin x \cdot \log(\cos x)$)	–	Exercise 7.11, Q7(iii)
86	Integrate a single inverse trig function by parts, treating it as (function)$\times 1$ exactly like Type 83 for $\log x$	–	Exercise 7.11, Q8(i)
87	A polynomial of degree 2 or higher multiplied by an inverse trig function	Yes	Exercise 7.11, Q9(i); Exercise 7.11, Q9(ii)
88	Prepare an integrand algebraically *before* by parts becomes visible	–	Exercise 7.11, Q10(i)
89	A quadratic (or higher) polynomial multiplied by an exponential or trig function	Yes	Exercise 7.11, Q11(i)
90	Integrate $(\log x)^2$ using by parts, treating the whole squared expression as u	–	Exercise 7.11, Q12(i)
91	The integrand doesn't look like a by-parts product at all until a substitution (usually for the argument inside a trig or exponential function) is made first	Yes	Exercise 7.11, Q12(ii)
92	Integrate an inverse trig or trig function of $\sqrt{x}$ by substituting $t = \sqrt{x}$ (or $x = t^2$) first, converting dx into $2t\,dt$	–	Exercise 7.11, Q14(i)
93	Simplify the argument of an inverse trig function using a triple-angle identity *before* attempting by parts	Yes	Exercise 7.11, Q16(i)

Type	Method / Pattern	★	Source(s)
94	Choose dv cleverly (not the "obvious" leftover piece) so that v turns out to be a simple algebraic expression rather than another inverse trig function	–	Exercise 7.11, Q17(i)
95	The "cyclic" by-parts technique: applying by parts twice to $e^{ax}\sin(bx)$ (or $\cos(bx)$) returns the *original* integral multiplied by a constant on the right-hand side	Yes	Exercise 7.11, Q19(i)
96	A self-referencing "reduction formula" style integral like $\text{cosec}^3 x$, where by parts produces the original integral back (with a different sign/coefficient) combined with a standard trig integral	–	Exercise 7.11, Q21(ii)
97	Combine substitution and the cyclic by-parts technique: substitute for an inverse trig argument first (converting to a standard angle variable), which reveals an $e^{a\theta}\times\text{trig}(\theta)$ structure that then needs Type 95's cyclic method to finish.	–	Exercise 7.11, Q21(i)
98	Recognise an integrand of the exact form $e^x[f(x)+f'(x)]$ where both $f(x)$ and its derivative are directly visible as separate terms	Yes	Exercise 7.12, Q1(ii)
99	The integrand is e^x times a single fraction that doesn't visibly split into $f(x)+f'(x)$	–	Exercise 7.12, Q11(i)
100	Given the general statement $\int e^x[\ldots]\,dx = e^x f(x) + C$, identify $f(x)$ by inspection	–	Exercise 7.12, Q6(i) (ISC 2024)
101	The base is $e^{g(x)}$ for a non-linear $g(x)$ (such as $\cot^{-1} x$) rather than plain e^x	–	Exercise 7.12, Q13(iii)
102	Direct application of the standard formula $\int \sqrt{a^2-x^2}\,dx = \frac{x}{2}\sqrt{a^2-x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$.	–	Exercise 7.13, Q1(i)
103	Direct application of the two companion standard formulas $\int \sqrt{x^2+a^2}\,dx$ and $\int \sqrt{x^2-a^2}\,dx$, which use a log term instead of $\sin^{-1}$ since the expression under the root doesn't have the bounded a^2-x^2 shape.	–	Exercise 7.13, Q1(ii)

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104	The same standard square-root formulas as Types 102–103, but with a non-unit coefficient on x^2	–	Exercise 7.13, Q3(i)
105	Prepare the integrand algebraically	–	Exercise 7.13, Q4(ii); Exercise 7.13, Q5(ii)
106	Substitute for an inner expression (like x^2 or $\log x$) first when the square-root structure is secretly built from that inner variable, then apply a standard sqrt formula in the new variable.	–	Exercise 7.13, Q6(ii); Exercise 7.13, Q6(i)
107	Integrate x (or a related algebraic expression) times an inverse trig function using by parts, where the resulting remainder integral is a standard square-root form	–	Exercise 7.13, Q8(i)
108	Complete the square under a square root (exactly as in Types 64–65), then apply the matching standard $\sqrt{x^2 \pm a^2}$ or $\sqrt{a^2 - x^2}$ formula from Types 102–103 to the shifted variable.	–	Exercise 7.14, Q1(i)
109	For (linear)$\times\sqrt{\text{quadratic}}$, combine the numerator-decomposition trick from Type 66 with completing the square: decompose the linear factor as $A \cdot$ (derivative of the quadratic) + B, giving a direct power-rule piece from A and a completing-the-square standard-formula piece from B.	Yes	Exercise 7.14, Q3(i)
110	Evaluate a definite integral by finding the antiderivative exactly as in indefinite integration, then applying the Fundamental Theorem of Calculus: substitute the upper limit, subtract the value at the lower limit, and drop the constant of integration entirely (it cancels).	–	Exercise 7.15, Q1(iii)
111	Evaluate a definite integral of $\lvert x - a\rvert$ (or the greatest integer function $[x]$) by splitting the interval at the point(s) where the expression inside changes sign or jumps, then integrating each piece separately with its own sign or constant value.	Yes	Exercise 7.17, Q1(ii) (ISC 2022)
112	Evaluate a definite integral requiring a substitution	–	Exercise 7.16, Q2(ii)

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113	Evaluate a definite integral of an algebraic rational function, applying whichever indefinite-integration technique (splitting, substitution, standard form) is needed, then evaluating at the converted or original limits.	–	Exercise 7.16, Q4(i)
114	Evaluate a definite integral involving an inverse trig or log function, typically via by parts, being careful that the "boundary term" $\left[u \cdot v\right]$ from by parts is itself evaluated at the limits alongside the remaining integral.	–	Exercise 7.16, Q10(ii)
115	Given a definite integral set equal to a numeric value, with the upper (or lower) limit as an unknown a, integrate first (leaving a symbolic), then solve the resulting equation in a.	–	Exercise 7.15, Q21(i)
116	Evaluate a definite integral of a trig power or product (requiring the same techniques as Types 24, 41–43 for the indefinite case), then apply the limits	–	Exercise 7.16, Q8(i)
117	Evaluate a definite integral requiring completing the square (or a standard sqrt/arctan form from Types 61–67), applying the converted or evaluated inverse trig values at the limits carefully.	–	Exercise 7.16, Q12(i)
118	A definite integral requiring both a substitution *and* an algebraic manipulation (such as partial fractions or a numerator split) combined	–	Exercise 7.16, Q17(i)
119	"King's Rule": $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. For a rational function of $\sin x$ and $\cos x$ integrated from 0 to $\pi/2$, applying this property and *adding* the original and transformed integrals often makes the trig functions cancel into a constant.	Yes	Exercise 7.17, Q8
120	The general form of King's Rule, $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$, applied to an interval that doesn't start at 0	–	Exercise 7.17, Q23 (King's Rule general form)

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121	Use the even/odd symmetric-interval property: $\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$ if f is even ($f(-x) = f(x)$), or $= 0$ if f is odd ($f(-x) = -f(x)$)	Yes	Exercise 7.17, Q3(iii)
122	Use the reflection-about-midpoint property $\int_{0}^{2a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$ if $f(2a-x) = f(x)$, or $= 0$ if $f(2a-x) = -f(x)$	–	Exercise 7.17, Q7 (property proof, applied)
123	Split $\int_{a}^{b} \lvert g(x)\rvert\,dx$ at every point within $[a,b]$ where $g(x) = 0$, integrating $g(x)$ or $-g(x)$ on each sub-piece as appropriate	–	Exercise 7.17, Q10(iii)
124	Integrate $\frac{g(x)}{\lvert g(x)\rvert}$ (a "sign function" in disguise, equal to $+1$ or -1 depending on the sign of $g(x)$) by splitting at $g(x) = 0$ exactly as in Type 123, but integrating the constant ± 1 on each piece instead of $g(x)$ itself.	–	Exercise 7.17, Q1(iv)
125	Integrate the greatest integer function $[x]$ by splitting the interval at every integer it contains, since $[x]$ is constant (equal to that integer) on each unit subinterval $[n, n+1)$.	–	Exercise 7.17, Q2(ii)
126	Integrate $\lvert\sin x\rvert$, $\lvert\cos x\rvert$, or $\lvert\cos kx\rvert$ over an interval spanning more than one sign-change of the trig function	–	Exercise 7.17, Q11(iii)
127	An odd power of $\sin x$ or $\cos x$ integrated over a full period or half-period often integrates to a clean fraction via the reduction technique from Type 42, applied directly at the definite-integral level using the standard "Walker" pattern (each factor drops the power by 2 and multiplies by a running fraction).	–	Exercise 7.16, Q18(ii)
128	Prove or apply the classical result $\int_{0}^{\pi/2} \log(\sin x)\,dx = \int_{0}^{\pi/2} \log(\cos x)\,dx = -\frac{\pi}{2}\log 2$, derived by combining King's Rule with the double-angle identity $\sin 2x = 2\sin x\cos x$ and a substitution.	Yes	Exercise 7.17, Q25

Type	Method / Pattern	★	Source(s)
129	Apply the general result $\int_0^a \frac{f(x)}{f(x)+f(a-x)}\,dx = \frac{a}{2}$, which follows directly from King's Rule	–	Exercise 7.17, Q24
130	A genuinely hard combined integral requiring King's Rule together with a nontrivial algebraic or trigonometric simplification	Yes	Exercise 7.17, Q21 (King's Rule, hard combination)
131	Prove one of the definite-integral properties itself from first principles (e.g. $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$), using only the substitution rule for definite integrals	–	Exercise 7.17, Q8 (property proof)
132	Divide by x^2 and substitute $t = x \mp \frac{1}{x}$ for palindromic quartic denominators like x^4+1	Yes	Extended (classic x^4+1 family)
133	Integrate a sum of several absolute-value terms, each with its own breakpoint	Yes	Composite Board-Style Question
134	Integrate $[g(x)]$ for a compound argument (e.g. $[x^2]$), solving $g(x) = n$ for the true split points	Yes	ISC Specimen Paper 2025
135	Integrate a combined $\lvert\cdot\rvert$ and $[\cdot]$ expression, merging breakpoints from both sources	–	Composite Board-Style Question
136	Reciprocal substitution $x = 1/t$ for a square root over a high negative power of x	Yes	Exercise 7.16, Q23(ii) (Exemplar)
137	LCM substitution $t = x^{1/n}$ to clear multiple different roots of x at once	Yes	Exercise 7.5, Q22(i)
138	Split $x^2 = (x^2+a^2) - a^2$ to avoid substitution entirely for $x^2/\sqrt{x^2+a^2}$	–	Exercise 7.13, Q4(i)
139	$\cos^{-1}\sqrt{x}$ (or $\sin^{-1}\sqrt{x}$) via $x = t^2$ then by parts, finishing with the Type 138 split	–	Exercise 7.13, Q8(ii)

Formula Sheet

Basic Standard Integrals

$$\int x^n\,dx = \frac{x^{n+1}}{n+1} + C \ (n \neq -1), \qquad \int \frac{1}{x}\,dx = \log|x| + C$$

Power rule and its $n = -1$ exception

Basic Standard Integrals

$$\int e^x\,dx = e^x + C, \qquad \int a^x\,dx = \frac{a^x}{\log a} + C$$

Exponential forms, $a > 0,\ a \neq 1$

Basic Trig Integrals

$$\int \sin x\,dx = -\cos x + C, \qquad \int \cos x\,dx = \sin x + C$$

Basic Trig Integrals

$$\int \sec^2 x\,dx = \tan x + C, \qquad \int \operatorname{cosec}^2 x\,dx = -\cot x + C$$

Basic Trig Integrals

$$\int \sec x \tan x\,dx = \sec x + C, \qquad \int \operatorname{cosec} x \cot x\,dx = -\operatorname{cosec} x + C$$

Log-Form Trig Integrals

$$\int \tan x\,dx = \log|\sec x| + C, \qquad \int \cot x\,dx = \log|\sin x| + C$$

Log-Form Trig Integrals

$$\int \sec x\,dx = \log|\sec x + \tan x| + C, \qquad \int \operatorname{cosec} x\,dx = \log|\operatorname{cosec} x - \cot x| + C$$

Special Log-Trig Derivatives

$$\frac{d}{dx}\log\tan\frac{x}{2} = \operatorname{cosec} x, \qquad \frac{d}{dx}\log(\operatorname{cosec} x - \cot x) = \operatorname{cosec} x$$

Less common but essential recognition results

Substitution Rule

$$\int f(ax+b)\,dx = \frac{1}{a}F(ax+b)+C, \qquad \int \frac{g'(x)}{g(x)}\,dx = \log|g(x)|+C$$

The two most-used substitution shortcuts in the chapter

Substitution Rule

$$\int [g(x)]^n g'(x)\,dx = \frac{[g(x)]^{n+1}}{n+1}+C, \qquad \int \frac{g'(x)}{\sqrt{g(x)}}\,dx = 2\sqrt{g(x)}+C$$

Power-Reduction & Product-to-Sum

$$\sin^2 x = \frac{1-\cos 2x}{2}, \qquad \cos^2 x = \frac{1+\cos 2x}{2}$$

Used to integrate even powers of sin/cos

Power-Reduction & Product-to-Sum

$$\cos A\cos B = \frac{1}{2}[\cos(A-B)+\cos(A+B)], \qquad \sin A\cos B = \frac{1}{2}[\sin(A+B)+\sin(A-B)]$$

Converts products of trig functions to sums

Half-Angle Identities

$$1+\cos x = 2\cos^2\frac{x}{2}, \qquad 1-\cos x = 2\sin^2\frac{x}{2}, \qquad 1+\sin x = \left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)^2$$

Standard Algebraic Forms

$$\int \frac{dx}{x^2-a^2} = \frac{1}{2a}\log\left|\frac{x-a}{x+a}\right|+C, \qquad \int \frac{dx}{a^2-x^2} =$$

$$\frac{1}{2a}\log\left|\frac{a+x}{a-x}\right|+C$$

Standard Algebraic Forms

$$\int\frac{dx}{x^2+a^2}=\frac{1}{a}\tan^{-1}\frac{x}{a}+C,\qquad\int\frac{dx}{\sqrt{a^2-x^2}}=\sin^{-1}\frac{x}{a}+C$$

Standard Algebraic Forms

$$\int\frac{dx}{x\sqrt{c^2x^2-b^2}}=\frac{1}{b}\sec^{-1}\left(\frac{cx}{b}\right)+C$$

Standard Algebraic Forms

$$\int\frac{dx}{\sqrt{x^2+a^2}}=\log\left|x+\sqrt{x^2+a^2}\right|+C,\qquad\int\frac{dx}{\sqrt{x^2-a^2}}=\log\left|x+\sqrt{x^2-a^2}\right|+C$$

Completing the Square

$$x^2+bx+c=\left(x+\frac{b}{2}\right)^2+\left(c-\frac{b^2}{4}\right)$$

Always the first step for a non-standard quadratic denominator

Standard Square-Root Formulas

$$\int\sqrt{a^2-x^2}\,dx=\frac{x}{2}\sqrt{a^2-x^2}+\frac{a^2}{2}\sin^{-1}\frac{x}{a}+C$$

Standard Square-Root Formulas

$$\int\sqrt{x^2+a^2}\,dx=\frac{x}{2}\sqrt{x^2+a^2}+\frac{a^2}{2}\log\left|x+\sqrt{x^2+a^2}\right|+C$$

Standard Square-Root Formulas

$$\int\sqrt{x^2-a^2}\,dx=\frac{x}{2}\sqrt{x^2-a^2}-\frac{a^2}{2}\log\left|x+\sqrt{x^2-a^2}\right|+C$$

Weierstrass Substitution

$$t = \tan\frac{x}{2}, \quad \sin x = \frac{2t}{1+t^2}, \quad \cos x = \frac{1-t^2}{1+t^2}, \quad dx = \frac{2\,dt}{1+t^2}$$

Universal substitution for any rational function of $\sin x, \cos x$

Partial Fractions Setups

$$\frac{f(x)}{(x-a)(x-b)} = \frac{A}{x-a} + \frac{B}{x-b}; \quad \frac{f(x)}{(x-a)^2} = \frac{A}{x-a} + \frac{B}{(x-a)^2}; \quad \frac{f(x)}{x^2+b^2} = \frac{Ax+B}{x^2+b^2}$$

Distinct, repeated, and irreducible-quadratic cases

Integration by Parts

$$\int u\,v\,dx = u\int v\,dx - \int\left[\frac{du}{dx}\int v\,dx\right]dx$$

Choose u using ILATE: Inverse trig, Log, Algebraic, Trig, Exponential

Standard By-Parts Results

$$\int \log x\,dx = x\log x - x + C, \qquad \int \tan^{-1}x\,dx = x\tan^{-1}x - \frac{1}{2}\log(1+x^2) + C$$

Standard By-Parts Results

$$\int \sin^{-1}x\,dx = x\sin^{-1}x + \sqrt{1-x^2} + C, \qquad \int \cos^{-1}x\,dx = x\cos^{-1}x - \sqrt{1-x^2} + C$$

Cyclic Exponential-Trig Integrals

$$\int e^{ax}\sin bx\,dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + C, \qquad \int e^{ax}\cos bx\,dx = \frac{e^{ax}(a\cos bx + b\sin bx)}{a^2+b^2} + C$$

Derived by applying by parts twice and solving algebraically

The e to the x times (f plus f prime) Special Form

$$\int e^x[f(x)+f'(x)]\,dx = e^x f(x)+C, \qquad \int e^{g(x)}[f(x)g'(x)+f'(x)]\,dx = e^{g(x)}f(x)+C$$

The generalised version applies when the base is $e^{g(x)}$

Triple-Angle Identities

$$\sin 3\theta = 3\sin\theta - 4\sin^3\theta, \qquad \tan 3\theta = \frac{3\tan\theta-\tan^3\theta}{1-3\tan^2\theta}$$

Used to collapse $\sin^{-1}(3x-4x^3) \to 3\sin^{-1}x$ and similar

Fundamental Theorem of Calculus

$$\int_a^b f(x)\,dx = F(b)-F(a), \text{ where } F \text{ is any antiderivative of } f$$

No constant of integration needed for a definite integral

Properties of Definite Integrals

$$\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx \quad \text{(King's Rule)}$$

Special case $a=0$: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Properties of Definite Integrals

$$\int_{-a}^a f(x)\,dx = 2\int_0^a f(x)\,dx \text{ if even}, \quad = 0 \text{ if odd}$$

Always check parity before integrating over a symmetric interval

Properties of Definite Integrals

$$\int_0^{2a} f(x)\,dx = 2\int_0^a f(x)\,dx \text{ if } f(2a-x)=f(x), \quad = 0 \text{ if } f(2a-x)=-f(x)$$

Properties of Definite Integrals

$$\int_0^a \frac{f(x)}{f(x)+f(a-x)}\,dx = \frac{a}{2}$$

Direct corollary of King's Rule

Celebrated Result

$$\int_0^{\pi/2} \log(\sin x)\,dx = \int_0^{\pi/2} \log(\cos x)\,dx = -\frac{\pi}{2}\log 2$$

Proved via King's Rule + double-angle + the $[0, 2a]$ property

Wallis' Formula

$$\int_0^{\pi/2} \sin^n x\,dx = \int_0^{\pi/2} \cos^n x\,dx = \frac{(n-1)(n-3)\cdots}{n(n-2)\cdots} \times \left(1 \text{ or } \frac{\pi}{2}\right)$$

End the product with 1 for odd n, $\pi/2$ for even n

I. Standard Integrals & Algebraic Manipulation (Ex 7.1–7.2)

Type 1

Direct application of the power rule $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C$ $(n \neq -1)$, including cases where a radical or reciprocal must first be rewritten as x raised to a fractional or negative power.

Formula Used: $\int x^n\,dx = \dfrac{x^{n+1}}{n+1} + C, \quad n \neq -1$

Source: Exercise 7.1, Q1(iii)

Given: $\int \dfrac{1}{\sqrt{y}}\,dy$

Solution:

We first rewrite the radical as a fractional power so the power rule can be applied directly: $\dfrac{1}{\sqrt{y}} = y^{-1/2}$.

Applying the power rule with $n = -\frac{1}{2}$: $\displaystyle\int y^{-1/2}\,dy = \frac{y^{1/2}}{1/2} + C$, since $n + 1 = \frac{1}{2}$.

Simplifying the fraction: $\dfrac{y^{1/2}}{1/2} = 2y^{1/2} = 2\sqrt{y}$.

Final Answer

$2\sqrt{y} + C$

Common Mistake: Forgetting to rewrite the radical as a power before integrating, and instead trying to integrate $1/\sqrt{y}$ as if it were a standard log or trig form – always convert to index (power) notation first.

Alternate Board-Style Example

Source: Exercise 7.1, Q2(i)

Given: $\int \dfrac{1}{\sqrt[3]{x}}\,dx$

Solution:

Rewrite as a power: $\frac{1}{\sqrt[3]{x}} = x^{-1/3}$.

Applying the power rule: $\int x^{-1/3}\,dx = \frac{x^{2/3}}{2/3} + C = \frac{3}{2}x^{2/3} + C$.

Final Answer

$\frac{3}{2}x^{2/3} + C$

Quick Recall: Any radical or reciprocal power of x can be rewritten as x^n with a fractional or negative n – do this conversion first, then apply the power rule mechanically.

Type 2

The integrand looks trig-heavy, but simplifies algebraically (via reciprocal or Pythagorean identities) to a single standard derivative you already know – $\sec x \tan x$, a constant, or similar – before any integration is actually needed.

Formula Used: $\frac{d}{dx}(\sec x) = \sec x \tan x, \qquad \operatorname{cosec} x = \frac{1}{\sin x}$

Source: Exercise 7.1, Q2(iii)

Given: $\int \sin x \sec^2 x\,dx$

Solution:

Rewrite $\sec^2 x$ in terms of sine and cosine to see what cancels: $\sin x \sec^2 x = \sin x \cdot \frac{1}{\cos^2 x} = \frac{\sin x}{\cos x} \cdot \frac{1}{\cos x} = \tan x \sec x$ – this reveals the integrand is exactly the derivative of $\sec x$.

Since $\frac{d}{dx}(\sec x) = \sec x \tan x$, the integral follows immediately by reversing this known derivative.

Final Answer

$\sec x + C$

Common Mistake: Trying to integrate $\sin x \sec^2 x$ term by term or via substitution without first noticing it algebraically collapses to the standard $\sec x \tan x$ form – always simplify trig products fully before reaching for a technique.

Alternate Board-Style Example

Source: Exercise 7.1, Q2(ii)

Given: $\int \sin^2 x \operatorname{cosec}^2 x\,dx$

Solution:

Rewrite $\operatorname{cosec}^2 x = \dfrac{1}{\sin^2 x}$, so the product becomes $\sin^2 x \cdot \dfrac{1}{\sin^2 x} = 1$ – the entire integrand collapses to the constant 1 before any integration technique is needed.

Integrating the constant: $\displaystyle\int 1\,dx = x + C$.

Final Answer

$x + C$

Quick Recall: Before choosing a technique, always rewrite every trig factor in terms of $\sin x$ and $\cos x$ (or use a reciprocal identity) and see what cancels – many "hard-looking" trig integrands are actually a standard derivative or even a constant in disguise.

Type 3

Integrate an exponential of the form a^x directly, including cases where a product or quotient of two different exponential bases must first be combined into a single base ($a^x \cdot b^x = (ab)^x$).

Formula Used: $\int a^x\,dx = \dfrac{a^x}{\log a} + C, \quad a > 0,\ a \neq 1$

Source: Exercise 7.1, Q3(iii)

Given: $\int \dfrac{4^x}{9^x}\,dx$

Solution:

We first combine the quotient of two exponential bases into a single base: $\dfrac{4^x}{9^x} = \left(\dfrac{4}{9}\right)^x$, since $\dfrac{a^x}{b^x} = (a/b)^x$.

Applying the standard exponential integral with $a = \frac{4}{9}$: $\displaystyle\int \left(\frac{4}{9}\right)^x dx = \frac{(4/9)^x}{\log(4/9)} + C$.

Final Answer

$\dfrac{(4/9)^x}{\log(4/9)} + C$

Common Mistake: Trying to integrate 4^x and 9^x separately (as if the quotient were a difference) – a quotient of two exponential bases combines multiplicatively into a single new base first, it does not split term by term.

Alternate Board-Style Example

Source: Exercise 7.1, Q3(ii)

Given: $\int 7^x e^x \, dx$

Solution:

Combine the product into a single base: $7^x e^x = (7e)^x$, since $a^x \cdot b^x = (ab)^x$.

Applying the standard exponential integral: $\displaystyle\int (7e)^x \, dx = \frac{(7e)^x}{\log(7e)} + C$.

Final Answer

$$\frac{(7e)^x}{\log(7e)} + C$$

Quick Recall: a^x integrates to $a^x / \log a$ – if you see a product or quotient of two different exponential bases with the same power x, combine them into one base first using $a^x b^x = (ab)^x$.

Type 4

Use a double-angle or half-angle identity to collapse a trigonometric numerator (or an entire integrand under a square root) to a constant or a single standard form before integrating – distinct from Type 2 because the simplification specifically relies on the $\cos 2x$ family of identities, not just reciprocal cancellation.

Formula Used: $\cos 2x = 1 - 2\sin^2 x, \qquad 1 + \cos 2x = 2\cos^2 x$

Source: Exercise 7.1, Q4(ii)

Given: $\int \frac{\cos 2x + 2\sin^2 x}{\cos^2 x} \, dx$

Solution:

We rewrite the numerator using $\cos 2x = 1 - 2\sin^2 x$: $\cos 2x + 2\sin^2 x = (1 - 2\sin^2 x) + 2\sin^2 x = 1$ – the $\sin^2 x$ terms cancel exactly, leaving just the constant 1 in the numerator.

The integrand simplifies to $\dfrac{1}{\cos^2 x} = \sec^2 x$, a standard derivative.

Integrating: $\displaystyle\int \sec^2 x \, dx = \tan x + C$.

Final Answer

$\tan x + C$

Common Mistake: Trying to split the original fraction into $\dfrac{\cos 2x}{\cos^2 x} + \dfrac{2\sin^2 x}{\cos^2 x}$ and integrate each piece separately – this works but is far more laborious than simplifying the numerator to a constant first; always check for numerator simplification before splitting a fraction.

Alternate Board-Style Example

Source: Exercise 7.1, Q4(iii)

Given: $\int \sqrt{\frac{1}{2}(1+\cos 2x)}\,dx$

Solution:

Using the half-angle identity $1+\cos 2x = 2\cos^2 x$: the expression under the root becomes $\dfrac{1}{2}(2\cos^2 x) = \cos^2 x$.

Taking the square root: $\sqrt{\cos^2 x} = |\cos x|$, which equals $\cos x$ on an interval where cosine is non-negative (the standard assumption for this type of question).

Integrating: $\displaystyle\int \cos x\,dx = \sin x + C$.

Final Answer

$\sin x + C$ (on an interval where $\cos x \geq 0$)

Quick Recall: Whenever you see $1+\cos 2x$ or $1-\cos 2x$ (or a numerator built from $\cos 2x$ and $\sin^2 x/\cos^2 x$), substitute the double-angle identity immediately – it almost always collapses the expression to something much simpler.

Type 5

Simplify an expression of the form $a^{k\log_b x}$ using the identity $a^{\log_a f(x)} = f(x)$ – sometimes the log base matches the outer base directly, and sometimes a change-of-base step is needed first.

Formula Used: $a^{\log_a f(x)} = f(x)$

Source: Exercise 7.1, Q5(i)

Given: $\int e^{2\log x}\,x^{-3}\,dx$

Solution:

Simplify the exponential-log term first: $e^{2\log x} = e^{\log(x^2)} = x^2$, using $e^{\log f(x)} = f(x)$ with $f(x) = x^2$.

Substituting, the integrand becomes $x^2 \cdot x^{-3} = x^{-1} = \dfrac{1}{x}$.

Integrating: $\int \frac{1}{x}\,dx = \log|x| + C$.

Final Answer

$\log|x| + C$

Common Mistake: Trying to integrate $e^{2\log x}$ directly as an exponential in x (using the a^x formula) – it isn't of that form at all; the base and the exponent's log must be simplified algebraically first using $e^{\log f(x)} = f(x)$.

Alternate Board-Style Example

Source: Exercise 7.1, Q5(iii)

Given: $\int 2^{\log_4 x}\,dx$

Solution:

Here the base of the exponent (4) does not match the outer base (2), so we need a change of base first: $\log_4 x = \frac{\log_2 x}{\log_2 4} = \frac{\log_2 x}{2}$, since $\log_2 4 = 2$.

Substituting: $2^{\log_4 x} = 2^{(\log_2 x)/2} = \left(2^{\log_2 x}\right)^{1/2} = x^{1/2} = \sqrt{x}$, using $2^{\log_2 x} = x$.

Integrating: $\int \sqrt{x}\,dx = \frac{2}{3}x^{3/2} + C$.

Final Answer

$\frac{2}{3}x^{3/2} + C$

Quick Recall: $a^{\log_a f(x)} = f(x)$ always – if the log's base doesn't match the outer exponential's base, convert the log to the matching base first using $\log_b x = \frac{\log_a x}{\log_a b}$, then apply the identity.

Type 6

Integrate an algebraic expression term by term using the power rule, where the expression is already a sum of separate power/radical terms (possibly with literal constants a, b, c) – the most basic pattern in this exercise, extended from Type 1 to multi-term expressions.

Formula Used: $\int x^n\,dx = \frac{x^{n+1}}{n+1} + C$

Source: Exercise 7.2, Q2(ii)

Given: $\int \left(\frac{2a}{\sqrt{x}} - \frac{b}{x^2} + 3c\sqrt[3]{x^2}\right) dx$

Solution:

Rewrite every term as a power of x: $\frac{2a}{\sqrt{x}} = 2ax^{-1/2}$, $\frac{b}{x^2} = bx^{-2}$, and $3c\sqrt[3]{x^2} = 3cx^{2/3}$ – this puts every term in a form the power rule can handle directly.

Integrate each term separately using the power rule (constants like a, b, c simply carry through unchanged): $\int 2ax^{-1/2}dx = \frac{2ax^{1/2}}{1/2} = 4a\sqrt{x}$, $\int bx^{-2}dx = \frac{bx^{-1}}{-1} = -\frac{b}{x}$, and $\int 3cx^{2/3}dx = \frac{3cx^{5/3}}{5/3} = \frac{9c}{5}x^{5/3}$.

Combine all three results with their original signs.

Final Answer

$4a\sqrt{x} + \frac{b}{x} + \frac{9c}{5}x^{5/3} + C$

Common Mistake: Losing track of a literal constant's sign or forgetting it carries through the whole integration unchanged – treat a, b, c exactly like any other constant multiplier.

Quick Recall: Rewrite every term as x^n first, then integrate term by term – literal constants ride along for the whole process without changing.

Type 7

Expand a product, square, or cube of a binomial *before* integrating, since the expanded form is a simple sum of power terms even though the original expression is not.

★ **Board Favorite** – This "expand-first" pattern is a standard high-yield warm-up type – ISC frequently opens integration papers with a question exactly like this to test whether students integrate term by term correctly after expanding, rather than trying to integrate a product directly.

Formula Used: $\int x^n \, dx = \frac{x^{n+1}}{n+1} + C$

Source: Exercise 7.2, Q8(ii)

Given: $\int (x + \sqrt{x})^3 \, dx$

Solution:

We cannot integrate a cubed binomial directly, so we expand it first using $(A + B)^3 = A^3+3A^2B+3AB^2+B^3$ with $A = x,\ B = \sqrt{x}$: $(x+\sqrt{x})^3 = x^3+3x^2\sqrt{x}+3x(\sqrt{x})^2+(\sqrt{x})^3 = x^3 + 3x^{5/2} + 3x^2 + x^{3/2}$.

Now integrate each of the four resulting power terms separately: $\displaystyle\int x^3 dx = \frac{x^4}{4}$, $\displaystyle\int 3x^{5/2} dx = 3 \cdot \frac{x^{7/2}}{7/2} = \frac{6}{7}x^{7/2}$, $\displaystyle\int 3x^2 dx = x^3$, and $\displaystyle\int x^{3/2} dx = \frac{2}{5}x^{5/2}$.

Final Answer

$\dfrac{x^4}{4} + \dfrac{6}{7}x^{7/2} + x^3 + \dfrac{2}{5}x^{5/2} + C$

Common Mistake: Trying to integrate $(x + \sqrt{x})^3$ using a substitution or the power rule directly on the whole bracket – the power rule only applies to x^n itself, not to a bracketed sum raised to a power; expansion is mandatory here.

Alternate Board-Style Example

Source: Exercise 7.2, Q6(ii)

Given: $\int \sqrt{x}\,(3x^2 + 2x + 3)\, dx$

Solution:

Distribute $\sqrt{x} = x^{1/2}$ into the bracket: $\sqrt{x}(3x^2 + 2x + 3) = 3x^{5/2} + 2x^{3/2} + 3x^{1/2}$.
Integrate term by term: $\displaystyle\int 3x^{5/2} dx = \frac{6}{7}x^{7/2}$, $\displaystyle\int 2x^{3/2} dx = \frac{4}{5}x^{5/2}$, $\displaystyle\int 3x^{1/2} dx = 2x^{3/2}$.

Final Answer

$\dfrac{6}{7}x^{7/2} + \dfrac{4}{5}x^{5/2} + 2x^{3/2} + C$

Quick Recall: If the integrand is a bracket raised to a power, or a product of two brackets, expand it fully into separate power terms first – there is no product or power rule for integration itself.

Type 8

Split a single algebraic fraction into several simpler fractions by dividing every term of the numerator by the denominator separately, turning one hard-looking integral into several easy power-rule integrals.

Formula Used: $\dfrac{A + B + C}{D} = \dfrac{A}{D} + \dfrac{B}{D} + \dfrac{C}{D}$

Source: Exercise 7.2, Q7(i)

Given: $\int \dfrac{x^3 + 5x^2 + 4x + 1}{x^2}\, dx$

Solution:

Divide each term of the numerator by x^2 separately: $\dfrac{x^3+5x^2+4x+1}{x^2} = \dfrac{x^3}{x^2}+\dfrac{5x^2}{x^2}+\dfrac{4x}{x^2}+\dfrac{1}{x^2} = x+5+\dfrac{4}{x}+x^{-2}$ – this converts one fraction into four separate power-rule-friendly terms.

Integrate each term: $\displaystyle\int x\,dx = \frac{x^2}{2}$, $\displaystyle\int 5\,dx = 5x$, $\displaystyle\int \frac{4}{x}\,dx = 4\log|x|$, $\displaystyle\int x^{-2}dx = -x^{-1} = -\frac{1}{x}$.

Final Answer

$$\frac{x^2}{2}+5x+4\log|x|-\frac{1}{x}+C$$

Common Mistake: Trying to cancel only part of the fraction, or attempting polynomial long division when simple term-by-term division already works – long division is only needed when the numerator's degree structure doesn't divide out cleanly term by term (see Type 12).

Quick Recall: If the denominator is a single term (a monomial), always split the fraction into one term per piece of the numerator – this is far faster than any other technique.

Type 9

Recognise a fraction that is already a standard trig derivative in disguise – most often $\cos x/\sin^2 x$ (the derivative of $-\operatorname{cosec} x$) or a ratio of reciprocal trig functions that reduces to $\tan^2 x$ or $\cot^2 x$ via a Pythagorean identity.

Formula Used: $\dfrac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x\cot x, \qquad \sec^2 x - 1 = \tan^2 x$

Source: Exercise 7.2, Q3(i)

Given: $\int \dfrac{2\cos x}{3\sin^2 x}\,dx$

Solution:

Pull out the constant and rewrite the trig ratio: $\dfrac{2\cos x}{3\sin^2 x} = \dfrac{2}{3}\cdot\dfrac{\cos x}{\sin x}\cdot\dfrac{1}{\sin x} = \dfrac{2}{3}\cot x\operatorname{cosec} x$ – this reveals the standard derivative of cosecant hiding inside.

Since $\dfrac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x\cot x$, we have $\displaystyle\int \operatorname{cosec} x\cot x\,dx = -\operatorname{cosec} x + C$, so our integral is $\dfrac{2}{3}$ of the negative of this.

Final Answer

$-\frac{2}{3}\operatorname{cosec} x + C$

Common Mistake: Not recognising $\cos x/\sin^2 x$ as a disguised standard derivative and instead attempting a substitution $u = \sin x$ from scratch – either approach works, but the direct recognition is much faster on a board exam.

Alternate Board-Style Example

Source: Exercise 7.2, Q11(ii)

Given: $\int \frac{\sec^2 x}{\operatorname{cosec}^2 x}\, dx$

Solution:

Rewrite the ratio of reciprocal trig functions in terms of sine and cosine: $\frac{\sec^2 x}{\operatorname{cosec}^2 x} = \frac{1/\cos^2 x}{1/\sin^2 x} = \frac{\sin^2 x}{\cos^2 x} = \tan^2 x$.

Using the Pythagorean identity $\tan^2 x = \sec^2 x - 1$: $\displaystyle\int (\sec^2 x - 1)\, dx = \tan x - x + C$.

Final Answer

$\tan x - x + C$

Quick Recall: A ratio of two "co-" trig functions (sec/cosec, tan/cot) almost always reduces to $\tan^2 x$ or $\cot^2 x$ after rewriting in sine/cosine – then swap in the matching Pythagorean identity to make it integrable.

Type 10

Use a double-angle identity to rewrite a trig numerator so the whole fraction collapses into one or more standard secant/cosecant-family integrals – distinct from Type 9 because multiple terms combine via an identity rather than a single ratio simplifying directly.

Formula Used: $\cos 2x = \cos^2 x - \sin^2 x$

Source: Exercise 7.2, Q15(ii)

Given: $\int \frac{\cos 2x}{\sin^2 x \cos^2 x}\, dx$

Solution:

Rewrite the numerator using $\cos 2x = \cos^2 x - \sin^2 x$, then split the fraction over the common denominator: $\dfrac{\cos^2 x - \sin^2 x}{\sin^2 x \cos^2 x} = \dfrac{\cos^2 x}{\sin^2 x \cos^2 x} - \dfrac{\sin^2 x}{\sin^2 x \cos^2 x} = \dfrac{1}{\sin^2 x} - \dfrac{1}{\cos^2 x} = \mathrm{cosec}^2 x - \sec^2 x$ – this is now a difference of two standard derivatives.

Integrating: $\displaystyle\int (\mathrm{cosec}^2 x - \sec^2 x)\,dx = -\cot x - \tan x + C$.

Final Answer

$-\cot x - \tan x + C$

Common Mistake: Trying to integrate $\cos 2x/(\sin^2 x \cos^2 x)$ directly via substitution without first splitting the numerator – the double-angle rewrite is what makes this tractable; skipping it leads to a much harder integral.

Alternate Board-Style Example

Source: Exercise 7.2, Q12(ii)

Given: $\int \dfrac{3 - 5\sin x}{\cos^2 x}\,dx$

Solution:

Split the fraction into two separate standard forms: $\dfrac{3 - 5\sin x}{\cos^2 x} = \dfrac{3}{\cos^2 x} - \dfrac{5\sin x}{\cos^2 x} = 3\sec^2 x - 5\sec x \tan x$, using $\dfrac{\sin x}{\cos^2 x} = \dfrac{\sin x}{\cos x} \cdot \dfrac{1}{\cos x} = \tan x \sec x$.

Integrating: $\displaystyle\int (3\sec^2 x - 5\sec x \tan x)\,dx = 3\tan x - 5\sec x + C$.

Final Answer

$3\tan x - 5\sec x + C$

Quick Recall: If a trig numerator has more than one term, look for a double-angle identity to rewrite it, then split the resulting fraction into separate standard secant/cosecant-family derivatives.

Type 11

Integrate a fraction built directly from the secant/cosecant family by expanding a product first – e.g. $\sec x(\sec x + \tan x)$ or $(1 - \sin x)/\cos^2 x$ – where distributing immediately reveals two standard integrable terms.

Formula Used: $\int \sec^2 x\,dx = \tan x + C, \qquad \int \sec x \tan x\,dx = \sec x + C$

Source: Exercise 7.2, Q11(i)

Given: $\int \sec x(\sec x + \tan x)\,dx$

Solution:

Distribute $\sec x$ into the bracket: $\sec x(\sec x + \tan x) = \sec^2 x + \sec x \tan x$ – both resulting terms are standard derivatives.
Integrating term by term: $\displaystyle\int \sec^2 x\, dx = \tan x$ and $\displaystyle\int \sec x \tan x\, dx = \sec x$.

Final Answer

$\tan x + \sec x + C$

Common Mistake: Trying to substitute or manipulate the product before simply distributing it – distributing first is always the fastest route when a bracket contains a sum being multiplied by a single trig factor.

Alternate Board-Style Example

Source: Exercise 7.2, Q10(i)

Given: $\int \dfrac{1 - \sin x}{\cos^2 x}\, dx$

Solution:

Split the fraction into two standard terms: $\dfrac{1 - \sin x}{\cos^2 x} = \dfrac{1}{\cos^2 x} - \dfrac{\sin x}{\cos^2 x} = \sec^2 x - \sec x \tan x$.
Integrating: $\displaystyle\int (\sec^2 x - \sec x \tan x)\, dx = \tan x - \sec x + C$.

Final Answer

$\tan x - \sec x + C$

Quick Recall: If you see a bracket containing $\sec x$ and $\tan x$ together (as a product or a fraction over $\cos^2 x$), distribute or split immediately – it always resolves into the two standard forms $\sec^2 x$ and $\sec x \tan x$.

Type 12

Rationalize a trig denominator by multiplying top and bottom by a conjugate expression (built from a Pythagorean identity), turning an otherwise unintegrable fraction into a standard secant/cosecant-family sum.

★ **Board Favorite** – Rationalizing trig denominators is a standard high-yield technique in this chapter – it reappears constantly (with $1 \pm \sin x$, $1 \pm \cos x$, $\sec x \pm \tan x$, $\operatorname{cosec} x \pm \cot x$), so recognising when to multiply by a conjugate is one of the most reusable skills from this exercise.

Formula Used: $1 - \sin^2 x = \cos^2 x, \qquad \sec^2 x - \tan^2 x = 1$

Source: Exercise 7.2, Q13(ii)

Given: $\int \dfrac{1}{1-\sin x}\,dx$

Solution:

Multiply numerator and denominator by the conjugate $1+\sin x$: $\dfrac{1}{1-\sin x}\cdot\dfrac{1+\sin x}{1+\sin x} = \dfrac{1+\sin x}{1-\sin^2 x} = \dfrac{1+\sin x}{\cos^2 x}$, using $1-\sin^2 x = \cos^2 x$.

Split the resulting fraction: $\dfrac{1+\sin x}{\cos^2 x} = \dfrac{1}{\cos^2 x} + \dfrac{\sin x}{\cos^2 x} = \sec^2 x + \sec x \tan x$.

Integrating: $\displaystyle\int (\sec^2 x + \sec x \tan x)\,dx = \tan x + \sec x + C$.

Final Answer

$\tan x + \sec x + C$

Common Mistake: Not recognising when a denominator needs rationalizing – if a trig denominator is a simple sum or difference of 1 and a single trig function, multiplying by its conjugate to trigger a Pythagorean identity is almost always the way in.

Alternate Board-Style Example

Source: Exercise 7.2, Q16(i)

Given: $\int \dfrac{\sec x}{\sec x + \tan x}\,dx$

Solution:

Multiply numerator and denominator by the conjugate $\sec x - \tan x$: $\dfrac{\sec x}{\sec x + \tan x}\cdot\dfrac{\sec x - \tan x}{\sec x - \tan x} = \dfrac{\sec^2 x - \sec x \tan x}{\sec^2 x - \tan^2 x} = \dfrac{\sec^2 x - \sec x \tan x}{1}$, using $\sec^2 x - \tan^2 x = 1$.

Integrating the resulting sum directly: $\displaystyle\int (\sec^2 x - \sec x \tan x)\,dx = \tan x - \sec x + C$.

Final Answer

$\tan x - \sec x + C$

Quick Recall: If a trig denominator is $1 \pm$ (something) or a sum/difference of two co-functions, multiply by the conjugate to trigger $\sin^2 + \cos^2 = 1$ or $\sec^2 - \tan^2 = 1$ – the denominator becomes 1 or a perfect square, and the rest falls into place.

Type 13

Use $\sin^2 x = 1 - \cos^2 x = (1-\cos x)(1+\cos x)$ (difference of squares) to factor a numerator so it cancels directly with a matching factor in the denominator – an algebraic factoring trick rather than a conjugate-multiplication trick.

Formula Used: $\sin^2 x = (1-\cos x)(1+\cos x)$

Source: Exercise 7.2, Q14(ii)

Given: $\int \dfrac{\sin^2 x}{1+\cos x}\,dx$

Solution:

Factor the numerator as a difference of squares: $\sin^2 x = 1-\cos^2 x = (1-\cos x)(1+\cos x)$.
Substituting, the $(1+\cos x)$ factor cancels directly with the denominator: $\dfrac{(1-\cos x)(1+\cos x)}{1+\cos x} = 1-\cos x$.
Integrating: $\displaystyle\int (1-\cos x)\,dx = x - \sin x + C$.

Final Answer

$x - \sin x + C$

Common Mistake: Reaching for a conjugate-multiplication trick (Type 12) here instead of noticing the numerator already factors to cancel with the denominator directly – always check for a direct factoring cancellation before multiplying by anything extra.

Quick Recall: If you see $\sin^2 x$ over $1 \pm \cos x$ (or the sine/cosine roles reversed), factor $\sin^2 x$ as a difference of squares first – it usually cancels a factor directly with the denominator, no conjugate needed.

Type 14

Simplify the argument of an inverse trig function using double-angle or half-angle identities until it collapses to the function's own inverse acting on a simple angle, i.e. reduces to a form like $\cot^{-1}(\cot x) = x$.

Formula Used: $\dfrac{\sin 2x}{1-\cos 2x} = \cot x, \qquad \sqrt{\dfrac{1-\cos 2x}{1+\cos 2x}} = |\tan x|$

Source: Exercise 7.2, Q19(i)

Given: $\int \cot^{-1}\left(\dfrac{\sin 2x}{1-\cos 2x}\right) dx$

Solution:

Simplify the argument using the half-angle forms $\sin 2x = 2\sin x\cos x$ and $1-\cos 2x = 2\sin^2 x$: $\dfrac{\sin 2x}{1-\cos 2x} = \dfrac{2\sin x\cos x}{2\sin^2 x} = \dfrac{\cos x}{\sin x} = \cot x$.

The integrand becomes $\cot^{-1}(\cot x) = x$ (on an interval within the principal branch).
Integrating: $\displaystyle\int x\,dx = \frac{x^2}{2} + C$.

Final Answer

$\dfrac{x^2}{2} + C$

Common Mistake: Trying to integrate the inverse trig expression directly (e.g. via integration by parts) without first simplifying its argument – always simplify what's *inside* an inverse trig function using double/half-angle identities before deciding on a technique.

Alternate Board-Style Example

Source: Exercise 7.2, Q20(ii)

Given: $\int \cos^{-1}\left(\dfrac{1-\tan^2 x}{1+\tan^2 x}\right) dx$

Solution:

Recognise the argument as the tangent form of the double-angle cosine identity: $\dfrac{1-\tan^2 x}{1+\tan^2 x} = \cos 2x$.
The integrand becomes $\cos^{-1}(\cos 2x) = 2x$ (on an interval within the principal branch).
Integrating: $\displaystyle\int 2x\,dx = x^2 + C$.

Final Answer

$x^2 + C$

Quick Recall: An inverse trig function wrapped around a messy trig fraction is almost always a disguised double-angle identity – simplify the argument fully first; it will collapse to a simple angle, turning the whole integral into a one-line power-rule problem.

Type 15

Find a function $f(x)$ given its derivative $f'(x)$ and an initial condition $f(x_0) = y_0$ – integrate $f'(x)$ to get the general antiderivative plus a constant C, then use the initial condition to pin down C.

Formula Used: $f(x) = \int f'(x)\,dx = F(x) + C$, then solve for C using the given condition

Source: Exercise 7.2, Q21

Given: $f'(x) = 4x^3 - 6,\ f(0) = 3;$ find $f(x)$

Solution:

Integrate $f'(x)$ term by term to get the general form of $f(x)$: $f(x) = \displaystyle\int (4x^3 - 6)\, dx = x^4 - 6x + C$.
Use the initial condition $f(0) = 3$ to find C: substituting $x = 0$ gives $f(0) = 0^4 - 6(0) + C = C$, so $C = 3$.

Final Answer

$f(x) = x^4 - 6x + 3$

Common Mistake: Forgetting to include the constant of integration C before applying the initial condition – without C, there is nothing for the given condition to solve for, and the answer will be wrong unless C happens to be zero.

Quick Recall: Integrate first (always include $+C$), then substitute the given point into the general antiderivative to solve for C – never skip the constant, since the initial condition exists specifically to pin it down.

Type 16

Integrate a "linear-inside" expression directly using the reverse chain rule – $(ax+b)^n$, e^{ax+b}, $\sec^2(ax+b)$, $\operatorname{cosec}(ax+b)\cot(ax+b)$, and similar, where the standard derivative formula is applied to a linear argument and divided by the coefficient of x.

Formula Used: $\int f(ax+b)\, dx = \dfrac{1}{a}F(ax+b) + C$, where F is the antiderivative of f

Source: Exercise 7.3, Q1(i)

Given: $(ax+b)^3$

Solution:

This matches the power rule applied to a linear argument rather than x alone – integrating $(ax+b)^n$ gives the usual power-rule antiderivative, but divided by a (the coefficient of x) to compensate for the chain rule that would appear if we differentiated back.
Applying this with $n = 3$: $\displaystyle\int (ax+b)^3\, dx = \frac{(ax+b)^4}{4a} + C$.

Final Answer

$\dfrac{1}{4a}(ax+b)^4 + C$

Common Mistake: Forgetting to divide by a (the coefficient of x inside the bracket) – this is the single most common slip across this entire "linear-inside" family of integrals.

Alternate Board-Style Example

Source: Exercise 7.3, Q3(ii)

Given: $5\,\text{cosec}^2(2-7x)$

Solution:

This matches the standard derivative $\dfrac{d}{dx}(-\cot x) = \text{cosec}^2 x$, applied to the linear argument $2-7x$ (coefficient of x is -7).
$\displaystyle\int 5\,\text{cosec}^2(2-7x)\,dx = 5\cdot\frac{-\cot(2-7x)}{-7} + C = \frac{5}{7}\cot(2-7x) + C.$

Final Answer

$\dfrac{5}{7}\cot(2-7x) + C$

Quick Recall: For any standard derivative applied to $(ax+b)$ instead of plain x, do the usual antiderivative and then divide by a – nothing else changes.

Type 17

Simplify an expression like $\sqrt{e^x}$ or $e^{k\log x}$ into a single power or exponential term before integrating – an index-law simplification that must happen before any integration formula applies.

Formula Used: $\sqrt{e^x} = e^{x/2}$

Source: Exercise 7.3, Q4(i)

Given: $\sqrt{e^x}$

Solution:

Rewrite the square root as a fractional exponent: $\sqrt{e^x} = (e^x)^{1/2} = e^{x/2}$, using the index law $(a^m)^n = a^{mn}$.

This is now a standard exponential in linear-argument form: $\displaystyle\int e^{x/2}\,dx = \frac{e^{x/2}}{1/2} + C = 2e^{x/2} + C.$

Final Answer

$2\sqrt{e^x} + C$

Common Mistake: Trying to integrate $\sqrt{e^x}$ directly without first rewriting it as $e^{x/2}$ – the square-root form doesn't match any standard integral pattern until it's converted to a single exponential.

Quick Recall: Any radical of an exponential is just the exponential with a fractional power – convert first, then apply the standard exponential integral.

Type 18

Simplify $\sqrt{1+\cos x}$ or $\sqrt{1+\sin x}$ using half-angle identities (writing $1+\cos x$ or $1+\sin x$ as a perfect square in terms of $x/2$) before integrating.

Formula Used: $1+\cos x = 2\cos^2\dfrac{x}{2}, \qquad 1+\sin x = \left(\cos\dfrac{x}{2}+\sin\dfrac{x}{2}\right)^2$

Source: Exercise 7.3, Q6(i)

Given: $\sqrt{1+\cos x},\ 0<x<\pi$

Solution:

Using the half-angle identity $1+\cos x = 2\cos^2\frac{x}{2}$: $\sqrt{1+\cos x} = \sqrt{2}\left|\cos\frac{x}{2}\right|$, and since $0<x<\pi$ means $0<\frac{x}{2}<\frac{\pi}{2}$ where cosine is positive, this simplifies to $\sqrt{2}\cos\frac{x}{2}$.
Integrating: $\displaystyle\int \sqrt{2}\cos\frac{x}{2}\,dx = \sqrt{2}\cdot\frac{\sin(x/2)}{1/2}+C = 2\sqrt{2}\sin\frac{x}{2}+C.$

Final Answer

$2\sqrt{2}\sin\dfrac{x}{2}+C$

Common Mistake: Not checking the given domain before dropping the absolute value – the half-angle expression is only positive (and hence simplifiable without a sign flip) on the specific interval given in the question.

Alternate Board-Style Example

Source: Exercise 7.3, Q6(ii)

Given: $\sqrt{1+\sin x},\ 0<x<\pi$

Solution:

Rewrite $1+\sin x$ as a perfect square: $1+\sin x = \cos^2\frac{x}{2}+2\sin\frac{x}{2}\cos\frac{x}{2}+\sin^2\frac{x}{2} = \left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)^2$, using $\sin x = 2\sin\frac{x}{2}\cos\frac{x}{2}$ and $\sin^2\frac{x}{2}+\cos^2\frac{x}{2}=1$.
Taking the square root (positive on this interval): $\sqrt{1+\sin x} = \cos\frac{x}{2}+\sin\frac{x}{2}$.
Integrating: $\displaystyle\int\left(\cos\frac{x}{2}+\sin\frac{x}{2}\right)dx = 2\sin\frac{x}{2}-2\cos\frac{x}{2}+C.$

Final Answer

$2\left(\sin\frac{x}{2} - \cos\frac{x}{2}\right) + C$

Quick Recall: $1 + \cos x$ is always $2\cos^2(x/2)$; $1 + \sin x$ is always a perfect square of $\cos(x/2) + \sin(x/2)$ – recognise these two exact patterns instantly rather than re-deriving them.

Type 19

Simplify a fraction by dividing the numerator by the denominator when the denominator is a monomial or when a linear divisor exactly divides the numerator (found by inspection or trial factoring), reducing to a simpler linear-plus-remainder or fully cancelling form.

Formula Used: $\frac{A}{B} = Q + \frac{R}{B}$, found by algebraic division

Source: Exercise 7.3, Q9(i)

Given: $\frac{x}{2x-3}$

Solution:

Since the numerator's degree equals the denominator's degree, we rewrite the numerator in terms of the denominator: $x = \frac{1}{2}(2x-3) + \frac{3}{2}$, so $\frac{x}{2x-3} = \frac{1}{2} + \frac{3/2}{2x-3}$ – this splits the fraction into a constant plus a simple reciprocal-linear term.

Integrating: $\int \left(\frac{1}{2} + \frac{3/2}{2x-3}\right) dx = \frac{1}{2}x + \frac{3}{2} \cdot \frac{1}{2} \log|2x-3| + C$.

Final Answer

$\frac{1}{2}x + \frac{3}{4}\log|2x-3| + C$

Common Mistake: Trying to integrate $x/(2x-3)$ directly as if it were already a log form – since the numerator isn't the exact derivative of the denominator, it must first be rewritten as constant + remainder over denominator.

Alternate Board-Style Example

Source: Exercise 7.3, Q9(ii)

Given: $\frac{x+2}{3x^2+5x-2}$

Solution:

Factor the denominator first: $3x^2 + 5x - 2 = (3x-1)(x+2)$.
The $(x+2)$ factor in the numerator cancels exactly with the matching factor in the denominator: $\dfrac{x+2}{(3x-1)(x+2)} = \dfrac{1}{3x-1}$.
Integrating: $\displaystyle\int \frac{1}{3x-1}\,dx = \frac{1}{3}\log|3x-1| + C$.

Final Answer

$\dfrac{1}{3}\log|3x-1| + C$

Quick Recall: Before doing any algebra, try factoring the denominator – if a factor cancels exactly with the numerator, the integral collapses to a single simple log form instantly.

Type 20

Substitute t =(the linear expression inside a square root or a squared bracket) to rewrite the whole numerator in terms of t, then integrate the resulting simpler expression in t.

Formula Used: Let $t =$ (linear expr.), $dt =$ (its deriv.) dx; rewrite x in t

Source: Exercise 7.3, Q10(i)

Given: $\dfrac{x}{(x+1)^2}$

Solution:

Let $t = x+1$, so $x = t-1$ and $dx = dt$ – this substitution is motivated by the repeated linear factor $(x+1)^2$ in the denominator.
Rewriting the numerator in terms of t: $\dfrac{x}{(x+1)^2} = \dfrac{t-1}{t^2} = \dfrac{1}{t} - \dfrac{1}{t^2}$, which splits into two simple power-rule terms.
Integrating in t: $\displaystyle\int\left(\frac{1}{t} - t^{-2}\right)dt = \log|t| + \frac{1}{t} + C$.
Converting back to x: $t = x+1$.

Final Answer

$\log|x+1| + \dfrac{1}{x+1} + C$

Common Mistake: Trying to integrate $x/(x+1)^2$ directly via a log-form recognition – the numerator isn't the derivative of the denominator, so a genuine substitution (rewriting x in terms of t) is required first.

Quick Recall: When the denominator is a power of a linear expression, substitute t for that linear expression and rewrite the entire numerator (including any lone x) in terms of t before integrating.

Type 21

Perform polynomial long division when the numerator's degree is at least as high as the denominator's and the two do not divide out cleanly by simple factoring – writing the fraction as a polynomial (quotient) plus a proper-fraction remainder.

Formula Used: $\dfrac{f(x)}{g(x)} = Q(x) + \dfrac{R(x)}{g(x)}$, found by polynomial long division

Source: Exercise 7.3, Q11(i)

Given: $\dfrac{x^3}{2x+1}$

Solution:

Since the numerator's degree (3) exceeds the denominator's degree (1), we perform polynomial long division of x^3 by $2x+1$, which gives a quotient of $\frac{1}{4}x^2 - \frac{1}{8}x + \frac{1}{16}$ with a remainder of $-\frac{1}{16}$.

So $\dfrac{x^3}{2x+1} = \dfrac{1}{4}x^2 - \dfrac{1}{8}x + \dfrac{1}{16} - \dfrac{1/16}{2x+1}$, splitting the improper fraction into a polynomial plus one simple reciprocal-linear term.

Integrating term by term: $\displaystyle\int \left(\frac{1}{4}x^2 - \frac{1}{8}x + \frac{1}{16} - \frac{1/16}{2x+1}\right) dx = \frac{x^3}{12} - \frac{x^2}{16} + \frac{x}{16} - \frac{1}{32}\log|2x+1| + C$.

Final Answer

$$\frac{x^3}{12} - \frac{x^2}{16} + \frac{x}{16} - \frac{1}{32}\log|2x+1| + C$$

Common Mistake: Attempting a substitution instead of long division when the numerator's degree is genuinely higher than the denominator's – long division is the correct and fastest tool whenever the fraction is "improper" (numerator degree $\geq$ denominator degree).

Quick Recall: If the top of a fraction has a higher (or equal) degree than the bottom, divide first – you'll get a polynomial (easy to integrate) plus a small proper-fraction remainder.

Type 22

Rationalize a denominator built from a sum or difference of two square roots by multiplying top and bottom by the conjugate surd, which uses the difference-of-squares identity to eliminate the roots from the denominator.

Formula Used: $(\sqrt{A}+\sqrt{B})(\sqrt{A}-\sqrt{B}) = A - B$

Source: Exercise 7.3, Q12(i)

Given: $\dfrac{1}{\sqrt{x+1}+\sqrt{x+2}}$

Solution:

Multiply numerator and denominator by the conjugate surd $\sqrt{x+2}-\sqrt{x+1}$: $\dfrac{1}{\sqrt{x+1}+\sqrt{x+2}} \cdot \dfrac{\sqrt{x+2}-\sqrt{x+1}}{\sqrt{x+2}-\sqrt{x+1}} = \dfrac{\sqrt{x+2}-\sqrt{x+1}}{(x+2)-(x+1)}$, using the difference-of-squares identity on the denominator.

The denominator simplifies to 1, leaving just $\sqrt{x+2}-\sqrt{x+1}$, a simple difference of two square-root terms.

Integrating each term using the power rule (each is $(\text{linear})^{1/2}$): $\displaystyle\int \left(\sqrt{x+2}-\sqrt{x+1}\right) dx = \frac{2}{3}(x+2)^{3/2} - \frac{2}{3}(x+1)^{3/2} + C.$

Final Answer

$\dfrac{2}{3}\left[(x+2)^{3/2}-(x+1)^{3/2}\right]+C$

Common Mistake: Trying to integrate the original fraction directly (with two surds added in the denominator) without rationalizing first – no standard formula applies to a sum of two surds in a denominator; rationalizing is mandatory.

Quick Recall: Whenever a denominator is a sum or difference of two square roots, multiply by the conjugate surd – the denominator becomes a simple linear difference (often just a constant), and what's left integrates directly.

Type 23

Use the product-to-sum formulas to convert a product of two trig functions of different (linear) arguments into a sum of two simple sine or cosine terms, each of which integrates directly.

★ **Board Favorite** – Product-to-sum conversion is a standard high-yield technique in this chapter – any integral of $\cos(mx)\cos(nx)$, $\sin(mx)\cos(nx)$, or similar with $m \neq n$ can only be integrated after this conversion, making it one of the most frequently reused formulas in the whole book.

Formula Used: $\cos A \cos B = \frac{1}{2}[\cos(A-B) + \cos(A+B)]$

Formula Used: $\sin A \cos B = \frac{1}{2}[\sin(A+B) + \sin(A-B)]$

Source: Exercise 7.3, Q17(i)

Given: $\cos 3x \cos 5x$

Solution:

Apply the product-to-sum formula for $\cos A \cos B$ with $A = 3x,\ B = 5x$: $\cos 3x \cos 5x = \frac{1}{2}[\cos(3x - 5x) + \cos(3x + 5x)] = \frac{1}{2}[\cos(-2x) + \cos 8x] = \frac{1}{2}[\cos 2x + \cos 8x]$, using $\cos(-\theta) = \cos\theta$.

Integrating each term (each is a standard $\cos(kx)$ form): $\int \frac{1}{2}(\cos 2x + \cos 8x)\,dx = \frac{1}{2}\left(\frac{\sin 2x}{2} + \frac{\sin 8x}{8}\right) + C$.

Final Answer

$\frac{1}{4}\sin 2x + \frac{1}{16}\sin 8x + C$

Common Mistake: Trying to integrate a product of two trig functions directly (e.g. via substitution or by parts) – a product of sine/cosine at two *different* linear arguments has no direct antiderivative; it must be converted to a sum first.

Alternate Board-Style Example

Source: Exercise 7.3, Q18(ii)

Given: $\sin x \sin 2x \sin 3x$

Solution:

Combine two of the three factors first using product-to-sum: $\sin x \sin 3x = \frac{1}{2}[\cos(x - 3x) - \cos(x + 3x)] = \frac{1}{2}[\cos 2x - \cos 4x]$ (using $\cos(A-B) - \cos(A+B)$ form for a product of sines).

Now multiply by the remaining factor $\sin 2x$: $\sin 2x \cdot \frac{1}{2}[\cos 2x - \cos 4x] = \frac{1}{2}[\sin 2x \cos 2x - \sin 2x \cos 4x]$, and apply product-to-sum again to each piece: $\sin 2x \cos 2x = \frac{1}{2}\sin 4x$, and

$\sin 2x \cos 4x = \frac{1}{2}[\sin 6x + \sin(-2x)] = \frac{1}{2}[\sin 6x - \sin 2x]$.

Combining and simplifying: the integrand becomes $\dfrac{1}{4}\sin 4x - \dfrac{1}{4}\sin 6x + \dfrac{1}{4}\sin 2x$, a sum of three standard sine terms.

Integrating: $\displaystyle\int \left(\frac{1}{4}\sin 4x - \frac{1}{4}\sin 6x + \frac{1}{4}\sin 2x\right) dx = -\frac{1}{16}\cos 4x + \frac{1}{24}\cos 6x - \frac{1}{8}\cos 2x + C$.

Final Answer

$$-\frac{1}{16}\cos 4x + \frac{1}{24}\cos 6x - \frac{1}{8}\cos 2x + C$$

Quick Recall: A product of two (or more) sines/cosines at different linear arguments always converts to a sum via the product-to-sum formulas – apply the conversion pairwise if there are three factors, then integrate each resulting simple sine/cosine term.

Type 24

Use repeated power-reduction (half-angle) identities to rewrite an even power of sine or cosine ($\sin^4 x$, $\cos^4 x$, $\sin^6 x$, etc.) as a sum of constant and $\cos(kx)$ terms, which then integrate directly.

Formula Used: $\sin^2 x = \dfrac{1-\cos 2x}{2}, \qquad \cos^2 x = \dfrac{1+\cos 2x}{2}$

Source: Exercise 7.3, Q14(ii)

Given: $\sin^4 x$

Solution:

Write $\sin^4 x = (\sin^2 x)^2$ and apply the power-reduction identity: $\sin^4 x = \left(\dfrac{1-\cos 2x}{2}\right)^2 = \dfrac{1 - 2\cos 2x + \cos^2 2x}{4}$, expanding the square.

The $\cos^2 2x$ term still has an even power, so we reduce it again using the same identity (with $2x$ in place of x): $\cos^2 2x = \dfrac{1+\cos 4x}{2}$.

Substituting back: $\sin^4 x = \dfrac{1}{4}\left(1 - 2\cos 2x + \dfrac{1+\cos 4x}{2}\right) = \dfrac{3}{8} - \dfrac{1}{2}\cos 2x + \dfrac{1}{8}\cos 4x$, after collecting like terms over a common denominator.

Integrating term by term: $\displaystyle\int \left(\frac{3}{8} - \frac{1}{2}\cos 2x + \frac{1}{8}\cos 4x\right) dx = \frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$.

Final Answer

$$\frac{3}{8}x - \frac{1}{4}\sin 2x + \frac{1}{32}\sin 4x + C$$

Common Mistake: Stopping after only one power-reduction step and leaving a $\cos^2 2x$ term un-integrated – any remaining even power of a trig function must itself be reduced again before the integral is fully in terms of simple $\cos(kx)$ and constant terms.

Quick Recall: Any even power of $\sin x$ or $\cos x$ reduces via the half-angle identity – apply it repeatedly (reducing power by half each time) until every term is a constant or a single $\cos(kx)$, then integrate term by term.

II. Integration by Substitution (Ex 7.3.2, 7.4–7.6)

Type 25

The integrand is a power of some inner function $g(x)$ multiplied by $g'(x)$ (the exact derivative of that inner function) – substituting $t = g(x)$ turns the whole integral into a simple power-rule problem in t.

★ **Board Favorite –** This is the single most common substitution pattern in the entire chapter – recognising "derivative of the inside is sitting right there as a factor" is the foundational substitution skill every other technique in this section builds on.

Formula Used: $\int [g(x)]^n \, g'(x) \, dx = \dfrac{[g(x)]^{n+1}}{n+1} + C$

Source: Exercise 7.4, Q1(ii)

Given: $(2x + 4)\sqrt{x^2 + 4x + 3}$

Solution:

Let $t = x^2 + 4x + 3$, so $\dfrac{dt}{dx} = 2x + 4$, i.e. $dt = (2x + 4)\, dx$ – we choose this substitution because $2x + 4$ is exactly the derivative of the expression under the square root.

Substituting, the integral becomes $\displaystyle\int \sqrt{t}\, dt = \int t^{1/2}\, dt = \frac{t^{3/2}}{3/2} + C = \frac{2}{3}t^{3/2} + C$.

Converting back to x: $t = x^2 + 4x + 3$.

Final Answer

$\dfrac{2}{3}(x^2 + 4x + 3)^{3/2} + C$

Common Mistake: Substituting t for the inner function but forgetting to check that its exact derivative (not just something proportional to it) is present in the integrand – if only a constant multiple is missing that's fine (adjust with a constant), but if the power of x doesn't match at all, this substitution won't work.

Alternate Board-Style Example

Source: Exercise 7.4, Q3(i)

Given: $\dfrac{3x + 1}{(3x^2 + 2x - 5)^3}$

Solution:

Let $t = 3x^2 + 2x - 5$, so $dt = (6x + 2)\,dx = 2(3x + 1)\,dx$ – the numerator $3x + 1$ is half of the exact derivative of the denominator's base.

Rewriting $(3x + 1)\,dx = \frac{1}{2}\,dt$, the integral becomes $\dfrac{1}{2}\displaystyle\int t^{-3}\,dt = \dfrac{1}{2} \cdot \dfrac{t^{-2}}{-2} + C = -\dfrac{1}{4t^2} + C$.

Converting back: $t = 3x^2 + 2x - 5$.

Final Answer

$$-\frac{1}{4(3x^2 + 2x - 5)^2} + C$$

Quick Recall: If you spot a power of an expression multiplied by something that looks like (a constant times) its derivative, substitute t for that expression – adjust for any missing constant factor, and the whole thing reduces to a one-line power-rule integral in t.

Type 26

The integrand is exactly $\dfrac{g'(x)}{g(x)}$ (the derivative of an inner function divided by the function itself) – this always integrates directly to $\log|g(x)|$, without needing to write out the substitution explicitly.

★ **Board Favorite** – The $f'(x)/f(x) \to \log|f(x)|$ pattern is a standard high-yield recognition skill tested throughout this chapter and reappears constantly inside harder integrals as a final simplifying step.

Formula Used: $\displaystyle\int \frac{g'(x)}{g(x)}\,dx = \log|g(x)| + C$

Source: Exercise 7.4, Q4(iv)

Given: $\dfrac{x^2}{1 + x^3}$

Solution:

Notice that the derivative of the denominator $1 + x^3$ is $3x^2$, and the numerator x^2 is exactly $\frac{1}{3}$ of that – this identifies the integrand as (a constant times) $g'(x)/g(x)$ with $g(x) = 1 + x^3$.

Rewrite the integrand to expose this ratio: $\dfrac{x^2}{1 + x^3} = \dfrac{1}{3} \cdot \dfrac{3x^2}{1 + x^3}$.

Applying the standard log-form result directly: $\dfrac{1}{3}\displaystyle\int \dfrac{3x^2}{1 + x^3}\,dx = \dfrac{1}{3}\log|1 + x^3| + C$.

Final Answer

$\frac{1}{3}\log|1+x^3|+C$

Common Mistake: Missing the constant-multiple adjustment – students sometimes see that the numerator is "close to" the denominator's derivative and either skip the needed constant entirely or misplace it.

Alternate Board-Style Example

Source: Exercise 7.4, Q9(ii)

Given: $\dfrac{\sin x}{1+\cos x}$

Solution:

The derivative of the denominator $1+\cos x$ is $-\sin x$, and the numerator is $\sin x$ – exactly the negative of that derivative.

Rewrite: $\dfrac{\sin x}{1+\cos x} = -\dfrac{-\sin x}{1+\cos x}$, matching the $g'(x)/g(x)$ pattern with a factor of -1.

Applying the log-form result: $\displaystyle\int \frac{\sin x}{1+\cos x}\,dx = -\log|1+\cos x|+C$.

Final Answer

$-\log|1+\cos x|+C$

Quick Recall: Whenever a fraction's numerator is (a constant multiple of) the exact derivative of its denominator, skip substitution notation entirely and write down log |denominator| directly, adjusting for the constant.

Type 27

Substitute t for an exponential expression (such as $a+be^x$) so the whole integral becomes a simple power-rule problem in t – a special case of Type 25 where the inner function is built from e^x.

Formula Used: Let $t=a+be^x,\ dt=be^x\,dx$

Source: Exercise 7.4, Q4(i)

Given: $e^x(a+be^x)^6$

Solution:

Let $t=a+be^x$, so $dt=be^x\,dx$, i.e. $e^x\,dx=\dfrac{dt}{b}$ – this substitution is motivated by the e^x factor sitting outside, which is (up to the constant b) exactly the derivative of what's inside the bracket.

Substituting, the integral becomes $\frac{1}{b}\int t^6\,dt = \frac{1}{b}\cdot\frac{t^7}{7} + C$.
Converting back: $t = a + be^x$.

Final Answer

$\frac{1}{7b}(a+be^x)^7 + C$

Common Mistake: Forgetting to solve for $e^x\,dx$ in terms of dt correctly (missing the $\frac{1}{b}$ factor) – always isolate exactly the piece of the original integrand ($e^x\,dx$ here) in terms of dt before substituting.

Quick Recall: If you see e^x multiplying a power of $(a + be^x)$, substitute t for the whole bracket – the lone e^x factor is exactly what's needed to convert dx into dt.

Type 28

Substitute $t = \log x$ when the integrand contains a power of $\log x$ together with a factor of $\frac{1}{x}$ (which is exactly the derivative of $\log x$) – reduces the problem to a simple power-rule integral in t.

Formula Used: Let $t = \log x,\ dt = \frac{1}{x}\,dx$

Source: Exercise 7.4, Q4(iii)

Given: $\frac{(\log x)^2}{x}$

Solution:

Let $t = \log x$, so $dt = \frac{1}{x}\,dx$ – the $\frac{1}{x}$ factor in the integrand is exactly what's needed to convert dx to dt.
Substituting, the integral becomes $\int t^2\,dt = \frac{t^3}{3} + C$.
Converting back: $t = \log x$.

Final Answer

$\frac{(\log x)^3}{3} + C$

Common Mistake: Trying to integrate $(\log x)^2$ using integration by parts instead of recognising the much faster substitution route available here because of the accompanying $\frac{1}{x}$ factor – always check for a ready-made dt factor before reaching for a harder technique.

Quick Recall: A power of $\log x$ multiplied by $\frac{1}{x}$ is always a substitution $t = \log x$ away from a one-line power-rule integral.

Type 29

Substitute t for an inverse trig function (such as $\tan^{-1} x$ or $\sin^{-1} 2x$) when its exact derivative appears as an accompanying factor – another special case of Type 25, built specifically from inverse trig derivatives.

Formula Used: $\dfrac{d}{dx}(\tan^{-1} x) = \dfrac{1}{1+x^2}$, $\quad \dfrac{d}{dx}(\sin^{-1} 2x) = \dfrac{2}{\sqrt{1-4x^2}}$

Source: Exercise 7.4, Q6(i)

Given: $\dfrac{\sqrt{\tan^{-1} x}}{1+x^2}$

Solution:

Let $t = \tan^{-1} x$, so $dt = \dfrac{1}{1+x^2}\,dx$ – the factor $\dfrac{1}{1+x^2}$ in the integrand is exactly the derivative of $\tan^{-1} x$.

Substituting, the integral becomes $\displaystyle\int \sqrt{t}\,dt = \frac{t^{3/2}}{3/2} + C = \frac{2}{3}t^{3/2} + C$.

Converting back: $t = \tan^{-1} x$.

Final Answer

$\dfrac{2}{3}(\tan^{-1} x)^{3/2} + C$

Common Mistake: Not recognising $\dfrac{1}{1+x^2}$ or $\dfrac{1}{\sqrt{1-4x^2}}$-type expressions as inverse-trig derivatives in disguise – memorise the derivatives of all six inverse trig functions so this pattern jumps out immediately.

Alternate Board-Style Example

Source: Exercise 7.4, Q6(ii)

Given: $\dfrac{(\sin^{-1} 2x)^3}{\sqrt{1-4x^2}}$

Solution:

Let $t = \sin^{-1} 2x$, so $dt = \dfrac{2}{\sqrt{1-4x^2}}\,dx$, i.e. $\dfrac{dx}{\sqrt{1-4x^2}} = \dfrac{dt}{2}$.

Substituting, the integral becomes $\displaystyle\frac{1}{2}\int t^3\,dt = \frac{1}{2}\cdot\frac{t^4}{4} + C = \frac{t^4}{8} + C$.

Converting back: $t = \sin^{-1} 2x$.

Final Answer

$$\frac{(\sin^{-1} 2x)^4}{8} + C$$

Quick Recall: Memorise all six inverse-trig derivatives – whenever one of those exact expressions appears as a multiplying factor, substitute t for the corresponding inverse trig function.

Type 30

Integrate a product of an odd power of one trig function and any power of another (e.g. $\sin x \cos^5 x$) by substituting t for the trig function with the *even* power, peeling off one factor of the odd-power function to supply dt.

Formula Used: Let $t = \cos x$ (or $\sin x$); peel one factor to form dt

Source: Exercise 7.4, Q7(i)

Given: $\sin x \cos^5 x$

Solution:

Let $t = \cos x$, so $dt = -\sin x\, dx$, i.e. $\sin x\, dx = -dt$ – we choose $t = \cos x$ because $\sin x$ (an odd, single power) is exactly what's needed to supply dt once we peel it off.

Substituting, the integral becomes $\displaystyle\int t^5\,(-dt) = -\frac{t^6}{6} + C$.

Converting back: $t = \cos x$.

Final Answer

$$-\frac{\cos^6 x}{6} + C$$

Common Mistake: Trying to substitute $t = \sin x$ instead – that choice leaves a $\cos^5 x$ factor with no matching dt available, since $\cos x\, dx = dt$ only supplies a single power of cosine, not five; always substitute for the function whose complementary factor is a clean single power.

Alternate Board-Style Example

Source: Exercise 7.4, Q8(i)

Given: $\sec^3 x \tan x$

Solution:

Rewrite $\sec^3 x \tan x = \sec^2 x \cdot (\sec x \tan x)$ – this exposes $\sec x \tan x$, the exact derivative

of $\sec x$, as one factor.
Let $t = \sec x$, so $dt = \sec x \tan x \, dx$.
Substituting: $\displaystyle\int t^2 \, dt = \frac{t^3}{3} + C$.
Converting back: $t = \sec x$.

Final Answer

$$\frac{\sec^3 x}{3} + C$$

Quick Recall: For a product of trig powers, substitute for whichever function has the accompanying single-power "leftover" factor available to form dt – peel off exactly one factor of the odd-power function to make this work.

Type 31

Substitute t for a trig expression sitting under a square root (or otherwise raised to a power) in the denominator, when the numerator supplies exactly the derivative needed – a variant of Type 25 where the "inner function" itself involves a trig term.

Formula Used: $\int \frac{g'(x)}{\sqrt{g(x)}} \, dx = 2\sqrt{g(x)} + C$

Source: Exercise 7.4, Q11(i)

Given: $\dfrac{\sin x}{\sqrt{3 + 2\cos x}}$

Solution:

Let $t = 3+2\cos x$, so $dt = -2\sin x \, dx$, i.e. $\sin x \, dx = -\dfrac{dt}{2}$ – chosen because the numerator $\sin x$ is (up to a constant) the derivative of what's under the root.
Substituting, the integral becomes $-\dfrac{1}{2}\displaystyle\int t^{-1/2} \, dt = -\frac{1}{2} \cdot \frac{t^{1/2}}{1/2} + C = -\sqrt{t} + C$.
Converting back: $t = 3 + 2\cos x$.

Final Answer

$$-\sqrt{3 + 2\cos x} + C$$

Common Mistake: Missing the sign flip that comes from differentiating $\cos x$ (giving $-\sin x$, not $+\sin x$) – always differentiate the substituted expression carefully rather than assuming the sign matches the original integrand.

Quick Recall: A trig function under a square root, with its derivative (up to a constant) sitting in the numerator, is always a direct substitution to a $2\sqrt{t}$-type result.

Type 32

Reduce an integral like $\tan^4 x$ or $\cot^4 x$ by repeatedly applying the Pythagorean identity to peel off a factor of $\sec^2 x$ (or $\operatorname{cosec}^2 x$), splitting the integral into simpler pieces that combine substitution with direct standard forms.

Formula Used: $\tan^2 x = \sec^2 x - 1$

Source: Exercise 7.4, Q14(i)

Given: $\tan^4 x$

Solution:

Split off a factor of $\tan^2 x$ and rewrite it using the Pythagorean identity: $\tan^4 x = \tan^2 x \cdot \tan^2 x = \tan^2 x(\sec^2 x - 1) = \tan^2 x \sec^2 x - \tan^2 x$ – this separates the integral into a substitution-friendly piece and a piece that needs the identity applied once more.

For $\int \tan^2 x \sec^2 x \, dx$: let $t = \tan x$, so $dt = \sec^2 x \, dx$, giving $\int t^2 \, dt = \frac{t^3}{3} = \frac{\tan^3 x}{3}$.

For $\int \tan^2 x \, dx = \int (\sec^2 x - 1) \, dx = \tan x - x$, applying the Pythagorean identity a second time.

Combining both pieces (note the subtraction from the original split): $\int \tan^4 x \, dx = \frac{\tan^3 x}{3} - (\tan x - x) + C$.

Final Answer

$\frac{1}{3}\tan^3 x - \tan x + x + C$

Common Mistake: Applying the Pythagorean identity only once and leaving a $\tan^2 x$ term un-integrated – any leftover even power of $\tan x$ (or $\cot x$) needs the same identity applied again until only $\sec^2 x$ (or $\operatorname{cosec}^2 x$) and constant terms remain.

Quick Recall: For any even power of $\tan x$ or $\cot x$, peel off $\tan^2 x = \sec^2 x - 1$ (or the cot/cosec analogue) repeatedly – each round either gives a direct substitution term or a lower power to reduce again.

Type 33

Recognise a mixed algebraic-trigonometric expression of the exact form $\frac{g'(x)}{g(x)}$ where $g(x)$ itself is a sum of an algebraic and a trig term (e.g. $x + \log \sec x$, or $x + \sin^2 x$) – the same log-form recognition as Type 26, applied to a compound inner function.

Formula Used: $\int \frac{g'(x)}{g(x)}\,dx = \log|g(x)| + C$

Source: Exercise 7.4, Q15(i)

Given: $\dfrac{1+\tan x}{x+\log\sec x}$

Solution:

Differentiate the denominator to check the match: $\frac{d}{dx}(x+\log\sec x) = 1 + \frac{\sec x\tan x}{\sec x} = 1+\tan x$, using $\frac{d}{dx}(\log\sec x) = \tan x$ – this is exactly the numerator.
Since the numerator is precisely the derivative of the denominator, the integral is a direct log form.

Final Answer

$\log|x+\log\sec x| + C$

Common Mistake: Not recognising $\frac{d}{dx}(\log\sec x) = \tan x$ as a standard derivative – memorising this (and its companions $\frac{d}{dx}\log\sin x = \cot x$, etc.) makes spotting this pattern instant rather than requiring re-derivation each time.

Alternate Board-Style Example

Source: Exercise 7.4, Q17(i)

Given: $\dfrac{2\cos x - 3\sin x}{6\cos x + 4\sin x}$

Solution:

We look for constants A, B such that the numerator equals $A\cdot$(derivative of denominator)$+$ $B\cdot$(denominator): the derivative of $6\cos x + 4\sin x$ is $-6\sin x + 4\cos x$.
Solving $2\cos x - 3\sin x = A(-6\sin x + 4\cos x) + B(6\cos x + 4\sin x)$ by matching $\cos x$ and $\sin x$ coefficients gives $4A + 6B = 2$ and $-6A + 4B = -3$; solving this pair simultaneously gives $A = \frac{1}{2},\ B = 0$.
So the numerator is exactly $\frac{1}{2}$ times the denominator's derivative, making this a direct log form: $\displaystyle\int \frac{2\cos x - 3\sin x}{6\cos x + 4\sin x}\,dx = \frac{1}{2}\log|6\cos x + 4\sin x| + C$.

Final Answer

$\frac{1}{2}\log|6\cos x + 4\sin x| + C$

Quick Recall: For a ratio of two linear trig combinations (or an algebraic-plus-trig sum), differentiate the denominator and compare with the numerator – if they match (exactly or up to a constant), it's an instant log form.

Type 34

Manipulate an expression algebraically first – dividing through by a common exponential factor, or rewriting using the definitions of a ratio of trig functions – before the $g'(x)/g(x)$ log-form pattern becomes visible.

Formula Used: $\dfrac{e^{2x}+1}{e^{2x}-1} = \dfrac{e^x+e^{-x}}{e^x-e^{-x}}$ (divide numerator and denominator by e^x)

Source: Exercise 7.4, Q19(ii)

Given: $\dfrac{e^{2x}+1}{e^{2x}-1}$

Solution:

Divide both numerator and denominator by e^x: $\dfrac{e^{2x}+1}{e^{2x}-1} = \dfrac{e^x+e^{-x}}{e^x-e^{-x}}$ – this rewriting is what reveals the log-form pattern, since the derivative of the new denominator matches the new numerator.

Checking: $\dfrac{d}{dx}(e^x-e^{-x}) = e^x+e^{-x}$, exactly the new numerator.

This is now a direct log form: $\displaystyle\int \frac{e^x+e^{-x}}{e^x-e^{-x}}\,dx = \log|e^x-e^{-x}|+C$.

Final Answer

$\log|e^x-e^{-x}|+C$

Common Mistake: Trying to integrate the original e^{2x}-based fraction directly without first dividing by e^x – the log-form pattern is completely hidden until this algebraic rewriting step is done.

Alternate Board-Style Example

Source: Exercise 7.4, Q21(i)

Given: $\dfrac{1-\cot x}{1+\cot x}$

Solution:

Rewrite $\cot x = \dfrac{\cos x}{\sin x}$ and multiply numerator and denominator by $\sin x$: $\dfrac{1-\cot x}{1+\cot x} = \dfrac{\sin x-\cos x}{\sin x+\cos x}$ – this algebraic rewriting exposes a log-form pattern.

Checking: $\dfrac{d}{dx}(\sin x + \cos x) = \cos x - \sin x = -(\sin x - \cos x)$, exactly the negative of the new numerator.
This is a direct log form with a sign flip: $\displaystyle\int \frac{\sin x - \cos x}{\sin x + \cos x}\,dx = -\log|\sin x + \cos x| + C$.

Final Answer

$-\log|\sin x + \cos x| + C$

Quick Recall: If the $g'(x)/g(x)$ pattern isn't visible, try a simple algebraic rewrite first – dividing by a common exponential factor, or converting cot/tan to sine/cosine form – the pattern is often hiding just one step away.

Type 35

The integrand is a trig function wrapped around a power of x (e.g. $\sin(x^2+1)$, $\cos(x^4)$), multiplied by exactly the derivative of that power – substitute t for the inner polynomial, turning the whole thing into a one-line standard trig integral.

Formula Used: $\int g'(x)\sin(g(x))\,dx = -\cos(g(x)) + C, \qquad \int g'(x)\cos(g(x))\,dx = \sin(g(x)) + C$

Source: Exercise 7.5, Q1(i)

Given: $2x\sin(x^2+1)$

Solution:

Let $t = x^2 + 1$, so $dt = 2x\,dx$ – the factor $2x$ in the integrand is exactly the derivative of the inner expression $x^2 + 1$.
Substituting, the integral becomes $\displaystyle\int \sin t\,dt = -\cos t + C$.
Converting back: $t = x^2 + 1$.

Final Answer

$-\cos(x^2+1) + C$

Common Mistake: Trying to integrate $\sin(x^2+1)$ as if it were $\sin x$ with a simple linear shift – since the inside is a genuine function of x (not linear), a real substitution (with a matching derivative factor) is required, not just the "linear-inside" shortcut from Type 16.

Quick Recall: A trig function of a polynomial in x, multiplied by that polynomial's exact derivative, substitutes directly – the trig integral of t falls out immediately.

Type 36

Substitute $t = \sqrt{x}$ when the integrand is a trig function of $\sqrt{x}$ divided by $\sqrt{x}$ – the accompanying $\dfrac{1}{\sqrt{x}}$ factor is (up to a constant) exactly the derivative of $\sqrt{x}$.

Formula Used: Let $t = \sqrt{x},\ dt = \dfrac{1}{2\sqrt{x}}\,dx$

Source: Exercise 7.5, Q2(i)

Given: $\dfrac{\cos\sqrt{x}}{\sqrt{x}}$

Solution:

Let $t = \sqrt{x}$, so $dt = \dfrac{1}{2\sqrt{x}}\,dx$, i.e. $\dfrac{dx}{\sqrt{x}} = 2\,dt$ – this substitution is motivated by the $\dfrac{1}{\sqrt{x}}$ factor, which matches (up to the constant 2) the derivative of $\sqrt{x}$.

Substituting, the integral becomes $\displaystyle\int \cos t \cdot 2\,dt = 2\sin t + C$.

Converting back: $t = \sqrt{x}$.

Final Answer

$2\sin\sqrt{x} + C$

Common Mistake: Forgetting the factor of 2 that comes from solving $dt = \frac{1}{2\sqrt{x}}dx$ for $dx/\sqrt{x}$ – this constant is easy to drop since it doesn't "look like" it belongs to the original integrand.

Quick Recall: A trig function of $\sqrt{x}$ divided by $\sqrt{x}$ always substitutes to $t = \sqrt{x}$ with an extra factor of 2 – memorise this pattern since it appears constantly.

Type 37

Substitute $t = \dfrac{1}{x}$ when the integrand is a trig function of $\dfrac{1}{x}$ divided by x^2 – the $\dfrac{1}{x^2}$ factor is exactly (up to a sign) the derivative of $\dfrac{1}{x}$.

Formula Used: Let $t = \dfrac{1}{x},\ dt = -\dfrac{1}{x^2}\,dx$

Source: Exercise 7.5, Q5(ii)

Given: $\dfrac{1}{x^2}\sin^2\left(\dfrac{1}{x}\right)$

Solution:

Let $t = \dfrac{1}{x}$, so $dt = -\dfrac{1}{x^2}\,dx$, i.e. $\dfrac{dx}{x^2} = -dt$ – chosen because $\dfrac{1}{x^2}$ is exactly the (negative of the) derivative of $\dfrac{1}{x}$.

Substituting: $\displaystyle\int \sin^2 t\,(-dt) = -\int \sin^2 t\,dt$.

Using the power-reduction identity $\sin^2 t = \dfrac{1-\cos 2t}{2}$: $\displaystyle -\int \frac{1-\cos 2t}{2}\,dt = -\frac{t}{2} + \frac{\sin 2t}{4} + C$.

Converting back: $t = \dfrac{1}{x}$.

Final Answer

$-\dfrac{1}{2x} + \dfrac{1}{4}\sin\dfrac{2}{x} + C$

Common Mistake: Forgetting the negative sign that comes from $dt = -\dfrac{1}{x^2}dx$ – since $\dfrac{d}{dx}\left(\dfrac{1}{x}\right) = -\dfrac{1}{x^2}$, this sign flip is essential and easy to lose track of.

Quick Recall: A trig function of $\dfrac{1}{x}$ divided by x^2 substitutes to $t = \dfrac{1}{x}$ with a sign flip – watch the negative sign carefully.

Type 38

Substitute $t = \log x$ when the integrand is a trig function of $\log x$ divided by x – the same substitution idea as Type 28, but with a trig outer function instead of a power.

Formula Used: Let $t = \log x,\ dt = \dfrac{1}{x}\,dx$

Source: Exercise 7.5, Q6(ii)

Given: $\dfrac{\operatorname{cosec}^2(\log x)}{x}$

Solution:

Let $t = \log x$, so $dt = \dfrac{1}{x}\,dx$ – the $\dfrac{1}{x}$ factor is exactly the derivative of $\log x$.

Substituting, the integral becomes $\displaystyle\int \operatorname{cosec}^2 t\,dt = -\cot t + C$.

Converting back: $t = \log x$.

Final Answer

$-\cot(\log x) + C$

Common Mistake: Confusing this with Type 28 and trying to apply the power rule instead of a standard trig integral once substituted – always re-check which formula matches the *outer* function after substitution, not just recognise "this is a $\log x$ substitution" and stop thinking.

Quick Recall: Any trig function of $\log x$, divided by x, is a direct $t = \log x$ substitution – what changes from Type 28 is only the standard integral you apply to t afterward.

Type 39

Substitute $t = \tan^{-1} x$ when the integrand is an exponential or trig function of $\tan^{-1} x$ divided by $1 + x^2$ – extends Type 29's substitution idea to a different outer function.

Formula Used: Let $t = \tan^{-1} x,\ dt = \dfrac{1}{1+x^2}\,dx$

Source: Exercise 7.5, Q7(i)

Given: $\dfrac{e^{\tan^{-1} x}}{1+x^2}$

Solution:

Let $t = \tan^{-1} x$, so $dt = \dfrac{1}{1+x^2}\,dx$ – the accompanying factor is exactly the derivative of $\tan^{-1} x$.

Substituting, the integral becomes $\displaystyle\int e^t\,dt = e^t + C$.

Converting back: $t = \tan^{-1} x$.

Final Answer

$e^{\tan^{-1} x} + C$

Common Mistake: Not recognising $\dfrac{1}{1+x^2}$ as the derivative of $\tan^{-1} x$ when it's paired with an unfamiliar outer function like e^t – the substitution trigger is always the accompanying derivative factor, regardless of what function wraps around $\tan^{-1} x$.

Alternate Board-Style Example

Source: Exercise 7.5, Q7(ii)

Given: $\dfrac{\sin(2\tan^{-1}x)}{1+x^2}$

Solution:

Let $t = \tan^{-1}x$, so $dt = \dfrac{1}{1+x^2}\,dx$.
Substituting: $\displaystyle\int \sin 2t\,dt = -\frac{\cos 2t}{2} + C$.
Converting back: $t = \tan^{-1}x$.

Final Answer

$-\dfrac{1}{2}\cos(2\tan^{-1}x) + C$

Quick Recall: $\dfrac{1}{1+x^2}$ paired with *any* function of $\tan^{-1}x$ is always a direct $t = \tan^{-1}x$ substitution – only the final standard integral applied to t changes.

Type 40

Substitute t for a compound inner expression (such as xe^x) whose derivative requires the product rule – the integrand supplies exactly this product-rule derivative as an accompanying factor.

Formula Used: Let $t = xe^x$, $dt = (e^x + xe^x)\,dx = (1+x)e^x\,dx$ (by the product rule)

Source: Exercise 7.5, Q9(ii)

Given: $\dfrac{(x+1)e^x}{\cot^2(xe^x)}$

Solution:

Let $t = xe^x$; differentiating using the product rule, $dt = (e^x + xe^x)\,dx = (1+x)e^x\,dx$ – this matches the numerator of the integrand exactly.
Rewrite the denominator using $\dfrac{1}{\cot^2\theta} = \tan^2\theta$: the integral becomes $\displaystyle\int \tan^2 t\,dt$.
Using $\tan^2 t = \sec^2 t - 1$: $\displaystyle\int (\sec^2 t - 1)\,dt = \tan t - t + C$.
Converting back: $t = xe^x$.

Final Answer

$\tan(xe^x) - xe^x + C$

Common Mistake: Trying to substitute $t = xe^x$ without checking whether its *product-rule* derivative actually matches the numerator – unlike simpler substitutions, a compound inner function like xe^x needs the product rule applied carefully before you can confirm the substitution works.

Quick Recall: If the inner expression is a product (like xe^x), differentiate it with the product rule before deciding whether the substitution will work – the numerator must match this product-rule derivative, not just one piece of it.

Type 41

Integrate an odd power of one trig function multiplied by another trig factor with a compound (non-x) argument, or paired with a square root – the same peeling technique as Type 30, applied to a disguised or compound argument.

Formula Used: Peel one odd-power factor to supply dt (Type 30)

Source: Exercise 7.5, Q12(i)

Given: $\sin^3(2x+1)$

Solution:

Split off one factor of sine and rewrite the rest using the Pythagorean identity: $\sin^3(2x+1) = \sin^2(2x+1)\cdot\sin(2x+1) = [1-\cos^2(2x+1)]\sin(2x+1)$.

Let $t = \cos(2x+1)$, so $dt = -2\sin(2x+1)\,dx$, i.e. $\sin(2x+1)\,dx = -\dfrac{dt}{2}$.

Substituting: $\displaystyle\int(1-t^2)\left(-\frac{dt}{2}\right) = -\frac{1}{2}\int(1-t^2)\,dt = -\frac{1}{2}\left(t-\frac{t^3}{3}\right)+C.$

Converting back: $t = \cos(2x+1)$.

Final Answer

$-\dfrac{1}{2}\cos(2x+1)+\dfrac{1}{6}\cos^3(2x+1)+C$

Common Mistake: Treating the compound argument $(2x+1)$ as if it were plain x and forgetting the extra factor of 2 that appears in dt from the chain rule – always differentiate the actual compound argument carefully.

Alternate Board-Style Example

Source: Exercise 7.5, Q14(ii)

Given: $\dfrac{\cos^3 x}{\sqrt{\sin x}}$

Solution:

Split off one cosine factor and rewrite: $\dfrac{\cos^3 x}{\sqrt{\sin x}} = \cos^2 x \cdot \dfrac{\cos x}{\sqrt{\sin x}} = (1-\sin^2 x)\cdot\dfrac{\cos x}{\sqrt{\sin x}}$.
Let $t = \sin x$, so $dt = \cos x\,dx$.
Substituting: $\displaystyle\int \frac{1-t^2}{\sqrt{t}}\,dt = \int \left(t^{-1/2} - t^{3/2}\right)dt = 2t^{1/2} - \frac{2}{5}t^{5/2} + C$.
Converting back: $t = \sin x$.

Final Answer

$2\sqrt{\sin x} - \dfrac{2}{5}\sin^{5/2} x + C$

Quick Recall: Odd powers of a trig function always peel off one factor for dt, no matter how the rest of the integrand looks (compound argument, square root, or otherwise) – rewrite the remaining even power using the Pythagorean identity before substituting.

Type 42

Integrate a higher odd power of sine or cosine (such as $\sin^5 x$, $\sin^7 x$) by peeling off one factor and rewriting the remaining even power fully via the Pythagorean identity, then expanding before integrating term by term – the same core idea as Type 30/41 but the expansion has more terms.

Formula Used: $\sin^2 x = 1 - \cos^2 x$, applied repeatedly before expanding

Source: Exercise 7.5, Q13(i)

Given: $\sin^7 x$

Solution:

Split off one factor of sine: $\sin^7 x = (\sin^2 x)^3 \sin x = (1-\cos^2 x)^3 \sin x$, using the Pythagorean identity on the remaining even power.
Let $t = \cos x$, so $dt = -\sin x\,dx$.
Substituting: $\displaystyle\int (1-t^2)^3(-dt) = -\int (1 - 3t^2 + 3t^4 - t^6)\,dt$, after expanding the cube using the binomial theorem.
Integrating term by term: $-\left(t - t^3 + \dfrac{3t^5}{5} - \dfrac{t^7}{7}\right) + C$.
Converting back: $t = \cos x$.

Final Answer

$-\cos x + \cos^3 x - \dfrac{3}{5}\cos^5 x + \dfrac{1}{7}\cos^7 x + C$

Common Mistake: Making an arithmetic error while expanding $(1-t^2)^3$ – with a cube, there are four terms to track carefully via the binomial theorem; rushing this step is the main source of error in this type.

Quick Recall: For any odd power beyond the cube, the method is unchanged from Type 30/41 – peel one factor, rewrite the (now even) remaining power using the Pythagorean identity, expand fully, then integrate term by term.

Type 43

Recognise the special standard derivatives $\dfrac{d}{dx}\log\left(\tan\dfrac{x}{2}\right) = \operatorname{cosec} x$ and $\dfrac{d}{dx}\log(\operatorname{cosec} x - \cot x) = \operatorname{cosec} x$ – less common than the basic six trig derivatives, but essential for recognising certain log-form integrals instantly.

★ **Board Favorite** – These two log-cosecant identities are a standard high-yield memorised result – without recognising them, the corresponding integral becomes very difficult to derive from scratch under exam time pressure.

Formula Used: $\dfrac{d}{dx}\log\left(\tan\dfrac{x}{2}\right) = \operatorname{cosec} x$

Source: Exercise 7.5, Q16(ii)

Given: $\dfrac{\log\left(\tan\frac{x}{2}\right)}{\sin x}$

Solution:

Recognise that $\dfrac{1}{\sin x} = \operatorname{cosec} x$, and that $\operatorname{cosec} x$ is exactly the derivative of $\log\left(\tan\frac{x}{2}\right)$ – this is a memorised standard result, not something derived on the spot.
So the integrand is of the exact form $g(x)\cdot g'(x)$ where $g(x) = \log\left(\tan\frac{x}{2}\right)$.
Let $t = \log\left(\tan\frac{x}{2}\right)$, so $dt = \operatorname{cosec} x\, dx$.
Substituting: $\displaystyle\int t\, dt = \frac{t^2}{2} + C$.
Converting back: $t = \log\left(\tan\frac{x}{2}\right)$.

Final Answer

$\dfrac{1}{2}\left[\log\left(\tan\dfrac{x}{2}\right)\right]^2 + C$

Common Mistake: Not having this derivative memorised and trying to differentiate $\log\tan(x/2)$ from scratch under exam pressure – this specific identity (along with its cosecant-cotangent companion) is worth memorising directly since re-deriving it wastes valuable time.

Alternate Board-Style Example

Source: Exercise 7.5, Q15(i)

Given: $\operatorname{cosec} x \log(\operatorname{cosec} x - \cot x)$

Solution:

Recognise that $\operatorname{cosec} x$ is exactly the derivative of $\log(\operatorname{cosec} x - \cot x)$ – the second of the two special log-trig identities.
Let $t = \log(\operatorname{cosec} x - \cot x)$, so $dt = \operatorname{cosec} x \, dx$.
Substituting: $\int t \, dt = \frac{t^2}{2} + C$.
Converting back: $t = \log(\operatorname{cosec} x - \cot x)$.

Final Answer

$\frac{1}{2} [\log(\operatorname{cosec} x - \cot x)]^2 + C$

Quick Recall: Memorise both $\frac{d}{dx} \log \tan(x/2) = \operatorname{cosec} x$ and $\frac{d}{dx} \log(\operatorname{cosec} x - \cot x) = \operatorname{cosec} x$ directly – whenever you see $\operatorname{cosec} x$ multiplying a log of a related trig expression, this is the pattern.

Type 44

Integrate $\sec^4 x$ (or $\tan^2 x \sec^4 x$) by splitting off a factor of $\sec^2 x$ and converting the remaining $\sec^2 x$ into $1 + \tan^2 x$, then substituting $t = \tan x$ – a companion technique to Type 32's $\tan^4 x$ reduction.

Formula Used: $\sec^2 x = 1 + \tan^2 x$

Source: Exercise 7.5, Q16(i)

Given: $\sec^4 x$

Solution:

Split off one factor of $\sec^2 x$ and rewrite the rest using the Pythagorean identity: $\sec^4 x = \sec^2 x \cdot \sec^2 x = (1 + \tan^2 x) \sec^2 x$ – this leaves one $\sec^2 x$ ready to supply dt.
Let $t = \tan x$, so $dt = \sec^2 x \, dx$.
Substituting: $\int (1 + t^2) \, dt = t + \frac{t^3}{3} + C$.
Converting back: $t = \tan x$.

Final Answer

$\tan x + \frac{1}{3} \tan^3 x + C$

Common Mistake: Splitting off the wrong pair of factors (e.g. trying to keep all four powers together) instead of isolating exactly one $\sec^2 x$ to serve as dt – the remaining $\sec^2 x$ must become $1 + \tan^2 x$ so everything is in terms of $\tan x$ before substituting.

Alternate Board-Style Example

Source: Exercise 7.5, Q11(ii)

Given: $\tan^3 2x \sec 2x$

Solution:

Split off a factor of $\sec 2x \tan 2x$ (the derivative of $\sec 2x$, up to a constant) and rewrite the rest: $\tan^3 2x \sec 2x = \tan^2 2x \cdot (\sec 2x \tan 2x) = (\sec^2 2x - 1)(\sec 2x \tan 2x)$.
Let $t = \sec 2x$, so $dt = 2 \sec 2x \tan 2x\, dx$, i.e. $\sec 2x \tan 2x\, dx = \dfrac{dt}{2}$.
Substituting: $\dfrac{1}{2} \displaystyle\int (t^2 - 1)\, dt = \dfrac{1}{2} \left(\dfrac{t^3}{3} - t \right) + C$.
Converting back: $t = \sec 2x$.

Final Answer

$\dfrac{1}{6} \sec^3 2x - \dfrac{1}{2} \sec 2x + C$

Quick Recall: For $\sec^4 x$-type integrals, peel off exactly one $\sec^2 x$, convert the rest to $1 + \tan^2 x$, and substitute $t = \tan x$ – for $\tan^3 x \sec x$-type integrals, peel off $\sec x \tan x$ instead and substitute $t = \sec x$.

Type 45

Integrate $\dfrac{N(x)}{D(x)}$ where $D(x)$ is a single shifted trig function like $\sin(x - \alpha)$ (or a linear combination of $\sin x, \cos x$) by writing the numerator as $A \cdot D(x) + B \cdot D'(x)$ for constants A, B – this splits the integral into an easy $\displaystyle\int A\, dx$ piece and a $\displaystyle\int B \cdot \dfrac{D'(x)}{D(x)}\, dx$ log-form piece.

★ **Board Favorite** – This numerator-decomposition technique ($N = A \cdot D + B \cdot D'$) is one of the most important reusable methods in the whole integration syllabus – it appears again in partial-fraction-adjacent problems and is a guaranteed board-exam pattern whenever a linear trig expression sits in the denominator.

Formula Used: $\sin x = \sin\big((x - \alpha) + \alpha\big) = \sin(x - \alpha) \cos \alpha + \cos(x - \alpha) \sin \alpha$

Source: Exercise 7.6, Q4(i)

Given: $\dfrac{\sin x}{\sin(x - \alpha)}$

Solution:

Rewrite the numerator by expanding $x = (x - \alpha) + \alpha$ inside the sine: $\sin x = \sin(x -$

$\alpha)\cos\alpha + \cos(x-\alpha)\sin\alpha$ – this decomposes $\sin x$ into a piece proportional to the denominator and a piece proportional to the denominator's derivative.

Dividing through by $\sin(x-\alpha)$: $\dfrac{\sin x}{\sin(x-\alpha)} = \cos\alpha + \sin\alpha \cdot \dfrac{\cos(x-\alpha)}{\sin(x-\alpha)} = \cos\alpha + \sin\alpha\cot(x-\alpha)$.

Integrating each piece: $\displaystyle\int \cos\alpha\, dx = x\cos\alpha$ (since α is a constant), and $\displaystyle\int \sin\alpha\cot(x-\alpha)\, dx = \sin\alpha\log|\sin(x-\alpha)|$, using the standard cot integral.

Final Answer

$x\cos\alpha + \sin\alpha\ \log|\sin(x-\alpha)| + C$

Common Mistake: Trying to integrate $\sin x/\sin(x-\alpha)$ via substitution directly – there is no clean substitution here; the numerator must first be algebraically rewritten in terms of the shifted angle $(x-\alpha)$.

Alternate Board-Style Example

Source: Exercise 7.6, Q9

Given: $\displaystyle\int \frac{\sqrt{2}\sin x}{\sin\left(x-\frac{\pi}{4}\right)}\, dx = ax + b\log\left|\sin\left(x-\frac{\pi}{4}\right)\right| + C,$ find a, b

Solution:

Following the same method, expand $\sqrt{2}\sin x$ in terms of $\left(x-\frac{\pi}{4}\right)$: since $\sin\left(x-\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}(\sin x - \cos x)$ and $\cos\left(x-\frac{\pi}{4}\right) = \frac{1}{\sqrt{2}}(\cos x + \sin x)$, adding these two gives $\sin\left(x-\frac{\pi}{4}\right) + \cos\left(x-\frac{\pi}{4}\right) = \sqrt{2}\sin x$ – exactly our numerator.

So $\dfrac{\sqrt{2}\sin x}{\sin(x-\pi/4)} = 1 + \cot\left(x-\dfrac{\pi}{4}\right)$, matching the pattern with $A = 1$ and the coefficient of the log term $B = 1$.

Integrating: $\displaystyle\int \left[1 + \cot\left(x-\frac{\pi}{4}\right)\right] dx = x + \log\left|\sin\left(x-\frac{\pi}{4}\right)\right| + C.$

Final Answer

$a = 1,\ \ b = 1$

Quick Recall: For $N(x)/D(x)$ with $D(x)$ a shifted or combined sine/cosine, always write $N(x)$ as (constant)$\times D(x)$ + (constant)$\times D'(x)$ – the first constant integrates to a plain x term, the second gives a $\log|D(x)|$ term.

Type 46

Integrate $\dfrac{1}{\sin(x-\alpha)\sin(x-\beta)}$ by writing the constant numerator as $\dfrac{1}{\sin(\alpha-\beta)} \cdot \sin\big((x-\beta)-(x-\alpha)\big)$, splitting the fraction into a difference of two simple cot/log-style integrals.

Formula Used: $1 = \dfrac{1}{\sin(\alpha-\beta)}\Big[\sin\big((x-\beta)-(x-\alpha)\big)\Big]$, using the sine subtraction formula inreverse

Source: Exercise 7.6, Q5(i)

Given: $\dfrac{1}{\sin(x-\alpha)\sin(x-\beta)}$

Solution:

Write 1 as $\dfrac{\sin[(x-\beta)-(x-\alpha)]}{\sin(\alpha-\beta)}$ (valid since $\sin(\alpha-\beta)$ is a nonzero constant), then expand the numerator using the sine subtraction formula: $\sin[(x-\beta)-(x-\alpha)] = \sin(x-\beta)\cos(x-\alpha)-\cos(x-\beta)\sin(x-\alpha)$.

Dividing by the original denominator $\sin(x-\alpha)\sin(x-\beta)$ splits the fraction into two pieces: $\dfrac{1}{\sin(\alpha-\beta)}\left[\dfrac{\cos(x-\alpha)}{\sin(x-\alpha)}-\dfrac{\cos(x-\beta)}{\sin(x-\beta)}\right] = \dfrac{1}{\sin(\alpha-\beta)}[\cot(x-\alpha)-\cot(x-\beta)]$.

Integrating each cotangent term: $\displaystyle\int \cot(x-\alpha)\,dx = \log|\sin(x-\alpha)|$, and similarly for the other term.

Final Answer

$$\frac{1}{\sin(\alpha-\beta)}\Big[\log|\sin(x-\alpha)| - \log|\sin(x-\beta)|\Big] + C$$

Common Mistake: Trying to combine the two shifted sines in the denominator using a product-to-sum formula instead – that path leads to a much harder integral; the "insert $\sin(\alpha-\beta)$ as 1" trick is specifically designed to split the denominator's product into a difference of two single-angle cotangent integrals.

Quick Recall: For $1/[\sin(x-\alpha)\sin(x-\beta)]$, insert $\sin(\alpha-\beta)/\sin(\alpha-\beta)$ and expand the numerator via the subtraction formula – this always splits into a difference of two $\log|\sin(\cdot)|$ terms.

Type 47

Convert a product of two shifted sines in the denominator into a sum using product-to-sum, when the numerator is itself a double-angle sine – distinct from Type 46 because here the numerator's structure calls for a product-to-sum conversion rather than the subtraction-formula trick.

Formula Used: $\sin 5x \sin 3x = \dfrac{1}{2}[\cos 2x - \cos 8x]$

Source: Exercise 7.6, Q7(i)

Given: $\dfrac{\sin 2x}{\sin 5x \sin 3x}$

Solution:

Convert the denominator's product to a sum: $\sin 5x \sin 3x = \frac{1}{2}[\cos(5x - 3x) - \cos(5x + 3x)] = \frac{1}{2}[\cos 2x - \cos 8x]$.

The integrand becomes $\frac{2\sin 2x}{\cos 2x - \cos 8x}$; recognising that $\cos 2x - \cos 8x = 2\sin 5x \sin 3x$ closes the loop, but the more useful route is to note the numerator $\sin 2x$ relates to the derivative of $(\cos 2x - \cos 8x)$ only indirectly, so instead we use the sum-to-product identity on the denominator directly: $\cos 2x - \cos 8x = 2\sin 5x \sin 3x$ (consistent), and separately write $\sin 2x = \sin(5x - 3x) = \sin 5x \cos 3x - \cos 5x \sin 3x$.

Dividing this expanded numerator by $\sin 5x \sin 3x$: $\frac{\sin 5x \cos 3x - \cos 5x \sin 3x}{\sin 5x \sin 3x} = \cot 3x - \cot 5x$.

Final Answer

$\frac{1}{3}\log|\sin 3x| - \frac{1}{5}\log|\sin 5x| + C$

Common Mistake: Getting lost trying to combine the product-to-sum conversion with the original numerator instead of switching to expanding $\sin 2x = \sin(5x - 3x)$ directly – when the numerator's angle is exactly the difference of the two denominator angles, expand it via the subtraction formula (as in Type 46) rather than forcing a product-to-sum route.

Quick Recall: If the numerator's angle equals the difference of the two denominator angles, expand the numerator via the sine subtraction formula and divide – it collapses to a simple difference of two cotangents.

Type 48

Integrate $\frac{1}{\sin x \cos^2 x}$ (or similar products of trig powers in the denominator) by writing $1 = \sin^2 x + \cos^2 x$ in the numerator, splitting the fraction into a sum of two simpler standard forms.

Formula Used: $1 = \sin^2 x + \cos^2 x$

Source: Exercise 7.6, Q7(ii)

Given: $\dfrac{1}{\sin x \cos^2 x}$

Solution:

Rewrite the numerator 1 as $\sin^2 x + \cos^2 x$: $\frac{\sin^2 x + \cos^2 x}{\sin x \cos^2 x} = \frac{\sin^2 x}{\sin x \cos^2 x} + \frac{\cos^2 x}{\sin x \cos^2 x} =$

$\dfrac{\sin x}{\cos^2 x} + \dfrac{1}{\sin x}$ – this splits one hard fraction into two standard forms.
The first term is $\sec x \tan x$ in disguise (as in Type 11), and the second is $\operatorname{cosec} x$.
Integrating: $\int (\sec x \tan x + \operatorname{cosec} x)\, dx = \sec x + \log|\operatorname{cosec} x - \cot x| + C$, using the standard integral of $\operatorname{cosec} x$.

Final Answer

$\sec x + \log|\operatorname{cosec} x - \cot x| + C$

Common Mistake: Not thinking to replace 1 with $\sin^2 x + \cos^2 x$ – this is a purely algebraic trick with no obvious visual cue, but it is the standard way to split any $1/(\sin^a x \cos^b x)$ integrand into manageable pieces.

Quick Recall: Whenever a fraction has 1 in the numerator and a product of sine and cosine powers in the denominator, replace the 1 with $\sin^2 x + \cos^2 x$ and split into two separate standard integrals.

Type 49

Simplify an algebraic-trig ratio using basic reciprocal/quotient identities until it collapses to a recognisable standard form – covers a family of "simplify first" integrands like $\dfrac{1+\tan^2 x}{2\tan x}$, $\dfrac{1}{\sin x \cos x}$, $\dfrac{\cos 2x}{\sin x}$, and $\dfrac{\sin 2x}{\sin 4x}$.

Formula Used: $1 + \tan^2 x = \sec^2 x, \qquad \sin 2x = 2\sin x \cos x$

Source: Exercise 7.6, Q2(i)

Given: $\dfrac{1+\tan^2 x}{2\tan x}$

Solution:

Rewrite the numerator using the Pythagorean identity: $1 + \tan^2 x = \sec^2 x = \dfrac{1}{\cos^2 x}$, and the denominator using $\tan x = \dfrac{\sin x}{\cos x}$.
Combining: $\dfrac{1/\cos^2 x}{2\sin x/\cos x} = \dfrac{1}{2\sin x \cos x} = \dfrac{1}{\sin 2x}$, using $\sin 2x = 2\sin x \cos x$.
Rewrite $\dfrac{1}{\sin 2x} = \operatorname{cosec} 2x$ and integrate using the linear-inside standard form: $\int \operatorname{cosec} 2x\, dx = \dfrac{1}{2}\log|\operatorname{cosec} 2x - \cot 2x| + C$.

Final Answer

$\dfrac{1}{2}\log|\operatorname{cosec} 2x - \cot 2x| + C$

Common Mistake: Trying to integrate the original fraction term by term without first simplifying it to a single trig function – always fully simplify an algebraic-trig ratio before deciding an integration technique; it often collapses to something with a direct standard formula.

Alternate Board-Style Example

Source: Exercise 7.6, Q3(i)

Given: $\dfrac{\cos 2x}{\sin x}$

Solution:

Rewrite the numerator using $\cos 2x = 1 - 2\sin^2 x$: $\dfrac{1-2\sin^2 x}{\sin x} = \dfrac{1}{\sin x} - 2\sin x = \operatorname{cosec} x - 2\sin x$.
Integrating: $\displaystyle\int (\operatorname{cosec} x - 2\sin x)\,dx = \log|\operatorname{cosec} x - \cot x| + 2\cos x + C$.

Final Answer

$\log|\operatorname{cosec} x - \cot x| + 2\cos x + C$

Quick Recall: Before choosing a technique, fully simplify any algebraic-trig ratio using reciprocal and double-angle identities – many of these problems reduce to a single standard $\sec x$, $\operatorname{cosec} x$, or $\sin / \cos$ term.

Type 50

Rationalize a square-root trig denominator like $\sqrt{1-\sin x}$ by first substituting $x = 2t$ so the expression under the root becomes a perfect square in t, $(\sin t - \cos t)^2$, then rewriting the result as a single shifted cosine to reach a standard sec integral.

Formula Used: $1 - \sin 2t = (\sin t - \cos t)^2, \qquad \cos t - \sin t = \sqrt{2}\cos\left(t + \dfrac{\pi}{4}\right)$

Source: Exercise 7.6, Q8(i)

Given: $\dfrac{1}{\sqrt{1-\sin x}}$

Solution:

Substitute $x = 2t$, so $dx = 2\,dt$: the expression under the root becomes $1 - \sin 2t = (\sin t - \cos t)^2$, a perfect square, using $\sin 2t = 2\sin t\cos t$ and $\sin^2 t + \cos^2 t = 1$.
Taking the square root (on an interval where $\cos t > \sin t$, so the quantity is positive as $\cos t - \sin t$): $\sqrt{1-\sin 2t} = \cos t - \sin t$.
The integral becomes $\displaystyle\int \frac{2\,dt}{\cos t - \sin t}$. Rewrite the denominator as a single shifted cosine:

$\cos t - \sin t = \sqrt{2}\cos\left(t + \frac{\pi}{4}\right)$.

So the integral is $\sqrt{2}\int \sec\left(t+\frac{\pi}{4}\right) dt = \sqrt{2}\log\left|\sec\left(t+\frac{\pi}{4}\right) + \tan\left(t+\frac{\pi}{4}\right)\right| + C$, using the standard sec integral.

Converting back to x using $t = \frac{x}{2}$.

Final Answer

$\sqrt{2}\,\log\left|\sec\left(\frac{x}{2}+\frac{\pi}{4}\right) + \tan\left(\frac{x}{2}+\frac{\pi}{4}\right)\right| + C$

Common Mistake: Trying to rationalize $\sqrt{1-\sin x}$ directly in terms of x using a conjugate multiplication – this leaves an unintegrable mixed sine/cosine expression; halving the angle first (so the double-angle identity produces a perfect square) is the necessary opening step.

Quick Recall: For $1/\sqrt{1 \pm \sin x}$ or $1/\sqrt{1 \pm \cos x}$, substitute $x = 2t$ first so the double-angle identity turns the root into a perfect square in t, then rewrite as a single shifted sine/cosine to reach a standard sec or cosec integral.

Type 51

Recognise $(\sin x + \cos x)^2 = 1 + \sin 2x$ to simplify a denominator built from $1+\sin 2x$, cancelling directly with a matching numerator.

Formula Used: $(\sin x + \cos x)^2 = 1 + \sin 2x$

Source: Exercise 7.6, Q8(ii)

Given: $\dfrac{\sin x + \cos x}{1 + \sin 2x}$

Solution:

Recognise the denominator as a perfect square: $1 + \sin 2x = (\sin x + \cos x)^2$, using the identity $\sin 2x = 2\sin x\cos x$ combined with $\sin^2 x + \cos^2 x = 1$.

The numerator $\sin x + \cos x$ cancels one factor of the squared denominator directly: $\dfrac{\sin x + \cos x}{(\sin x + \cos x)^2} = \dfrac{1}{\sin x + \cos x}$.

Multiply numerator and denominator by $\dfrac{1}{\sqrt{2}}$ to write the denominator as a single shifted sine: $\sin x + \cos x = \sqrt{2}\sin\left(x + \frac{\pi}{4}\right)$, giving $\dfrac{1}{\sqrt{2}}\text{cosec}\left(x+\frac{\pi}{4}\right)$.

Integrating using the standard cosecant integral (linear-inside form): $\dfrac{1}{\sqrt{2}}\int \text{cosec}\left(x+\frac{\pi}{4}\right) dx = \dfrac{1}{\sqrt{2}}\log\left|\text{cosec}\left(x+\frac{\pi}{4}\right) - \cot\left(x+\frac{\pi}{4}\right)\right| + C$.

Final Answer

$$\frac{1}{\sqrt{2}} \log \left| \operatorname{cosec}\left(x + \frac{\pi}{4}\right) - \cot\left(x + \frac{\pi}{4}\right) \right| + C$$

Common Mistake: Not recognising $1+\sin 2x$ as a perfect square and instead trying to integrate the original fraction directly – this identity is exactly what allows the numerator to cancel one power of the denominator.

Quick Recall: $1 + \sin 2x$ is always $(\sin x + \cos x)^2$ – if you see this pattern with a matching linear numerator, it cancels immediately; rewrite the remaining single factor as $\sqrt{2}\sin(x + \pi/4)$ to finish with the standard cosecant integral.

III. Standard Forms & Completing the Square (Ex 7.7–7.9)

Type 52

Direct application of the two standard log-form integrals $\int \frac{dx}{x^2 - a^2}$ and $\int \frac{dx}{a^2 - x^2}$ – memorised formulas that must be matched carefully since the two results look similar but have the numerator and denominator terms swapped.

Formula Used: $\int \frac{dx}{x^2 - a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$

Formula Used: $\int \frac{dx}{a^2 - x^2} = \frac{1}{2a} \log \left| \frac{a+x}{a-x} \right| + C$

Source: Exercise 7.7, Q1(i)

Given: $\frac{1}{x^2 - 16}$

Solution:

This matches the standard form $\int \frac{dx}{x^2 - a^2}$ with $a^2 = 16$, so $a = 4$.

Applying the formula directly: $\int \frac{dx}{x^2 - 16} = \frac{1}{2(4)} \log \left| \frac{x-4}{x+4} \right| + C$.

Final Answer

$\frac{1}{8} \log \left| \frac{x-4}{x+4} \right| + C$

Common Mistake: Mixing up the two formulas – when a^2 is subtracted from x^2, the log fraction is $\frac{x-a}{x+a}$; when x^2 is subtracted from a^2, it flips to $\frac{a+x}{a-x}$. Writing the wrong one is the single most common slip in this whole section.

Alternate Board-Style Example

Source: Exercise 7.7, Q1(ii)

Given: $\frac{1}{9 - x^2}$

Solution:

This matches $\displaystyle\int \frac{dx}{a^2 - x^2}$ with $a^2 = 9$, so $a = 3$.

Applying the formula: $\displaystyle\int \frac{dx}{9 - x^2} = \frac{1}{2(3)} \log\left|\frac{3+x}{3-x}\right| + C.$

Final Answer

$\displaystyle\frac{1}{6}\log\left|\frac{3+x}{3-x}\right| + C$

Quick Recall: Identify a^2 from the constant term, check which term is subtracted from which, and pick the matching one of the two log formulas – write both formulas out before starting if there's any doubt about which applies.

Type 53

Direct application of the two standard inverse-trig-form integrals $\displaystyle\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a} + C$ and $\displaystyle\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C.$

Formula Used: $\displaystyle\int \frac{dx}{\sqrt{a^2 - x^2}} = \sin^{-1}\frac{x}{a} + C$

Formula Used: $\displaystyle\int \frac{dx}{x^2 + a^2} = \frac{1}{a}\tan^{-1}\frac{x}{a} + C$

Source: Exercise 7.7, Q1(iii)

Given: $\displaystyle\frac{1}{\sqrt{1 - x^2}}$

Solution:

This matches $\displaystyle\int \frac{dx}{\sqrt{a^2 - x^2}}$ with $a = 1$.

Applying the formula directly: $\displaystyle\int \frac{dx}{\sqrt{1 - x^2}} = \sin^{-1} x + C.$

Final Answer

$\sin^{-1} x + C$

Common Mistake: Confusing the $\sqrt{a^2 - x^2}$ (arcsin) form with the $x^2 + a^2$ (arctan) form – one has a square root in the denominator and gives an inverse sine, the other has no square root and gives an inverse tangent; check the denominator's exact shape before picking a formula.

Alternate Board-Style Example

Source: Exercise 7.7, Q1(iv)

Given: $\dfrac{1}{x^2+16}$

Solution:

This matches $\displaystyle\int \frac{dx}{x^2+a^2}$ with $a = 4$.

Applying the formula: $\displaystyle\int \frac{dx}{x^2+16} = \frac{1}{4}\tan^{-1}\frac{x}{4} + C$.

Final Answer

$\dfrac{1}{4}\tan^{-1}\dfrac{x}{4} + C$

Quick Recall: Four standard forms exist for quadratic denominators: $x^2 - a^2$ and $a^2 - x^2$ give logs; $\sqrt{a^2 - x^2}$ gives $\sin^{-1}$; $x^2 + a^2$ gives $\tan^{-1}$ – match the exact shape of the denominator to the exact formula before integrating.

Type 54

The same four standard forms as Types 52–53, but with a coefficient in front of x^2 – factor that coefficient out first so the expression matches $x^2 \pm a^2$ or $a^2 \pm x^2$ exactly, picking up an extra constant along the way.

Formula Used: Factor the x^2 coefficient out before matching a standard form

Source: Exercise 7.7, Q2(i)

Given: $\dfrac{1}{\sqrt{9-4x^2}}$

Solution:

Factor 4 out from under the square root: $9-4x^2 = 4\left(\dfrac{9}{4} - x^2\right)$, so $\sqrt{9-4x^2} = 2\sqrt{\dfrac{9}{4} - x^2}$.

The integral becomes $\displaystyle\frac{1}{2}\int \frac{dx}{\sqrt{(3/2)^2 - x^2}}$, matching the standard arcsin form with $a = \dfrac{3}{2}$.

Applying the formula: $\dfrac{1}{2}\sin^{-1}\left(\dfrac{x}{3/2}\right) + C = \dfrac{1}{2}\sin^{-1}\left(\dfrac{2x}{3}\right) + C$.

Final Answer

$\dfrac{1}{2}\sin^{-1}\left(\dfrac{2x}{3}\right) + C$

Common Mistake: Forgetting to pull the coefficient's square root out from under the radical correctly (e.g. writing $\sqrt{9-4x^2} = 3\sqrt{1-4x^2}$ instead of factoring 4 out to get $2\sqrt{9/4-x^2}$) – always factor so what's left under the root is exactly $(\text{constant})^2 - x^2$, with a plain 1 coefficient on x^2.

Alternate Board-Style Example

Source: Exercise 7.7, Q2(ii)

Given: $\dfrac{1}{25+9x^2}$

Solution:

Factor 9 out of the denominator: $25 + 9x^2 = 9\left(\dfrac{25}{9} + x^2\right)$.

The integral becomes $\dfrac{1}{9}\displaystyle\int \dfrac{dx}{x^2 + (5/3)^2}$, matching the arctan form with $a = \dfrac{5}{3}$.

Applying the formula: $\dfrac{1}{9} \cdot \dfrac{1}{5/3} \tan^{-1}\left(\dfrac{x}{5/3}\right) + C = \dfrac{1}{15}\tan^{-1}\left(\dfrac{3x}{5}\right) + C$.

Final Answer

$\dfrac{1}{15}\tan^{-1}\left(\dfrac{3x}{5}\right) + C$

Quick Recall: If x^2 has a coefficient, factor it out first (from the whole denominator, or from under the root) so what remains is a plain x^2 against a constant squared – then match one of the four standard forms and don't forget the extra constant multiplier left outside.

Type 55

Direct application of the standard $\sec^{-1}$-form integral $\displaystyle\int \dfrac{dx}{x\sqrt{x^2a^2 - b^2}}$, after factoring so the expression matches the exact pattern $x\sqrt{(cx)^2 - b^2}$.

Formula Used: $\displaystyle\int \dfrac{dx}{x\sqrt{c^2x^2 - b^2}} = \dfrac{1}{b}\sec^{-1}\left(\dfrac{cx}{b}\right) + C$

Source: Exercise 7.7, Q3(i)

Given: $\dfrac{1}{x\sqrt{9x^2-16}}$

Solution:

This matches the standard $\sec^{-1}$ pattern $\dfrac{1}{x\sqrt{c^2x^2 - b^2}}$ with $c = 3$ and $b = 4$ (since

$9x^2 = (3x)^2$ and $16 = 4^2$).

Applying the formula: $\displaystyle\int \frac{dx}{x\sqrt{9x^2-16}} = \frac{1}{4}\sec^{-1}\left(\frac{3x}{4}\right) + C$.

Final Answer

$\frac{1}{4}\sec^{-1}\left(\frac{3x}{4}\right) + C$

Common Mistake: Trying to force this into the arcsin or log form instead of recognising the distinctive $x\sqrt{(\cdot)^2 - (\cdot)^2}$ shape that signals $\sec^{-1}$ – the presence of a lone x multiplying the square root is the giveaway for this specific standard form.

Quick Recall: Whenever you see a lone x multiplying a square root of the form $\sqrt{(cx)^2 - b^2}$, this is the $\sec^{-1}$ standard form – identify b and c carefully and plug directly into the formula.

Type 56

Direct application of the two standard log-form integrals $\displaystyle\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left|x + \sqrt{x^2+a^2}\right| + C$ and $\displaystyle\int \frac{dx}{\sqrt{x^2-a^2}} = \log\left|x + \sqrt{x^2-a^2}\right| + C$.

Formula Used: $\int \frac{dx}{\sqrt{x^2+a^2}} = \log\left|x + \sqrt{x^2+a^2}\right| + C$

Source: Exercise 7.7, Q3(ii)

Given: $\frac{1}{\sqrt{x^2+5}}$

Solution:

This matches $\displaystyle\int \frac{dx}{\sqrt{x^2+a^2}}$ with $a^2 = 5$.

Applying the formula directly: $\displaystyle\int \frac{dx}{\sqrt{x^2+5}} = \log\left|x + \sqrt{x^2+5}\right| + C$.

Final Answer

$\log\left|x + \sqrt{x^2+5}\right| + C$

Common Mistake: Confusing this log-form standard integral (denominator $\sqrt{x^2 \pm a^2}$, no lone x outside) with the $\sec^{-1}$ form from Type 55 (which specifically needs a lone x multiplying the root) – check carefully whether there's an extra x factor outside the square root.

Quick Recall: $\dfrac{1}{\sqrt{x^2 \pm a^2}}$ (with nothing extra multiplying the root) always integrates to a log of x plus the square root itself – memorise this exact form since it reappears constantly, including inside harder completing-the-square problems later in the chapter.

Type 57

When the numerator's degree is at least the denominator's degree (an "improper" rational function), divide first to get a polynomial plus a proper-fraction remainder, then apply a standard arctan or log form to what remains.

Formula Used: $\dfrac{f(x)}{g(x)} = Q(x) + \dfrac{R(x)}{g(x)}$ (divide first, then use a standard form)

Source: Exercise 7.7, Q6(ii)

Given: $\dfrac{x^4+1}{x^2+1}$

Solution:

Since the numerator's degree (4) exceeds the denominator's degree (2), divide first: $x^4 + 1 = (x^2+1)(x^2-1) + 2$, so $\dfrac{x^4+1}{x^2+1} = x^2 - 1 + \dfrac{2}{x^2+1}$.

Integrating term by term: the polynomial part gives $\displaystyle\int (x^2-1)\,dx = \frac{x^3}{3} - x$, and the remainder is a direct arctan standard form: $\displaystyle\int \frac{2}{x^2+1}\,dx = 2\tan^{-1} x$.

Final Answer

$\dfrac{x^3}{3} - x + 2\tan^{-1} x + C$

Common Mistake: Trying to apply a standard form (arctan, log, etc.) directly to an improper fraction without dividing first – the standard forms only apply to *proper* fractions (numerator degree strictly less than denominator degree); always check this before matching a formula.

Quick Recall: Before matching any standard quadratic-denominator formula, check the numerator's degree against the denominator's – if it's equal or higher, divide first; only the proper-fraction remainder gets the standard-form treatment.

Type 58

Substitute $t = x^2$ (or x^3, x^4, depending on the powers present) when the integrand is an odd power of x times a function of an even power of x – reduces the problem to one of the standard quadratic-denominator forms in t.

Formula Used: Let $t = x^2$ (or x^3, x^4), $dt =$ matching deriv. dx

Source: Exercise 7.7, Q7(i)

Given: $\dfrac{x}{x^4 - 1}$

Solution:

Let $t = x^2$, so $dt = 2x\,dx$, i.e. $x\,dx = \dfrac{dt}{2}$ – chosen because the denominator $x^4 - 1 = (x^2)^2 - 1$ is a function of x^2, and the numerator supplies exactly the right power of x to form dt.

Substituting, the integral becomes $\dfrac{1}{2}\displaystyle\int \dfrac{dt}{t^2 - 1}$, a direct standard log form.

Applying the standard formula: $\dfrac{1}{2} \cdot \dfrac{1}{2} \log\left|\dfrac{t-1}{t+1}\right| + C = \dfrac{1}{4}\log\left|\dfrac{t-1}{t+1}\right| + C.$

Converting back: $t = x^2$.

Final Answer

$\dfrac{1}{4}\log\left|\dfrac{x^2-1}{x^2+1}\right| + C$

Common Mistake: Not noticing that the denominator is secretly quadratic *in* x^2 (or x^3/x^4) – always check whether every power of x present is a multiple of some common power before deciding a substitution isn't available.

Alternate Board-Style Example

Source: Exercise 7.7, Q7(ii)

Given: $\dfrac{3x^2}{1 + x^6}$

Solution:

Let $t = x^3$, so $dt = 3x^2\,dx$ – chosen because the denominator is a function of x^3 (since $x^6 = (x^3)^2$), and the numerator exactly supplies dt.

Substituting: $\displaystyle\int \dfrac{dt}{1 + t^2}$, a direct arctan standard form.

Applying the formula: $\tan^{-1} t + C$.

Converting back: $t = x^3$.

Final Answer

$\tan^{-1}(x^3) + C$

Quick Recall: If the denominator is built entirely from one power of x (like x^2, x^3, or x^4) and the numerator supplies exactly the matching derivative, substitute for that power – the problem reduces to a standard quadratic-denominator form in the new variable.

Type 59

Split an algebraic fraction into two pieces using partial-fraction-style numerator manipulation before applying two different standard forms – distinct from a single substitution because the numerator itself needs to be broken into two separate pieces first.

Formula Used: Split the numerator into two pieces, each its own standard form

Source: Exercise 7.7, Q9(ii)

Given: $\dfrac{x(1+x^2)}{1+x^4}$

Solution:

Split the numerator: $x(1+x^2) = x + x^3$, so the integrand becomes $\dfrac{x}{1+x^4} + \dfrac{x^3}{1+x^4}$ – two separate fractions, each suited to a different substitution.

For $\displaystyle\int \frac{x}{1+x^4}\,dx$: let $u = x^2$, $du = 2x\,dx$, giving $\displaystyle\frac{1}{2}\int \frac{du}{1+u^2} = \frac{1}{2}\tan^{-1}(x^2)$.

For $\displaystyle\int \frac{x^3}{1+x^4}\,dx$: let $v = x^4$, $dv = 4x^3\,dx$, giving $\displaystyle\frac{1}{4}\int \frac{dv}{1+v} = \frac{1}{4}\log(1+x^4)$, a direct log form.

Final Answer

$\dfrac{1}{2}\tan^{-1}(x^2) + \dfrac{1}{4}\log(1+x^4) + C$

Common Mistake: Trying to find one single substitution that handles the whole fraction at once – when the numerator naturally splits into pieces that need different substitutions, split first rather than forcing one method on the whole thing.

Quick Recall: If a numerator is a sum of terms with different natural substitutions, split the fraction into separate pieces first, then handle each piece with its own matching substitution and standard form.

Type 60

Substitute t for a trig function (usually $\sin x$ or $\tan x$) to convert a trig-algebraic mixed integrand into one of the standard quadratic-denominator forms in t.

Formula Used: Let $t = \sin x$ (or $\cos x, \tan x$), $dt =$ matching deriv. dx

Source: Exercise 7.7, Q11(i)

Given: $\dfrac{\sin x}{16 - 9\cos^2 x}$

Solution:

Let $t = \cos x$, so $dt = -\sin x\, dx$, i.e. $\sin x\, dx = -dt$ – chosen because the denominator is a function of $\cos x$, and $\sin x$ supplies exactly what's needed for dt (up to sign).

Substituting: $-\displaystyle\int \frac{dt}{16 - 9t^2}$, matching the standard $a^2 - x^2$ log form (factoring out 9 first: $16 - 9t^2 = 9\left(\frac{16}{9} - t^2\right)$).

Applying the standard formula with $a = \frac{4}{3}$: $-\dfrac{1}{9} \cdot \dfrac{1}{2(4/3)} \log\left|\dfrac{4/3 + t}{4/3 - t}\right| + C = -\dfrac{1}{24} \log\left|\dfrac{4 + 3t}{4 - 3t}\right| + C$.

Converting back: $t = \cos x$.

Final Answer

$$-\frac{1}{24} \log\left|\frac{4 + 3\cos x}{4 - 3\cos x}\right| + C$$

Common Mistake: Trying to integrate the trig-algebraic mix directly without substituting – once a trig function's derivative is spotted as an available factor, always substitute for that trig function first; what remains is then a routine standard-form integral in the new variable.

Quick Recall: A trig-algebraic mixed integrand with the trig function's own derivative present is always a substitution away from one of the four standard quadratic-denominator forms – substitute, match the form in the new variable, then convert back.

Type 61

Complete the square in a quadratic denominator (rewriting $x^2 + bx + c$ as $(x + p)^2 + q$ or $(x + p)^2 - q$) so it matches one of the standard forms from Types 52–53, then apply the arctan or log formula directly.

★ **Board Favorite** – Completing the square before matching a standard form is one of the most frequently tested techniques in this whole chapter – almost every "quadratic in the denominator" question that isn't already a perfect $x^2 \pm a^2$ shape starts with this exact step.

Formula Used: $x^2 + bx + c = \left(x + \frac{b}{2}\right)^2 + \left(c - \frac{b^2}{4}\right)$

Source: Exercise 7.8, Q1(i)

Given: $\dfrac{1}{x^2 + 4x + 8}$

Solution:

Complete the square in the denominator: $x^2 + 4x + 8 = (x + 2)^2 + 4$, obtained by halving the coefficient of x (giving 2), squaring it (giving 4), and adjusting the constant so the identity still holds.

With the substitution $u = x + 2$ understood implicitly, this now matches the standard form $\int \frac{du}{u^2 + a^2}$ with $a = 2$.

Applying the arctan formula: $\int \frac{dx}{(x + 2)^2 + 4} = \frac{1}{2}\tan^{-1}\left(\frac{x + 2}{2}\right) + C$.

Final Answer

$\frac{1}{2}\tan^{-1}\left(\frac{x + 2}{2}\right) + C$

Common Mistake: Making an arithmetic slip while completing the square – always double-check by expanding $(x+p)^2+q$ back out and confirming it matches the original quadratic exactly before proceeding.

Alternate Board-Style Example

Source: Exercise 7.8, Q2(i)

Given: $\dfrac{1}{x^2 - 6x + 13}$

Solution:

Complete the square: $x^2 - 6x + 13 = (x - 3)^2 + 4$.

This matches the arctan standard form with $a = 2$: $\int \frac{dx}{(x - 3)^2 + 4} = \frac{1}{2}\tan^{-1}\left(\frac{x - 3}{2}\right) + C$.

Final Answer

$$\frac{1}{2}\tan^{-1}\left(\frac{x-3}{2}\right)+C$$

Quick Recall: Complete the square by halving and squaring the x-coefficient, treat the result as a shifted variable $u = x + p$, then match one of the four standard quadratic-denominator forms exactly as before.

Type 62

For $\frac{\text{linear}}{\text{quadratic}}$, first decompose the numerator as $A\cdot(\text{derivative of the quadratic})+B$, splitting into a log-form piece (from the A part) and a completed-square arctan piece (from the B part).

★ **Board Favorite** – This numerator-decomposition-plus-completing-the-square combination is a standard high-yield technique that combines two separate skills into one question, making it one of the most commonly tested patterns for linear-over-quadratic integrals.

Formula Used: Numerator $= A(2x+b)+B,$ matching $(x^2+bx+c)'$

Source: Exercise 7.8, Q2(ii)

Given: $\dfrac{x+3}{x^2-2x-5}$

Solution:

The derivative of the denominator is $2x-2$. Write the numerator as $x+3 = A(2x-2)+B$: matching coefficients, $2A = 1 \Rightarrow A = \frac{1}{2}$, and $-2A+B = 3 \Rightarrow B = 4$.

Split the integral using this decomposition: $\displaystyle\int \frac{x+3}{x^2-2x-5}\,dx = \frac{1}{2}\int \frac{2x-2}{x^2-2x-5}\,dx + 4\int \frac{dx}{x^2-2x-5}$.

The first piece is a direct log form: $\dfrac{1}{2}\log|x^2-2x-5|$. For the second, complete the square: $x^2-2x-5 = (x-1)^2-6$, giving the standard log-form result $\dfrac{1}{2\sqrt{6}}\log\left|\dfrac{x-1-\sqrt{6}}{x-1+\sqrt{6}}\right|$.

Final Answer

$$\frac{1}{2}\log|x^2-2x-5| + \frac{4}{2\sqrt{6}}\log\left|\frac{x-1-\sqrt{6}}{x-1+\sqrt{6}}\right| + C$$

Common Mistake: Trying to complete the square first and only then dealing with the numerator – the numerator decomposition should happen *before* completing the square, since it's based on the denominator's derivative in its original (un-completed) form.

Quick Recall: For linear-over-quadratic integrals, always decompose the numerator into (derivative of denominator) + constant first – this splits the problem into an easy log piece and a "just completing the square" piece.

Type 63

Substitute for an exponential or trig expression first (recognising it as the natural inner variable), then complete the square in the resulting algebraic quadratic before applying a standard form.

Formula Used: Substitute $t = e^x$ (or a trig fn), then complete the square in t

Source: Exercise 7.8, Q4(i)

Given: $\dfrac{e^x}{2e^{2x} + 3e^x + 5}$

Solution:

Let $t = e^x$, so $dt = e^x\,dx$ – the numerator e^x supplies exactly dt.
Substituting: $\displaystyle\int \frac{dt}{2t^2 + 3t + 5} = \frac{1}{2}\int \frac{dt}{t^2 + \frac{3}{2}t + \frac{5}{2}}$.
Complete the square: $t^2 + \frac{3}{2}t + \frac{5}{2} = \left(t + \frac{3}{4}\right)^2 + \frac{31}{16}$.
Applying the arctan standard form with $a = \dfrac{\sqrt{31}}{4}$: $\dfrac{1}{2} \cdot \dfrac{4}{\sqrt{31}} \tan^{-1}\left(\dfrac{4t+3}{\sqrt{31}}\right) + C = \dfrac{2}{\sqrt{31}} \tan^{-1}\left(\dfrac{4t+3}{\sqrt{31}}\right) + C$.
Converting back: $t = e^x$.

Final Answer

$$\frac{2}{\sqrt{31}} \tan^{-1}\left(\frac{4e^x + 3}{\sqrt{31}}\right) + C$$

Common Mistake: Trying to complete the square in terms of x directly, before substituting – since the quadratic is really in e^x (or the trig function), the substitution must come first to expose the true algebraic structure.

Quick Recall: If a quadratic-looking denominator is actually built from e^x or a trig function (not x itself), substitute for that inner expression first – then complete the square in the new variable exactly as usual.

Type 64

Complete the square under a square root (rewriting $bx - x^2$ or $c + bx - x^2$ as $a^2 - (x - p)^2$) to match the standard arcsin form.

Formula Used: $\int \frac{dx}{\sqrt{a^2 - (x-p)^2}} = \sin^{-1}\left(\frac{x-p}{a}\right) + C$

Source: Exercise 7.8, Q5(i)

Given: $\frac{1}{\sqrt{4x - x^2}}$

Solution:

Complete the square under the root: $4x - x^2 = -(x^2 - 4x) = -\left[(x-2)^2 - 4\right] = 4 - (x-2)^2$. This matches the standard arcsin form with $a = 2$ and the shifted variable $x - 2$.
Applying the formula: $\displaystyle\int \frac{dx}{\sqrt{4 - (x-2)^2}} = \sin^{-1}\left(\frac{x-2}{2}\right) + C.$

Final Answer

$\sin^{-1}\left(\frac{x-2}{2}\right) + C$

Common Mistake: Forgetting to factor out the negative sign correctly before completing the square – when x^2 has a negative coefficient under the root, always factor -1 out first so you're completing the square on $x^2 - (\text{positive})x$, then flip back to get $a^2 - (\text{shifted } x)^2$.

Quick Recall: Under a square root, a quadratic with a negative x^2 coefficient always completes to $a^2 - (x-p)^2$ (an arcsin form) – factor out the -1 first, complete the square on what remains, then flip the sign back.

Type 65

Complete the square under a square root where x^2 has a coefficient other than ± 1 – factor that coefficient out first, then complete the square, picking up an extra constant multiplier for the whole integral.

Formula Used: Factor out the x^2 coefficient, then complete the square

Source: Exercise 7.8, Q7(i)

Given: $\frac{1}{\sqrt{5 - 4x - 2x^2}}$

Solution:

Factor -2 out of the x-terms under the root: $5-4x-2x^2 = -2\left(x^2+2x\right)+5 = -2\left[(x+1)^2-1\right]+5 = -2(x+1)^2+7$.

Rewrite as $7-2(x+1)^2 = 2\left[\frac{7}{2}-(x+1)^2\right]$, factoring out the 2 to isolate a clean $a^2-(x+1)^2$ pattern inside.

So $\sqrt{5-4x-2x^2} = \sqrt{2}\sqrt{\left(\sqrt{7/2}\right)^2-(x+1)^2}$, matching the standard arcsin form with $a=\sqrt{7/2}$.

Applying the formula: $\dfrac{1}{\sqrt{2}}\displaystyle\int \frac{dx}{\sqrt{(7/2)-(x+1)^2}} = \frac{1}{\sqrt{2}}\sin^{-1}\left(\frac{x+1}{\sqrt{7/2}}\right)+C$.

Final Answer

$$\frac{1}{\sqrt{2}}\sin^{-1}\left(\frac{x+1}{\sqrt{7/2}}\right)+C$$

Common Mistake: Losing track of the extra constant multiplier that appears outside the integral once the leading coefficient is factored out from under the square root – this constant must be carried through to the final answer.

Quick Recall: With a non-unit coefficient on x^2 under a square root, factor it out completely first (constant and all), complete the square on the simplified quadratic, then don't forget the leftover constant multiplier on the whole integral.

Type 66

For $\dfrac{\text{linear}}{\sqrt{\text{quadratic}}}$, decompose the numerator as $A\cdot(\text{derivative of the quadratic})+B$ exactly as in Type 62, but here the A-piece becomes a square-root standard form and the B-piece becomes an arcsin (or log) form after completing the square.

Formula Used: $\int \dfrac{g'(x)}{\sqrt{g(x)}}\,dx = 2\sqrt{g(x)}+C$

Source: Exercise 7.8, Q8(ii)

Given: $\dfrac{5x+3}{\sqrt{x^2+4x+10}}$

Solution:

The derivative of the expression under the root is $2x+4$. Write the numerator as $5x+3 = A(2x+4)+B$: matching coefficients, $2A=5 \Rightarrow A=\frac{5}{2}$, and $4A+B = 3 \Rightarrow B = 3-10 = -7$.

Split the integral: $\displaystyle\int \frac{5x+3}{\sqrt{x^2+4x+10}}\,dx = \frac{5}{2}\int \frac{2x+4}{\sqrt{x^2+4x+10}}\,dx - 7\int \frac{dx}{\sqrt{x^2+4x+10}}$.

The first piece is a direct square-root standard form: $\frac{5}{2}\cdot 2\sqrt{x^2+4x+10} = 5\sqrt{x^2+4x+10}$. For the second, complete the square: $x^2+4x+10 = (x+2)^2+6$, giving the log-form result $\log\left|(x+2)+\sqrt{x^2+4x+10}\right|$.

Final Answer

$5\sqrt{x^2+4x+10} - 7\log\left|(x+2)+\sqrt{x^2+4x+10}\right| + C$

Common Mistake: Applying the numerator decomposition correctly but then forgetting the completing-the-square step is still needed for the second piece – both halves of this technique (decomposition *and* completing the square) are required, not just one.

Quick Recall: Linear-over-square-root-quadratic always splits into a direct $2\sqrt{\text{quadratic}}$ piece plus a completing-the-square log or arcsin piece – the same decomposition idea as Type 62, adapted for a square-root denominator.

Type 67

Substitute for an expression like $\log x$ or x^3 first, then complete the square in the resulting quadratic before applying a standard form – the algebraic completing-the-square counterpart to Type 63's exponential/trig version.

Formula Used: Substitute for the inner variable first, then complete the square

Source: Exercise 7.8, Q11

Given: $\dfrac{1}{x\left[(\log x)^2 - 3\log x - 4\right]}$

Solution:

Let $t = \log x$, so $dt = \frac{1}{x}\,dx$ – the accompanying $\frac{1}{x}$ factor supplies exactly dt.
Substituting: $\displaystyle\int \frac{dt}{t^2-3t-4}$.
Factor the denominator directly (since it factors nicely): $t^2-3t-4 = (t-4)(t+1)$, matching the standard x^2-a^2-style log form after completing the square, or factored directly: $\displaystyle\int \frac{dt}{(t-4)(t+1)} = \frac{1}{5}\log\left|\frac{t-4}{t+1}\right| + C$ (using the same standard-form logic as Type 52, since $t-4$ and $t+1$ differ by the same constant spread as a completed-square form).
Converting back: $t = \log x$.

Final Answer

$\dfrac{1}{5}\log\left|\dfrac{\log x - 4}{\log x + 1}\right| + C$

Common Mistake: Trying to complete the square in terms of x directly, forgetting that the true quadratic structure is in $\log x$, not x itself – always substitute for the actual inner variable before treating the expression as a quadratic.

Quick Recall: If the "quadratic" you're looking at is actually built from $\log x$, x^3, or another inner expression, substitute first – then complete the square (or factor directly, if it factors nicely) in the new variable exactly as usual.

Type 68

Integrate $\dfrac{1}{a + b\cos x}$ (or the analogous form with $\sin x$) using the Weierstrass substitution $t = \tan\dfrac{x}{2}$, which converts *every* rational function of $\sin x$ and $\cos x$ into an ordinary rational function of t – the universal trig substitution.

★ **Board Favorite** – The Weierstrass substitution is one of the most powerful and frequently tested techniques in the entire integration syllabus – it works on any rational combination of $\sin x, \cos x$, making it a guaranteed fallback whenever no other trig identity simplifies the integrand.

Formula Used: $t = \tan\dfrac{x}{2}, \quad \cos x = \dfrac{1-t^2}{1+t^2}, \quad \sin x = \dfrac{2t}{1+t^2}, \quad dx = \dfrac{2\,dt}{1+t^2}$

Source: Exercise 7.9, Q1(ii)

Given: $\dfrac{1}{5 + 4\cos x}$

Solution:

Substitute $t = \tan\dfrac{x}{2}$, so $\cos x = \dfrac{1-t^2}{1+t^2}$ and $dx = \dfrac{2\,dt}{1+t^2}$.

The denominator becomes $5 + 4\cdot\dfrac{1-t^2}{1+t^2} = \dfrac{5(1+t^2)+4(1-t^2)}{1+t^2} = \dfrac{t^2+9}{1+t^2}$, after combining over the common denominator and simplifying.

The integral becomes $\displaystyle\int \frac{1+t^2}{t^2+9}\cdot\frac{2\,dt}{1+t^2} = 2\int\frac{dt}{t^2+9}$ – the $(1+t^2)$ factors cancel completely, leaving a direct standard arctan form.

Applying the standard formula with $a = 3$: $2\cdot\dfrac{1}{3}\tan^{-1}\dfrac{t}{3} + C = \dfrac{2}{3}\tan^{-1}\dfrac{t}{3} + C$.

Converting back: $t = \tan\dfrac{x}{2}$.

Final Answer

$$\frac{2}{3}\tan^{-1}\left(\frac{1}{3}\tan\frac{x}{2}\right) + C$$

Common Mistake: Forgetting that dx itself changes to $\frac{2\,dt}{1+t^2}$, not just $\cos x$ – both substitutions must be made together, and it's easy to substitute for $\cos x$ correctly but leave dx unconverted.

Quick Recall: The Weierstrass substitution always converts $\cos x \to \frac{1-t^2}{1+t^2}$, $\sin x \to \frac{2t}{1+t^2}$, and $dx \to \frac{2\,dt}{1+t^2}$ simultaneously – after substituting, the $(1+t^2)$ in the denominator from dx very often cancels with a matching factor, leaving a routine standard-form integral in t.

Type 69

Integrate $\frac{1}{a+b\sin x+c\cos x}$ (mixing both sine and cosine, or sine alone) using the same Weierstrass substitution – the algebra after substituting is a little heavier since both $\sin x$ and $\cos x$ convert, but the method is identical to Type 68.

Formula Used: $t=\tan\frac{x}{2}$, $\quad \sin x=\frac{2t}{1+t^2}$, $\quad \cos x=\frac{1-t^2}{1+t^2}$, $\quad dx=\frac{2\,dt}{1+t^2}$

Source: Exercise 7.9, Q3(ii)

Given: $\frac{1}{1-\sin x+\cos x}$

Solution:

Substitute $t=\tan\frac{x}{2}$: the denominator becomes $1-\frac{2t}{1+t^2}+\frac{1-t^2}{1+t^2}=\frac{(1+t^2)-2t+(1-t^2)}{1+t^2}=\frac{2-2t}{1+t^2}$, after combining and simplifying (the t^2 terms cancel).

The integral becomes $\int \frac{1+t^2}{2-2t}\cdot\frac{2\,dt}{1+t^2}=\int\frac{2\,dt}{2-2t}=\int\frac{dt}{1-t}$ – again the $(1+t^2)$ factors cancel completely.

Integrating: $\int\frac{dt}{1-t}=-\log|1-t|+C$.

Converting back: $t=\tan\frac{x}{2}$.

Final Answer

$-\log\left|1-\tan\frac{x}{2}\right|+C$

Common Mistake: Making an algebra slip while combining the three fractions in the denominator (from 1, $-\sin x$, and $\cos x$ all converting separately) – write each converted term over the same denominator $1+t^2$ explicitly before combining, rather than trying to combine mentally.

Quick Recall: With more than one trig term in the denominator, convert every single one using the Weierstrass formulas, combine carefully over the common denominator $1+t^2$, and watch for cancellation with the $(1+t^2)$ that comes from dx.

Type 70

Integrate $\dfrac{1}{1 \pm \tan x}$ or $\dfrac{1}{1+\cot x}$ by multiplying numerator and denominator by $\cos x$ (or $\sin x$), converting the denominator into a $\sin x \pm \cos x$ combination – faster than the full Weierstrass substitution for this specific shape.

Formula Used: $\dfrac{1}{1+\tan x} = \dfrac{\cos x}{\cos x + \sin x}$

Source: Exercise 7.9, Q4(ii)

Given: $\dfrac{1}{1+\cot x}$

Solution:

Multiply numerator and denominator by $\sin x$: $\dfrac{1}{1+\cot x} = \dfrac{\sin x}{\sin x + \cos x}$, using $\cot x = \dfrac{\cos x}{\sin x}$.

Write $\sin x$ as a combination of $(\sin x + \cos x)$ and its derivative: $\sin x = \frac{1}{2}(\sin x + \cos x) + \frac{1}{2}(\sin x - \cos x)$, matching the Type 45/62 numerator-decomposition idea with $D(x) = \sin x + \cos x$.

Splitting: $\dfrac{\sin x}{\sin x + \cos x} = \dfrac{1}{2} + \dfrac{1}{2} \cdot \dfrac{\sin x - \cos x}{\sin x + \cos x} = \dfrac{1}{2} - \dfrac{1}{2} \cdot \dfrac{\cos x - \sin x}{\sin x + \cos x}$, and noting $\cos x - \sin x$ is (the negative of) the derivative of $\sin x + \cos x$.

Integrating: $\displaystyle\int \left[\frac{1}{2} - \frac{1}{2} \cdot \frac{-(D)'}{D}\right] dx = \frac{x}{2} + \frac{1}{2} \log|\sin x + \cos x| + C$, where the second piece is a direct log form.

Final Answer

$\dfrac{x}{2} + \dfrac{1}{2} \log|\sin x + \cos x| + C$

Common Mistake: Reaching for the full Weierstrass substitution here instead of the much faster $\times \cos x$ (or $\times \sin x$) trick – Weierstrass always works, but for this specific $1/(1 \pm \tan x)$ or $1/(1+\cot x)$ shape, the direct multiplication is significantly less algebra.

Quick Recall: For $1/(1 \pm \tan x)$ or $1/(1 \pm \cot x)$ specifically, multiply through by $\cos x$ or $\sin x$ first to reach a $\sin x \pm \cos x$ denominator, then apply the numerator-decomposition trick from Type 45 – save full Weierstrass for when nothing simpler is available.

Type 71

Given $\displaystyle\int \frac{k \tan x}{\tan x - c}\,dx = ax + b \log|\sin x - c\cos x| + C$, find the constants a, b by converting to $\sin x, \cos x$ and applying the numerator-decomposition technique from Type 45 to a tan-based denominator.

Formula Used: $\dfrac{\tan x}{\tan x - c} = \dfrac{\sin x}{\sin x - c\cos x}$

Source: Exercise 7.9, Q5

Given: If $\int \dfrac{5\tan x}{\tan x - 2}\,dx = ax + b\log|\sin x - 2\cos x| + C,$ find a, b

Solution:

Convert to sine and cosine: $\dfrac{5\tan x}{\tan x - 2} = \dfrac{5\sin x}{\sin x - 2\cos x}$, dividing numerator and denominator by $\cos x$.

Let $D(x) = \sin x - 2\cos x$, so $D'(x) = \cos x + 2\sin x$. Write the numerator as $5\sin x = A \cdot D(x) + B \cdot D'(x) = A(\sin x - 2\cos x) + B(\cos x + 2\sin x)$.

Matching coefficients of $\sin x$ and $\cos x$: from $\sin x$, $A + 2B = 5$; from $\cos x$, $-2A + B = 0 \Rightarrow B = 2A$. Substituting: $A + 4A = 5 \Rightarrow A = 1,\ B = 2$.

So $\dfrac{5\sin x}{\sin x - 2\cos x} = 1 + 2 \cdot \dfrac{D'(x)}{D(x)}$, splitting the integral into $\displaystyle\int 1\,dx + 2\int \frac{D'(x)}{D(x)}\,dx = x + 2\log|D(x)| + C$.

Final Answer

$a = 1,\ b = 2$

Common Mistake: Trying to match coefficients by inspection instead of setting up and solving the two simultaneous equations properly – with two unknowns and two trig terms to match, the systematic approach avoids sign errors.

Quick Recall: For $\dfrac{k\tan x}{\tan x - c}$, convert to $\sin x, \cos x$ first, then apply the same $A \cdot D + B \cdot D'$ numerator decomposition as Type 45 – the coefficient of x in the final answer is always A, and the coefficient of the log term is always B.

IV. Partial Fractions, Integration by Parts & Special Forms (Ex 7.10–7.12)

Type 72

Decompose a proper rational function with a denominator that factors into distinct (non-repeated) linear factors into a sum of simple fractions, one per factor, then integrate each piece as a standard log form.

★ **Board Favorite** – Partial fractions with distinct linear factors is the foundational case that every other partial-fraction type builds on, and appears in some form on almost every ISC board paper's integration section.

Formula Used: $\dfrac{f(x)}{(x-a)(x-b)} = \dfrac{A}{x-a} + \dfrac{B}{x-b}$

Source: Exercise 7.10, Q2(i)

Given: $\dfrac{2x+7}{x^2-x-2}$

Solution:

Factor the denominator: $x^2 - x - 2 = (x-2)(x+1)$.
Write $\dfrac{2x+7}{(x-2)(x+1)} = \dfrac{A}{x-2} + \dfrac{B}{x+1}$. Multiplying through by the denominator: $2x+7 = A(x+1) + B(x-2)$.
Substitute $x = 2$ (chosen to make the B term vanish): $4+7 = A(3) \Rightarrow A = \dfrac{11}{3}$. Substitute $x = -1$ (to make the A term vanish): $-2+7 = B(-3) \Rightarrow B = -\dfrac{5}{3}$.
So the integrand splits as $\dfrac{11/3}{x-2} - \dfrac{5/3}{x+1}$; integrating each term as a standard log form: $\displaystyle\int \left(\frac{11/3}{x-2} - \frac{5/3}{x+1}\right) dx = \frac{11}{3}\log|x-2| - \frac{5}{3}\log|x+1| + C$.

Final Answer

$\dfrac{11}{3}\log|x-2| - \dfrac{5}{3}\log|x+1| + C$

Common Mistake: Solving the two simultaneous equations for A and B by expanding and comparing coefficients instead of using the much faster "substitute the root" trick – plugging in $x = 2$ and $x = -1$ (the roots of each factor) isolates each constant instantly.

Quick Recall: Factor the denominator, write one unknown constant per linear factor, then substitute each factor's root in turn to solve for that constant directly – integrate each resulting simple fraction as a log.

Type 73

Decompose a proper rational function with a *repeated* linear factor in the denominator – the repeated factor needs one term for each power up to its multiplicity, not just one term for the whole repeated factor.

Formula Used: $\dfrac{f(x)}{(x-a)^2(x-b)} = \dfrac{A}{x-a} + \dfrac{B}{(x-a)^2} + \dfrac{C}{x-b}$

Source: Exercise 7.10, Q5(ii)

Given: $\dfrac{x}{(x-1)^2(x+2)}$

Solution:

Since $(x-1)$ is repeated, we need a term for both $(x-1)$ and $(x-1)^2$: $\dfrac{x}{(x-1)^2(x+2)} = \dfrac{A}{x-1} + \dfrac{B}{(x-1)^2} + \dfrac{C}{x+2}$.

Multiplying through: $x = A(x-1)(x+2) + B(x+2) + C(x-1)^2$.

Substitute $x = 1$ (isolates B): $1 = B(3) \Rightarrow B = \dfrac{1}{3}$. Substitute $x = -2$ (isolates C): $-2 = C(9) \Rightarrow C = -\dfrac{2}{9}$.

For A, compare the coefficient of x^2 on both sides (since substituting roots only found B, C): the x^2 coefficient on the right is $A + C$, and on the left it's 0, so $A = -C = \dfrac{2}{9}$.

Integrating each term: $\displaystyle\int \left(\frac{2/9}{x-1} + \frac{1/3}{(x-1)^2} - \frac{2/9}{x+2} \right) dx = \frac{2}{9}\log|x-1| - \frac{1}{3(x-1)} - \frac{2}{9}\log|x+2| + C$.

Final Answer

$\dfrac{2}{9}\log|x-1| - \dfrac{1}{3(x-1)} - \dfrac{2}{9}\log|x+2| + C$

Common Mistake: Writing only a single term $\dfrac{A}{(x-1)^2}$ for the repeated factor instead of the required *two* terms $\dfrac{A}{x-1} + \dfrac{B}{(x-1)^2}$ – every power of a repeated factor up to its multiplicity needs its own separate term.

Quick Recall: A repeated factor $(x-a)^k$ contributes k separate terms, $\dfrac{A_1}{x-a} + \dfrac{A_2}{(x-a)^2} + \cdots + \dfrac{A_k}{(x-a)^k}$ – substitute roots to find the constants tied to the highest powers directly, and compare leading coefficients for the rest.

Type 74

Decompose a proper rational function where the denominator includes an irreducible quadratic factor (one that doesn't factor over the reals) – this factor gets a linear numerator $Ax + B$, not just a constant, since a constant alone can't match all the necessary terms.

Formula Used: $\dfrac{f(x)}{(x-a)(x^2+b^2)} = \dfrac{A}{x-a} + \dfrac{Bx+C}{x^2+b^2}$

Source: Exercise 7.10, Q18(i)

Given: $\dfrac{1}{(x^2+1)(x^2+4)}$

Solution:

Here both factors are irreducible quadratics, so each gets a linear numerator: $\dfrac{1}{(x^2+1)(x^2+4)} = \dfrac{Ax+B}{x^2+1} + \dfrac{Cx+D}{x^2+4}$.

Since the original expression is even in x (unchanged if $x \to -x$), the odd-power terms Ax and Cx must both vanish, simplifying the setup to $\dfrac{1}{(x^2+1)(x^2+4)} = \dfrac{B}{x^2+1} + \dfrac{D}{x^2+4}$.

Multiplying through: $1 = B(x^2+4) + D(x^2+1)$. Comparing the constant term: $4B + D = 1$. Comparing the x^2 coefficient: $B + D = 0 \Rightarrow D = -B$.

Substituting: $4B - B = 1 \Rightarrow B = \dfrac{1}{3},\ D = -\dfrac{1}{3}$.

Integrating each term using the standard arctan form: $\displaystyle\int \left(\frac{1/3}{x^2+1} - \frac{1/3}{x^2+4} \right) dx = \frac{1}{3}\tan^{-1} x - \frac{1}{6}\tan^{-1}\frac{x}{2} + C$.

Final Answer

$\dfrac{1}{3}\tan^{-1} x - \dfrac{1}{6}\tan^{-1}\dfrac{x}{2} + C$

Common Mistake: Including the unnecessary Ax and Cx terms in the setup and doing more algebra than needed – recognising the original expression is even in x immediately tells you those odd-power numerator terms must be zero, saving significant work.

Quick Recall: An irreducible quadratic factor in the denominator always needs a linear (not constant) numerator in general – but check first whether the original expression is even in x, which lets you drop the odd-power numerator terms immediately.

Type 75

When the numerator's degree is at least the denominator's degree, divide first to get a polynomial plus a proper-fraction remainder – *then* apply partial fractions to the remainder, exactly as Type 57 did for standard-form denominators.

Formula Used: $\dfrac{f(x)}{g(x)} = Q(x) + \dfrac{R(x)}{g(x)}$ (divide first, then decompose the remainder)

Source: Exercise 7.10, Q2(ii)

Given: $\dfrac{x^2+1}{x^2-5x+6}$

Solution:

Since numerator and denominator have equal degree, divide first: $x^2+1 = (x^2-5x+6)+5x-5$, so $\dfrac{x^2+1}{x^2-5x+6} = 1 + \dfrac{5x-5}{x^2-5x+6}$.

Factor the remaining denominator: $x^2-5x+6 = (x-2)(x-3)$, and decompose $\dfrac{5x-5}{(x-2)(x-3)} = \dfrac{A}{x-2} + \dfrac{B}{x-3}$.

Substitute $x=2$: $5 = A(-1) \Rightarrow A = -5$. Substitute $x=3$: $10 = B(1) \Rightarrow B = 10$.

Integrating: $\displaystyle\int \left(1 - \frac{5}{x-2} + \frac{10}{x-3}\right) dx = x - 5\log|x-2| + 10\log|x-3| + C.$

Final Answer

$x - 5\log|x-2| + 10\log|x-3| + C$

Common Mistake: Trying to apply partial fractions directly to the original improper fraction – partial fraction decomposition only works on a *proper* fraction; skipping the division step produces an incorrect setup.

Quick Recall: Always check the degrees first – if the numerator's degree is not strictly less than the denominator's, divide before attempting partial fractions, exactly as with any other rational-function integral.

Type 76

Substitute $t = x^2$ (or another even power) first when the rational function is genuinely a function of x^2 alone (odd powers of x appear only as a single multiplying factor), reducing to an ordinary partial-fraction problem in t.

Formula Used: Let $t = x^2,\ dt = 2x\,dx$ when the denominator is a function of x^2

Source: Exercise 7.10, Q9(i)

Given: $\dfrac{1}{1-x^3}$

Solution:

This denominator isn't purely a function of x^2, so instead factor directly: $1 - x^3 = (1-x)(1+x+x^2)$, using the standard difference-of-cubes factorisation.

Decompose: $\dfrac{1}{(1-x)(1+x+x^2)} = \dfrac{A}{1-x} + \dfrac{Bx+C}{1+x+x^2}$ (the quadratic factor is irreducible, needing a linear numerator).

Multiplying through and substituting $x = 1$: $1 = A(3) \Rightarrow A = \dfrac{1}{3}$. Comparing coefficients for the remaining constants gives $B = \dfrac{1}{3},\ C = \dfrac{2}{3}$ (matching x^2 and constant terms after substituting A), so the quadratic piece is $\dfrac{1}{3} \cdot \dfrac{x+2}{1+x+x^2}$.

Decompose this piece's numerator using the derivative of the denominator $(2x+1)$: $x+2 = \dfrac{1}{2}(2x+1) + \dfrac{3}{2}$, splitting $\dfrac{1}{3} \cdot \dfrac{x+2}{1+x+x^2}$ into a log-form piece and an arctan piece after completing the square $(1+x+x^2 = \left(x+\frac{1}{2}\right)^2 + \frac{3}{4})$.

Combining all pieces and integrating: the $\dfrac{A}{1-x}$ term gives a log, and the quadratic piece gives a log-plus-arctan combination.

Final Answer

$$-\frac{1}{3}\log|x-1| + \frac{1}{6}\log(x^2+x+1) + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + C$$

Common Mistake: Trying to force a $t = x^2$ substitution onto a denominator that isn't actually a pure function of x^2 – always check the actual structure of the denominator (does it factor nicely as a sum/difference of cubes, or is it genuinely built from x^2?) before choosing a substitution.

Alternate Board-Style Example (genuine x^2 case)

Source: Exercise 7.10, Q11(ii)

Given: $\dfrac{1}{x(x^2+1)}$

Solution:

This denominator is genuinely built from x and x^2+1; decompose directly (no substitution needed here since the x factor is separate): $\dfrac{1}{x(x^2+1)} = \dfrac{A}{x} + \dfrac{Bx+C}{x^2+1}$.
Multiplying through and substituting $x = 0$: $1 = A(1) \Rightarrow A = 1$. Comparing x^2 coefficients: $0 = A + B \Rightarrow B = -1$. Comparing x coefficients: $0 = C$.
Integrating: $\displaystyle\int \left(\frac{1}{x} - \frac{x}{x^2+1}\right) dx = \log|x| - \frac{1}{2}\log(x^2+1) + C$.

Final Answer

$\log|x| - \dfrac{1}{2}\log(x^2+1) + C$

Quick Recall: Before reaching for a substitution, check whether the denominator factors directly (sum/difference of cubes, or an obvious common factor) – a genuine $t = x^2$ substitution is only needed when odd powers of x appear *exclusively* as a single accompanying factor throughout.

Type 77

Substitute for a trig or exponential expression first (recognising the "true" variable), then apply partial fractions to the resulting algebraic rational function in the new variable.

Formula Used: Substitute (e.g. $t = \sin x$ or $t = e^x$), then use partial fractions

Source: Exercise 7.10, Q13(ii)

Given: $\dfrac{1}{e^{2x}+1}$

Solution:

Let $t = e^x$, so $dt = e^x\,dx$, i.e. $dx = \dfrac{dt}{t}$ – chosen because the denominator is naturally a function of e^x.
Substituting: $\displaystyle\int \frac{1}{t^2+1}\cdot\frac{dt}{t} = \int \frac{dt}{t(t^2+1)}$.
Decompose using partial fractions: $\dfrac{1}{t(t^2+1)} = \dfrac{A}{t} + \dfrac{Bt+C}{t^2+1}$. Substituting $t = 0$: $A = 1$.
Comparing t^2 coefficients: $B = -1$; comparing t coefficients: $C = 0$.
Integrating: $\displaystyle\int \left(\frac{1}{t} - \frac{t}{t^2+1}\right) dt = \log|t| - \frac{1}{2}\log(t^2+1) + C$.
Converting back: $t = e^x$.

Final Answer

$x - \frac{1}{2}\log(e^{2x}+1) + C$

Common Mistake: Trying to apply partial fractions directly in terms of x (treating e^{2x} and e^x as if they were algebraically independent) – the substitution to a genuine algebraic variable t must come first before partial fractions can be applied at all.

Alternate Board-Style Example

Source: Exercise 7.10, Q14(ii)

Given: $\dfrac{\sin 2x}{(1+\sin x)(2+\sin x)}$

Solution:

Rewrite $\sin 2x = 2\sin x\cos x$, so the integral is $\displaystyle\int \frac{2\sin x\cos x}{(1+\sin x)(2+\sin x)}\,dx$.

Let $t = \sin x$, so $dt = \cos x\,dx$; substituting: $\displaystyle\int \frac{2t}{(1+t)(2+t)}\,dt$.

Decompose: $\dfrac{2t}{(1+t)(2+t)} = \dfrac{A}{1+t} + \dfrac{B}{2+t}$. Substituting $t=-1$: $-2 = A(1) \Rightarrow A = -2$.
Substituting $t=-2$: $-4 = B(-1) \Rightarrow B = 4$.

Integrating: $\displaystyle\int \left(\frac{-2}{1+t} + \frac{4}{2+t}\right) dt = -2\log|1+t| + 4\log|2+t| + C$.

Converting back: $t = \sin x$.

Final Answer

$-2\log|1+\sin x| + 4\log|2+\sin x| + C$

Quick Recall: Whenever a rational-looking expression is secretly built from e^x, $\sin x$, or another trig/exponential function (not x itself), substitute for that inner function first – then partial fractions applies exactly as in the purely algebraic case.

Type 78

For a denominator with a symmetric or reciprocal structure (like $x^4 - x^2 - 12$, secretly quadratic in x^2), substitute $t = x^2$ directly since every power of x present is already even – the cleanest version of the $t = x^2$ idea, with no odd-power factor to worry about.

Formula Used: Let $t = x^2$ when every term has an even power of x

Source: Exercise 7.10, Q19(ii)

Given: $\dfrac{x^2}{x^4 - x^2 - 12}$

Solution:

Every power of x present (x^2 and x^4) is even, so let $t = x^2$, giving $dt = 2x\,dx$ – but here we need to handle the $x^2\,dx$ combination carefully since there's no lone x factor; instead, treat the whole expression as a function of t directly via $x^4 = t^2$.
Rewrite the integral using $x^2 = t$ in both numerator and denominator conceptually, then factor the denominator as a quadratic in x^2: $x^4 - x^2 - 12 = (x^2 - 4)(x^2 + 3)$.
Decompose $\dfrac{x^2}{(x^2-4)(x^2+3)} = \dfrac{A}{x^2-4} + \dfrac{B}{x^2+3}$ treating x^2 as the variable: multiplying through, $x^2 = A(x^2+3) + B(x^2-4)$; matching coefficients, $A + B = 1$ and $3A - 4B = 0$, giving $A = \dfrac{4}{7}$, $B = \dfrac{3}{7}$.
Integrating each piece using the standard forms for $1/(x^2 - a^2)$ and $1/(x^2 + a^2)$:

$$\int \left(\frac{4/7}{x^2-4} + \frac{3/7}{x^2+3} \right) dx = \frac{4}{7} \cdot \frac{1}{4} \log \left| \frac{x-2}{x+2} \right| + \frac{3}{7} \cdot \frac{1}{\sqrt{3}} \tan^{-1} \frac{x}{\sqrt{3}} + C.$$

Final Answer

$\dfrac{1}{7} \log \left| \dfrac{x-2}{x+2} \right| + \dfrac{\sqrt{3}}{7} \tan^{-1} \dfrac{x}{\sqrt{3}} + C$

Common Mistake: Trying to substitute $t = x^2$ and carry the substitution all the way through the integration (including dx) when there's no lone factor of x to absorb into dt – for this shape, it's actually cleaner to treat x^2 as "the variable" only for the *partial-fraction decomposition step*, then integrate the resulting pieces directly in x using the standard forms from Type 52–53.

Quick Recall: When a denominator is a quadratic in x^2 (i.e. only even powers of x appear throughout, with no lone x multiplying), factor and decompose treating x^2 as the variable, then integrate each resulting piece in x directly using the standard $1/(x^2 \pm a^2)$ forms – no explicit substitution or extra dx conversion is needed.

Type 79

The most basic integration-by-parts pattern: a single polynomial factor x multiplied by a standard exponential or trig function, chosen as the "first function" using the ILATE priority order (Inverse trig, Log, Algebraic, Trig, Exponential – pick whichever comes first in this list as u).

Formula Used: $\int u\,v\,dx = u \int v\,dx - \int \left[\dfrac{du}{dx} \int v\,dx \right] dx$

Source: Exercise 7.11, Q1(i)

Given: xe^x

Solution:

By ILATE, Algebraic (x) comes before Exponential (e^x), so let $u = x$ (differentiates to something simpler) and $dv = e^x\,dx$ (integrates easily).
Then $du = dx$ and $v = e^x$.
Applying the by-parts formula: $\displaystyle\int xe^x\,dx = xe^x - \int e^x\,dx = xe^x - e^x + C$.

Final Answer

$(x-1)e^x + C$

Common Mistake: Choosing $u = e^x$ and $dv = x\,dx$ instead – this technically "works" through the formula but produces a harder integral on the right side ($\int \frac{x^2}{2} e^x\,dx$) instead of an easier one; always pick u so that differentiating simplifies it.

Quick Recall: ILATE order (Inverse trig, Log, Algebraic, Trig, Exponential) tells you which factor to differentiate (u) – whichever type appears earlier in this list, differentiating it should make the problem simpler, not harder.

Type 80

The same basic by-parts pattern as Type 79, but the trig/exponential factor has a linear (compound) argument like $\sin 3x$ or $\cos 2x$ – an extra layer of care is needed when integrating v, since it picks up a $\frac{1}{a}$ factor from the chain rule.

Formula Used: $\int f(ax+b)\,dx = \dfrac{1}{a}F(ax+b) + C$, applied to the dv piece

Source: Exercise 7.11, Q3(ii)

Given: $x\cos 2x$

Solution:

By ILATE, let $u = x$ and $dv = \cos 2x\,dx$.
Then $du = dx$ and $v = \displaystyle\int \cos 2x\,dx = \frac{\sin 2x}{2}$ (remembering the $\frac{1}{2}$ from the compound argument).
Applying by parts: $\displaystyle\int x\cos 2x\,dx = x\cdot\frac{\sin 2x}{2} - \int \frac{\sin 2x}{2}\,dx = \frac{x\sin 2x}{2} - \frac{1}{2}\cdot\left(-\frac{\cos 2x}{2}\right) + C$.

Final Answer

$\dfrac{x\sin 2x}{2} + \dfrac{\cos 2x}{4} + C$

Common Mistake: Forgetting the $\frac{1}{2}$ factor when integrating $\cos 2x$ to get v – this single missing constant throws off both the main term and the remaining integral.

Quick Recall: When the trig/exponential factor has a compound linear argument, integrate it exactly as in Type 16 (divide by the coefficient of x) when finding v – everything else about the by-parts process is unchanged.

Type 81

The "second function" dv is itself not a standard integral, but a recognisable derivative-of-a-standard-form pattern (like $\sec^2 x \tan x$) – integrate dv using a quick substitution first, *then* proceed with by parts as usual.

Formula Used: $\int \sec^2 x \tan x\, dx = \dfrac{\sec^2 x}{2} + C$ (via $t = \sec x$, or equivalently $\dfrac{\tan^2 x}{2} + C$)

Source: Exercise 7.11, Q4(i)

Given: $x \sec^2 x \tan x$

Solution:

By ILATE, let $u = x$ and $dv = \sec^2 x \tan x\, dx$.
Before proceeding, integrate dv separately: let $t = \sec x$, so $dt = \sec x \tan x\, dx$; rewriting $\sec^2 x \tan x\, dx = \sec x \cdot (\sec x \tan x\, dx) = t\, dt$, giving $\displaystyle\int t\, dt = \frac{t^2}{2} = \frac{\sec^2 x}{2}$. So $v = \dfrac{\sec^2 x}{2}$.
Applying by parts: $\displaystyle\int x \sec^2 x \tan x\, dx = x \cdot \frac{\sec^2 x}{2} - \int \frac{\sec^2 x}{2}\, dx = \frac{x \sec^2 x}{2} - \frac{\tan x}{2} + C.$

Final Answer

$$\frac{x \sec^2 x}{2} - \frac{\tan x}{2} + C$$

Common Mistake: Trying to guess v without actually working out $\int \sec^2 x \tan x\, dx$ properly first – always fully solve the dv integral as its own small problem before plugging it into the by-parts formula.

Quick Recall: If dv isn't a standard form you recognise immediately, stop and integrate it separately (often via a quick substitution) before continuing with the by-parts formula – don't try to do both steps simultaneously.

Type 82

A polynomial multiplied by $\log x$ – by ILATE, $\log x$ is always chosen as u (Logarithmic comes before Algebraic), even though it looks like the "harder" factor, since $\log x$

differentiates to something simpler ($\frac{1}{x}$) while the polynomial integrates easily.

★ **Board Favorite** – Polynomial-times-$\log x$ is one of the most frequently tested by-parts patterns on ISC papers, precisely because it's the clearest illustration of why ILATE order matters – picking $u = \log x$ (not the polynomial) is what makes the problem tractable.

Formula Used: $\dfrac{d}{dx}(\log x) = \dfrac{1}{x}$

Source: Exercise 7.11, Q6(i)

Given: $x^4 \log x$

Solution:

By ILATE, Logarithmic ($\log x$) comes before Algebraic (x^4), so let $u = \log x$ and $dv = x^4\,dx$.

Then $du = \dfrac{1}{x}\,dx$ and $v = \dfrac{x^5}{5}$.

Applying by parts: $\displaystyle\int x^4 \log x\,dx = \log x \cdot \frac{x^5}{5} - \int \frac{x^5}{5} \cdot \frac{1}{x}\,dx = \frac{x^5 \log x}{5} - \frac{1}{5}\int x^4\,dx.$

Completing the remaining integral: $\dfrac{x^5 \log x}{5} - \dfrac{1}{5} \cdot \dfrac{x^5}{5} + C.$

Final Answer

$\dfrac{x^5}{5}\log x - \dfrac{x^5}{25} + C$

Common Mistake: Choosing $u = x^4$ (the polynomial) instead of $u = \log x$ – this leaves $\log x$ needing to be integrated for v, which is itself a by-parts problem (Type 83), making the whole thing needlessly circular.

Quick Recall: Whenever $\log x$ appears multiplied by a polynomial, $\log x$ is always u – it simplifies on differentiating ($\to \frac{1}{x}$), while the polynomial integrates easily as v.

Type 83

Integrate $\log x$ on its own by treating it as $(\log x) \cdot 1$ – a special single-function case of by parts where $dv = 1\,dx$ is the "invisible" second function.

Formula Used: $\int \log x\,dx = x \log x - x + C$

Source: Exercise 7.11, Q6(ii)

Given: $\log x$

Solution:

Rewrite as a product: $\log x = (\log x) \cdot 1$, treating the constant 1 as the second function $dv = 1\,dx$ – this is the key trick that lets by parts apply even though only one function is visibly present.

Let $u = \log x$, $dv = 1\,dx$, so $du = \frac{1}{x}\,dx$ and $v = x$.

Applying by parts: $\int \log x\,dx = x \log x - \int x \cdot \frac{1}{x}\,dx = x \log x - \int 1\,dx = x \log x - x + C$.

Final Answer

$x \log x - x + C$

Common Mistake: Not recognising that a single function can still be integrated by parts by treating it as multiplied by 1 – this trick applies to $\log x$, $\tan^{-1} x$, $\sin^{-1} x$, and any other "lone" function whose derivative is simpler than the function itself.

Quick Recall: A single non-standard function (with no obvious antiderivative) can almost always be integrated by writing it as (function) $\times 1$ and applying by parts with $dv = dx$.

Type 84

A polynomial multiplied by log of a *compound* expression (like $\log(1+x)$, not just $\log x$) – the same ILATE priority applies, but differentiating $u = \log(1+x)$ now requires the chain rule, and the resulting $\frac{du}{dx}$ combines with the polynomial in a way that needs further algebraic simplification.

Formula Used: $\frac{d}{dx} \log(1+x) = \frac{1}{1+x}$

Source: Exercise 7.11, Q7(ii)

Given: $x^2 \log(1+x)$

Solution:

By ILATE, let $u = \log(1+x)$ and $dv = x^2\,dx$.

Then $du = \frac{1}{1+x}\,dx$ and $v = \frac{x^3}{3}$.

Applying by parts: $\int x^2 \log(1+x)\,dx = \frac{x^3}{3} \log(1+x) - \frac{1}{3} \int \frac{x^3}{1+x}\,dx$.

The remaining integral is an improper fraction, so divide first: $\frac{x^3}{1+x} = x^2 - x + 1 - \frac{1}{1+x}$ (polynomial long division).

Integrating this remainder: $\int \left(x^2 - x + 1 - \frac{1}{1+x}\right) dx = \frac{x^3}{3} - \frac{x^2}{2} + x - \log(1+x)$.

Final Answer

$\frac{x^3}{3}\log(1+x) - \frac{1}{3}\left[\frac{x^3}{3} - \frac{x^2}{2} + x - \log(1+x)\right] + C$

Common Mistake: Stopping after applying by parts once and not noticing that the remaining integral $\int \frac{x^3}{1+x}dx$ is itself improper and needs division before it can be completed – by-parts problems with a compound-log u very often need a second technique (division, substitution, or partial fractions) to finish the remaining integral.

Quick Recall: When u is log of a compound expression, du will be a fraction – after by parts, check whether the resulting integral is a proper fraction; if not, divide before continuing.

Type 85

A trig function multiplied by log of a *related* trig function (like $\sin x \cdot \log(\cos x)$) – by ILATE, the log factor is still u, and dv integrates to a standard trig form, but differentiating $\log(\cos x)$ produces $-\tan x$, which then combines with the v piece to need a further substitution.

Formula Used: $\frac{d}{dx}\log(\cos x) = -\tan x$

Source: Exercise 7.11, Q7(iii)

Given: $\sin x \cdot \log(\cos x)$

Solution:

By ILATE, let $u = \log(\cos x)$ and $dv = \sin x\,dx$.
Then $du = -\tan x\,dx$ and $v = -\cos x$ (since $\int \sin x\,dx = -\cos x$).
Applying by parts: $\displaystyle\int \sin x \log(\cos x)\,dx = -\cos x\log(\cos x) - \int(-\cos x)(-\tan x)\,dx = -\cos x\log(\cos x) - \int \cos x\tan x\,dx.$

Simplify the remaining integral: $\cos x\tan x = \sin x$, so $\displaystyle\int \sin x\,dx = -\cos x$.

Final Answer

$-\cos x\log(\cos x) + \cos x + C$

Common Mistake: Leaving the remaining integral as $\int \cos x\tan x\,dx$ without simplifying it to $\int \sin x\,dx$ first – always simplify algebraically before assuming a second by-parts round or substitution is needed; sometimes it collapses to something trivial.

Quick Recall: When u is a log of a trig function, its derivative is a standard trig ratio (memorise $\frac{d}{dx}\log\sin x = \cot x$, $\frac{d}{dx}\log\cos x = -\tan x$) – after by parts, always simplify the resulting product before deciding what to do next.

Type 86

Integrate a single inverse trig function by parts, treating it as (function)×1 exactly like Type 83 for $\log x$ – the derivative of the inverse trig function becomes the new integrand's second piece, usually requiring one more substitution to finish.

Formula Used: $\dfrac{d}{dx}\tan^{-1} x = \dfrac{1}{1+x^2}$

Source: Exercise 7.11, Q8(i)

Given: $\tan^{-1} x$

Solution:

Rewrite as $(\tan^{-1} x)\cdot 1$, letting $u = \tan^{-1} x$ and $dv = 1\,dx$, so $du = \dfrac{1}{1+x^2}\,dx$ and $v = x$.
Applying by parts: $\displaystyle\int \tan^{-1} x\,dx = x\tan^{-1} x - \int \frac{x}{1+x^2}\,dx$.
The remaining integral is a direct log form (numerator is half the denominator's derivative): $\displaystyle\int \frac{x}{1+x^2}\,dx = \frac{1}{2}\log(1+x^2)$.

Final Answer

$x\tan^{-1} x - \dfrac{1}{2}\log(1+x^2) + C$

Common Mistake: Not recognising the remaining integral $\int \frac{x}{1+x^2}dx$ as a direct log form and instead attempting an unnecessary substitution from scratch – always check for the $g'(x)/g(x)$ pattern first (Type 26) before reaching for a heavier technique.

Quick Recall: Any lone inverse trig function integrates by writing it as (function)×1; the resulting second integral (from $u\,dv$) is almost always a direct log or standard form using the inverse trig function's own derivative.

Type 87

A polynomial of degree 2 or higher multiplied by an inverse trig function – by ILATE the inverse trig function is u, but the resulting remainder integral is now a genuine rational function (not just a simple log form), often needing division or partial fractions to finish.

★ **Board Favorite** – Polynomial-times-inverse-trig by parts is a standard high-yield pattern precisely because it combines by parts with a second full technique (division or partial fractions) in the remainder – ISC frequently tests this two-technique combination directly.

Formula Used: $\dfrac{d}{dx}\sin^{-1}x = \dfrac{1}{\sqrt{1-x^2}}$, $\quad \dfrac{d}{dx}\tan^{-1}x = \dfrac{1}{1+x^2}$

Source: Exercise 7.11, Q9(i)

Given: $x^3\tan^{-1}x$

Solution:

By ILATE, let $u = \tan^{-1}x$ and $dv = x^3\,dx$, so $du = \dfrac{1}{1+x^2}\,dx$ and $v = \dfrac{x^4}{4}$.

Applying by parts: $\displaystyle\int x^3\tan^{-1}x\,dx = \frac{x^4}{4}\tan^{-1}x - \frac{1}{4}\int\frac{x^4}{1+x^2}\,dx$.

The remaining fraction is improper; divide: $\dfrac{x^4}{1+x^2} = x^2 - 1 + \dfrac{1}{1+x^2}$.

Integrating this remainder: $\displaystyle\int\left(x^2 - 1 + \frac{1}{1+x^2}\right)dx = \frac{x^3}{3} - x + \tan^{-1}x$.

Final Answer

$$\frac{x^4}{4}\tan^{-1}x - \frac{1}{4}\left[\frac{x^3}{3} - x + \tan^{-1}x\right] + C$$

Common Mistake: Trying to integrate $\dfrac{x^4}{1+x^2}$ directly using a substitution instead of recognising it as an improper fraction needing division first – always check the degree comparison before choosing a technique for the remainder.

Alternate Board-Style Example

Source: Exercise 7.11, Q9(ii)

Given: $x^2\sin^{-1}x$

Solution:

Let $u = \sin^{-1}x$, $dv = x^2\,dx$, so $du = \dfrac{1}{\sqrt{1-x^2}}\,dx$ and $v = \dfrac{x^3}{3}$.

Applying by parts: $\displaystyle\int x^2\sin^{-1}x\,dx = \frac{x^3}{3}\sin^{-1}x - \frac{1}{3}\int\frac{x^3}{\sqrt{1-x^2}}\,dx$.

For the remaining integral, let $t = 1-x^2$ so $x^2 = 1-t$ and $x\,dx = -\frac{1}{2}dt$: $\displaystyle\int\frac{x^3}{\sqrt{1-x^2}}\,dx = \int\frac{x^2\cdot x\,dx}{\sqrt{1-x^2}} = -\frac{1}{2}\int\frac{1-t}{\sqrt{t}}\,dt = -\frac{1}{2}\left(2\sqrt{t} - \frac{2}{3}t^{3/2}\right)$.

Converting back and simplifying: $-\sqrt{1-x^2}+\frac{1}{3}(1-x^2)^{3/2}$.

Final Answer

$\frac{x^3}{3}\sin^{-1}x-\frac{1}{3}\left[-\sqrt{1-x^2}+\frac{1}{3}(1-x^2)^{3/2}\right]+C$

Quick Recall: Polynomial-times-inverse-trig always leaves a rational or algebraic remainder after one round of by parts – check whether it's improper (divide) or needs a substitution (as with the $\sqrt{1-x^2}$ family), then finish with that second technique.

Type 88

Prepare an integrand algebraically *before* by parts becomes visible – an expression like $\frac{\sin^{-1}x}{\sqrt{1-x}}$ needs a substitution or rewriting first to reveal the product structure that by parts can act on.

Formula Used: $\sqrt{1-x}=\sqrt{1-x}$, rewrite to expose a by-parts-friendlyproduct

Source: Exercise 7.11, Q10(i)

Given: $\frac{\sin^{-1}x}{\sqrt{1-x}}$

Solution:

Rewrite the fraction to expose two separate factors: $\frac{\sin^{-1}x}{\sqrt{1-x}}=\sin^{-1}x\cdot\frac{1}{\sqrt{1-x}}$ – this is now a genuine product, ready for by parts once we find the antiderivative of the second piece.

Let $u=\sin^{-1}x$, $dv=\frac{1}{\sqrt{1-x}}dx$; integrating dv: $v=-2\sqrt{1-x}$ (a direct chain-rule reverse).

Then $du=\frac{1}{\sqrt{1-x^2}}dx=\frac{1}{\sqrt{(1-x)(1+x)}}dx$.

Applying by parts:

$$\int\frac{\sin^{-1}x}{\sqrt{1-x}}dx=-2\sqrt{1-x}\sin^{-1}x+2\int\frac{\sqrt{1-x}}{\sqrt{(1-x)(1+x)}}dx=-2\sqrt{1-x}\sin^{-1}x+2\int\frac{dx}{\sqrt{1+x}}.$$

The remaining integral is direct: $2\int(1+x)^{-1/2}dx=4\sqrt{1+x}$.

Final Answer

$-2\sqrt{1-x}\sin^{-1}x+4\sqrt{1+x}+C$

Common Mistake: Not noticing that the fraction can be split into a clean product before starting by parts – always look for a way to expose a genuine "two-factor" structure algebraically before deciding a technique doesn't apply.

Quick Recall: If an integrand doesn't look like an obvious product, try rewriting a fraction as (numerator)×(reciprocal of denominator) – this often reveals a by-parts structure that wasn't visible in the original fraction form.

Type 89

A quadratic (or higher) polynomial multiplied by an exponential or trig function – by parts must be applied *twice* (or more), since each round reduces the polynomial's degree by one but doesn't eliminate it entirely until it reaches degree zero.

★ **Board Favorite** – Repeated by-parts for a quadratic-times-exponential/trig is a standard high-yield pattern – it directly tests whether students can correctly manage the by-parts formula across multiple rounds without losing a sign or a term.

Formula Used: Apply by parts once per degree, until the polynomial is constant

Source: Exercise 7.11, Q11(i)

Given: x^2e^{3x}

Solution:

By ILATE, let $u = x^2$, $dv = e^{3x}dx$, so $du = 2x\,dx$ and $v = \dfrac{e^{3x}}{3}$.

First round: $\displaystyle\int x^2e^{3x}\,dx = \frac{x^2e^{3x}}{3} - \frac{2}{3}\int xe^{3x}\,dx$.

The remaining integral $\int xe^{3x}dx$ is itself a by-parts problem: let $u = x$, $dv = e^{3x}dx$, so $v = \dfrac{e^{3x}}{3}$, giving $\displaystyle\int xe^{3x}dx = \frac{xe^{3x}}{3} - \frac{1}{3}\int e^{3x}dx = \frac{xe^{3x}}{3} - \frac{e^{3x}}{9}$.

Substituting this back into the first round's result: $\dfrac{x^2e^{3x}}{3} - \dfrac{2}{3}\left[\dfrac{xe^{3x}}{3} - \dfrac{e^{3x}}{9}\right] = \dfrac{x^2e^{3x}}{3} - \dfrac{2xe^{3x}}{9} + \dfrac{2e^{3x}}{27}$.

Final Answer

$$e^{3x}\left(\frac{x^2}{3} - \frac{2x}{9} + \frac{2}{27}\right) + C$$

Common Mistake: Losing track of the accumulated constant factors (like the $-\frac{2}{3}$ carried from the first round) when substituting the second round's result back in – write out each round as a complete, separate mini-calculation, then substitute carefully rather than trying to track everything in your head at once.

Quick Recall: For polynomial degree n times exponential/trig, expect exactly n rounds of by parts – keep each round's result clearly labelled so substituting back into the previous round doesn't lose a constant.

Type 90

Integrate $(\log x)^2$ using by parts, treating the whole squared expression as u – this reduces to a Type 82-style integral (a single $\log x$ times a polynomial) after one round, requiring a second round to finish.

Formula Used: $\dfrac{d}{dx}(\log x)^2 = \dfrac{2\log x}{x}$

Source: Exercise 7.11, Q12(i)

Given: $(\log x)^2$

Solution:

Rewrite as $(\log x)^2 \cdot 1$, letting $u = (\log x)^2$ and $dv = 1\,dx$, so $du = \dfrac{2\log x}{x}\,dx$ and $v = x$.

Applying by parts: $\displaystyle\int (\log x)^2\,dx = x(\log x)^2 - \int x \cdot \frac{2\log x}{x}\,dx = x(\log x)^2 - 2\int \log x\,dx$.

The remaining integral is the standard $\log x$ result from Type 83: $\displaystyle\int \log x\,dx = x\log x - x$.

Final Answer

$x(\log x)^2 - 2x\log x + 2x + C$

Common Mistake: Trying to apply by parts to $(\log x)^2$ as if it were two separate factors of $\log x$ multiplied together (which would need the product rule for u, not standard by parts) – treat the whole square as a single function u and differentiate it as one unit using the chain rule.

Quick Recall: $(\log x)^2$, $(\sin^{-1} x)^2$, and similar squared functions integrate the same way as Type 83/86 – write as (function)×1, differentiate the whole square via the chain rule, and expect one extra round of by parts to finish off the resulting simpler log/inverse-trig term.

Type 91

The integrand doesn't look like a by-parts product at all until a substitution (usually for the argument inside a trig or exponential function) is made first – after substituting, the new integral in the substituted variable reveals the by-parts structure.

★ **Board Favorite** – Substitution-then-by-parts is a standard high-yield combination pattern – ISC frequently disguises a by-parts problem behind a substitution specifically to test whether students recognise the need for both techniques in sequence.

Formula Used: Substitute first to reveal a $u\,dv$ product, then use by parts

Source: Exercise 7.11, Q12(ii)

Given: $x^3 \sin(x^2)$

Solution:

Let $t = x^2$, so $dt = 2x\,dx$; rewrite $x^3 \sin(x^2)\,dx = x^2 \sin(x^2) \cdot x\,dx = t \sin t \cdot \frac{dt}{2}$, splitting off one factor of x to supply dt and leaving $x^2 = t$ as a genuine algebraic factor.

The integral becomes $\frac{1}{2}\int t \sin t\,dt$ – now a standard by-parts problem in t.

Let $u = t$, $dv = \sin t\,dt$, so $du = dt$, $v = -\cos t$: $\int t \sin t\,dt = -t\cos t + \int \cos t\,dt = -t\cos t + \sin t$.

So the integral is $\frac{1}{2}(-t\cos t + \sin t) + C$. Converting back: $t = x^2$.

Final Answer

$\frac{1}{2}\left[\sin(x^2) - x^2\cos(x^2)\right] + C$

Common Mistake: Trying to apply by parts directly in terms of x (treating x^3 and $\sin(x^2)$ as the two factors) – this doesn't work cleanly because $\sin(x^2)$'s antiderivative isn't a standard form in x; the substitution must happen first to convert the trig argument to a plain variable.

Quick Recall: If a trig or exponential function has a non-linear argument (like x^2) and there's a spare power of x available to form dt, substitute first – by parts only becomes visible after this conversion to the new variable.

Type 92

Integrate an inverse trig or trig function of $\sqrt{x}$ by substituting $t = \sqrt{x}$ (or $x = t^2$) first, converting dx into $2t\,dt$ – this conversion typically creates the extra polynomial factor needed for by parts in the new variable.

Formula Used: Let $x = t^2$, $dx = 2t\,dt$ when the argument is $\sqrt{x}$

Source: Exercise 7.11, Q14(i)

Given: $\tan^{-1}\sqrt{x}$

Solution:

Let $x = t^2$, so $\sqrt{x} = t$ and $dx = 2t\,dt$.

Substituting: $\int \tan^{-1}\sqrt{x}\,dx = \int \tan^{-1} t \cdot 2t\,dt = 2\int t\tan^{-1} t\,dt$ – this is now a standard polynomial-times-inverse-trig by-parts problem (Type 87) in t.

Let $u = \tan^{-1} t$, $dv = t\,dt$, so $du = \frac{1}{1+t^2}dt$, $v = \frac{t^2}{2}$: $\int t\tan^{-1} t\,dt = \frac{t^2}{2}\tan^{-1} t - \frac{1}{2}\int \frac{t^2}{1+t^2}\,dt$.

The remaining fraction is improper: $\frac{t^2}{1+t^2} = 1 - \frac{1}{1+t^2}$, integrating to $t - \tan^{-1} t$.

Combining: $2\left[\frac{t^2}{2}\tan^{-1} t - \frac{1}{2}(t - \tan^{-1} t)\right] = t^2\tan^{-1} t - t + \tan^{-1} t$. Converting back: $t = \sqrt{x}$.

Final Answer

$x\tan^{-1}\sqrt{x} - \sqrt{x} + \tan^{-1}\sqrt{x} + C$

Common Mistake: Forgetting to convert dx into $2t\,dt$ (not just replacing $\sqrt{x}$ with t inside the function) – both the function's argument *and* the differential must be rewritten in the new variable before by parts can be applied.

Quick Recall: Any inverse trig or trig function of $\sqrt{x}$ substitutes to $x = t^2$ first – the resulting $2t\,dt$ almost always supplies exactly the polynomial factor needed to turn the problem into a standard by-parts pattern in t.

Type 93

Simplify the argument of an inverse trig function using a triple-angle identity *before* attempting by parts – expressions like $\sin^{-1}(3x - 4x^3)$ or $\tan^{-1}\left(\frac{3x - x^3}{1 - 3x^2}\right)$ collapse to $3\sin^{-1} x$ or $3\tan^{-1} x$, turning a seemingly hard integral into a trivial one.

★ **Board Favorite** – Recognising the triple-angle collapse is a standard high-yield pattern – without spotting it, students attempt by parts directly on an extremely messy derivative; with it, the problem becomes almost immediate.

Formula Used: $\sin 3\theta = 3\sin\theta - 4\sin^3\theta \implies \sin^{-1}(3x - 4x^3) = 3\sin^{-1} x$ (on a restricted domain)

Source: Exercise 7.11, Q16(i)

Given: $\sin^{-1}(3x - 4x^3)$

Solution:

Recognise the triple-angle sine identity in reverse: if $x = \sin\theta$, then $3x - 4x^3 = 3\sin\theta - 4\sin^3\theta = \sin 3\theta$, so $\sin^{-1}(3x - 4x^3) = 3\theta = 3\sin^{-1}x$ (valid on the restricted domain $|x| \leq \frac{1}{2}$ where the principal branches align).

The integral collapses to $\displaystyle\int 3\sin^{-1}x\,dx = 3\int \sin^{-1}x\,dx$, a direct Type 86-style integral.

Using the standard result $\displaystyle\int \sin^{-1}x\,dx = x\sin^{-1}x + \sqrt{1-x^2}$: multiply through by 3.

Final Answer

$3x\sin^{-1}x + 3\sqrt{1-x^2} + C$

Common Mistake: Attempting to differentiate $\sin^{-1}(3x-4x^3)$ directly by the chain rule and proceeding with by parts on the resulting messy expression – this is technically possible but extremely long; always check first whether the argument matches a double- or triple-angle identity.

Quick Recall: $3x - 4x^3$ inside $\sin^{-1}$, or $\dfrac{3x - x^3}{1 - 3x^2}$ inside $\tan^{-1}$, are always the triple-angle identities in disguise – they collapse to $3\sin^{-1}x$ or $3\tan^{-1}x$ (on a restricted domain), turning a hard-looking problem into a simple constant multiple of a standard integral.

Type 94

Choose dv cleverly (not the "obvious" leftover piece) so that v turns out to be a simple algebraic expression rather than another inverse trig function – for $\dfrac{\sin^{-1}x}{(1-x^2)^{3/2}}$, recognise that this whole fraction is actually $\sin^{-1}x$ times the derivative of $\dfrac{x}{\sqrt{1-x^2}}$.

Formula Used: $\dfrac{d}{dx}\left(\dfrac{x}{\sqrt{1-x^2}}\right) = \dfrac{1}{(1-x^2)^{3/2}}$

Source: Exercise 7.11, Q17(i)

Given: $\dfrac{\sin^{-1}x}{(1-x^2)^{3/2}}$

Solution:

Recognise that $\dfrac{1}{(1-x^2)^{3/2}}$ is exactly the derivative of $\dfrac{x}{\sqrt{1-x^2}}$ – this identifies our choice for dv.

Let $u = \sin^{-1}x$, $dv = \dfrac{1}{(1-x^2)^{3/2}}\,dx$, so $du = \dfrac{1}{\sqrt{1-x^2}}\,dx$ and $v = \dfrac{x}{\sqrt{1-x^2}}$.

Applying by parts: $\displaystyle\int \frac{\sin^{-1}x}{(1-x^2)^{3/2}}\,dx = \frac{x\sin^{-1}x}{\sqrt{1-x^2}} - \int \frac{x}{\sqrt{1-x^2}} \cdot \frac{1}{\sqrt{1-x^2}}\,dx = \frac{x\sin^{-1}x}{\sqrt{1-x^2}} -$

$\int \frac{x}{1-x^2}\, dx$.

The remaining integral is a direct log form: $\int \frac{x}{1-x^2}\, dx = -\frac{1}{2}\log(1-x^2)$.

Final Answer

$\frac{x\sin^{-1}x}{\sqrt{1-x^2}} + \frac{1}{2}\log(1-x^2) + C$

Common Mistake: Trying the "obvious" split $u = \sin^{-1}x,\ dv = (1-x^2)^{-3/2}dx$ without first recognising what v actually is – knowing $v = \frac{x}{\sqrt{1-x^2}}$ in advance (rather than integrating $(1-x^2)^{-3/2}$ from scratch) is what makes this problem tractable at all.

Quick Recall: Unusual-looking denominators like $(1-x^2)^{3/2}$ are frequently the derivative of a simpler algebraic expression – check standard derivative tables mentally before assuming a fraction needs a hard integration technique.

Type 95

The "cyclic" by-parts technique: applying by parts twice to $e^{ax}\sin(bx)$ (or $\cos(bx)$) returns the *original* integral multiplied by a constant on the right-hand side – treat the whole equation as algebra and solve for the unknown integral directly, rather than continuing by parts forever.

★ **Board Favorite** – The cyclic $e^{ax}\sin bx$ technique is one of the most important and frequently tested integration methods in the entire syllabus – it's the standard way to handle any exponential-times-trig integral, and appears on nearly every board paper in some form.

Formula Used: $\int e^{ax}\sin bx\, dx = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + C$

Source: Exercise 7.11, Q19(i)

Given: $e^{ax}\sin bx$

Solution:

Let $I = \int e^{ax}\sin bx\, dx$. Apply by parts with $u = \sin bx$, $dv = e^{ax}dx$: $I = \frac{e^{ax}\sin bx}{a} - \frac{b}{a}\int e^{ax}\cos bx\, dx$.

Apply by parts again to the remaining integral, with $u = \cos bx$, $dv = e^{ax}dx$: $\int e^{ax}\cos bx\, dx = \frac{e^{ax}\cos bx}{a} + \frac{b}{a}\int e^{ax}\sin bx\, dx = \frac{e^{ax}\cos bx}{a} + \frac{b}{a}I$ – crucially, the original integral I has reappeared.

Substituting back: $I = \frac{e^{ax}\sin bx}{a} - \frac{b}{a}\left[\frac{e^{ax}\cos bx}{a} + \frac{b}{a}I\right] = \frac{e^{ax}\sin bx}{a} - \frac{be^{ax}\cos bx}{a^2} - \frac{b^2}{a^2}I$.

Collect all I terms on the left: $I + \frac{b^2}{a^2}I = \frac{e^{ax}\sin bx}{a} - \frac{be^{ax}\cos bx}{a^2}$, so $I\left(\frac{a^2+b^2}{a^2}\right) = \frac{ae^{ax}\sin bx - be^{ax}\cos bx}{a^2}$.

Solving for I: multiply both sides by $\frac{a^2}{a^2+b^2}$.

Final Answer

$I = \frac{e^{ax}(a\sin bx - b\cos bx)}{a^2+b^2} + C$

Common Mistake: Continuing to apply by parts a third time instead of recognising that the original integral I has reappeared on the right-hand side – once you see the same integral you started with, stop integrating and switch to solving the equation algebraically for I.

Quick Recall: For e^{ax} times $\sin bx$ or $\cos bx$, apply by parts twice keeping the same choice pattern (always differentiate the trig factor) each time – the original integral will reappear; collect it on one side and divide to solve, never apply by parts a third time.

Type 96

A self-referencing "reduction formula" style integral like $\operatorname{cosec}^3 x$, where by parts produces the original integral back (with a different sign/coefficient) combined with a standard trig integral – solved the same way as Type 95's cyclic method, by collecting the repeated integral algebraically.

Formula Used: $\int \operatorname{cosec} x\, dx = \log|\operatorname{cosec} x - \cot x| + C$

Source: Exercise 7.11, Q21(ii)

Given: $\operatorname{cosec}^3 x$

Solution:

Write $\operatorname{cosec}^3 x = \operatorname{cosec} x \cdot \operatorname{cosec}^2 x$, and let $I = \int \operatorname{cosec}^3 x\, dx$.

Apply by parts with $u = \operatorname{cosec} x$ (differentiates to $-\operatorname{cosec} x \cot x$) and $dv = \operatorname{cosec}^2 x\, dx$ (integrates to $v = -\cot x$): $I = -\operatorname{cosec} x \cot x - \int \cot x \cdot \operatorname{cosec} x \cot x\, dx = -\operatorname{cosec} x \cot x - \int \operatorname{cosec} x \cot^2 x\, dx$.

Rewrite $\cot^2 x = \operatorname{cosec}^2 x - 1$: $\int \operatorname{cosec} x \cot^2 x\, dx = \int \operatorname{cosec} x(\operatorname{cosec}^2 x - 1)\, dx = \int \operatorname{cosec}^3 x\, dx - \int \operatorname{cosec} x\, dx = I - \int \operatorname{cosec} x\, dx$ – the original integral I has reappeared.

Substituting back: $I = -\operatorname{cosec} x \cot x - I + \displaystyle\int \operatorname{cosec} x\, dx$, so $2I = -\operatorname{cosec} x \cot x + \log|\operatorname{cosec} x - \cot x|$.

Final Answer

$I = \dfrac{1}{2}\left[-\operatorname{cosec} x \cot x + \log|\operatorname{cosec} x - \cot x|\right] + C$

Common Mistake: Not recognising $\int \operatorname{cosec} x \cot^2 x\, dx$ as I minus a standard integral after rewriting $\cot^2 x$ via the Pythagorean identity – this rewriting step is exactly what reveals the self-referencing structure; skipping it leaves the problem looking unsolvable.

Quick Recall: $\sec^3 x$ and $\operatorname{cosec}^3 x$ both follow the same cyclic pattern as $e^{ax} \sin bx$: by parts once, use a Pythagorean identity to reveal the original integral hiding inside the remainder, then collect and solve algebraically – never try a second full round of by parts.

Type 97

Combine substitution and the cyclic by-parts technique: substitute for an inverse trig argument first (converting to a standard angle variable), which reveals an $e^{a\theta} \times \text{trig}(\theta)$ structure that then needs Type 95's cyclic method to finish.

Formula Used: Let $\theta = \tan^{-1} x,\ x = \tan\theta$ (converts to cyclic form in θ)

Source: Exercise 7.11, Q21(i)

Given: $\dfrac{e^{m\tan^{-1} x}}{(1+x^2)^{3/2}}$

Solution:

Let $\theta = \tan^{-1} x$, so $x = \tan\theta$ and $dx = \sec^2\theta\, d\theta$; also $1+x^2 = \sec^2\theta$, so $(1+x^2)^{3/2} = \sec^3\theta$.

Substituting: $\displaystyle\int \frac{e^{m\theta}}{\sec^3\theta} \cdot \sec^2\theta\, d\theta = \int e^{m\theta}\cos\theta\, d\theta$ – exactly the cyclic $e^{ax}\cos bx$ pattern from Type 95, with $a = m,\ b = 1$.

Applying the standard cyclic result: $\displaystyle\int e^{m\theta}\cos\theta\, d\theta = \frac{e^{m\theta}(m\cos\theta + \sin\theta)}{m^2+1} + C$.

Converting back to x: $\theta = \tan^{-1} x$, so $\cos\theta = \dfrac{1}{\sqrt{1+x^2}}$ and $\sin\theta = \dfrac{x}{\sqrt{1+x^2}}$ (from the right triangle for $\tan\theta = x$).

Final Answer

$\dfrac{e^{m\tan^{-1} x}(m+x)}{(m^2+1)\sqrt{1+x^2}} + C$

Common Mistake: Trying to apply by parts directly in terms of x without first substituting to the angle variable θ – the cyclic pattern is completely hidden until the substitution converts $(1+x^2)^{3/2}$ into the clean $\sec^3\theta$ that cancels against $dx = \sec^2\theta\, d\theta$.

Quick Recall: Whenever you see $e^{m\tan^{-1}x}$ combined with a power of $(1+x^2)$, substitute $\theta = \tan^{-1}x$ first – this very often converts the whole problem into the standard $e^{a\theta}\sin\theta$ or $e^{a\theta}\cos\theta$ cyclic pattern from Type 95.

Type 98

Recognise an integrand of the exact form $e^x[f(x) + f'(x)]$ where both $f(x)$ and its derivative are directly visible as separate terms – this always integrates instantly to $e^x f(x) + C$, with no by-parts calculation actually needed.

★ **Board Favorite** – The $e^x[f(x) + f'(x)]$ recognition is one of the fastest possible integration techniques when spotted correctly, and ISC frequently tests it precisely because the alternative (expanding and integrating term by term, or attempting by parts) is drastically slower.

Formula Used: $\int e^x[f(x) + f'(x)]\, dx = e^x f(x) + C$

Source: Exercise 7.12, Q1(ii)

Given: $e^x(\tan x + \sec^2 x)$

Solution:

Recognise $f(x) = \tan x$, since its derivative $f'(x) = \sec^2 x$ is exactly the second term present.

Applying the standard result directly, with no further work needed: $\displaystyle\int e^x(\tan x+\sec^2 x)\, dx = e^x \tan x + C$.

Final Answer

$e^x \tan x + C$

Common Mistake: Trying to expand the product and integrate $e^x \tan x$ and $e^x \sec^2 x$ separately (each of which individually has no elementary antiderivative) – the whole point of this pattern is that the *sum* integrates directly; never split it into two separate integrals.

Quick Recall: As soon as you see e^x multiplying a sum of two terms, check whether one term is the derivative of the other – if so, the answer is instantly $e^x\times$(the term that is NOT a derivative of the other, i.e. the "original" function).

Type 99

The integrand is e^x times a single fraction that doesn't visibly split into $f(x) + f'(x)$ – algebraic manipulation (splitting the fraction, or applying a trig identity) is needed first to reveal the two separate pieces.

Formula Used: Split the fraction algebraically into $f(x) + f'(x)$

Source: Exercise 7.12, Q11(i)

Given: $e^x \cdot \dfrac{1 + \sin x}{1 + \cos x}$

Solution:

Use the half-angle identities $1+\sin x = \left(\sin\frac{x}{2} + \cos\frac{x}{2}\right)^2$ and $1+\cos x = 2\cos^2\frac{x}{2}$ to rewrite the fraction: $\dfrac{1+\sin x}{1+\cos x} = \dfrac{\left(\sin\frac{x}{2}+\cos\frac{x}{2}\right)^2}{2\cos^2\frac{x}{2}}$.

This is a disguised form; testing $f(x) = \tan\frac{x}{2}$ (whose derivative is $\frac{1}{2}\sec^2\frac{x}{2}$), we can verify by direct computation that $f(x) + f'(x) = \tan\dfrac{x}{2} + \dfrac{1}{2}\sec^2\dfrac{x}{2}$ indeed equals $\dfrac{1+\sin x}{1+\cos x}$ after simplifying both over the common denominator $2\cos^2(x/2)$.

Applying the standard result with this $f(x)$: $\displaystyle\int e^x \cdot \frac{1+\sin x}{1+\cos x}\,dx = e^x \tan\frac{x}{2} + C$.

Final Answer

$e^x \tan\dfrac{x}{2} + C$

Common Mistake: Giving up on finding $f(x)$ after the fraction doesn't obviously split – when a single fraction is given (not a visible sum), try rewriting using half-angle or double-angle identities first; the $f + f'$ structure is often hidden inside a single simplified-looking ratio.

Quick Recall: If the integrand is e^x times one fraction rather than an obvious sum of two terms, suspect a disguised $f(x) + f'(x)$ pattern – try half-angle substitutions or splitting the fraction into two pieces to reveal it.

Type 100

Given the general statement $\displaystyle\int e^x[\ldots]\,dx = e^x f(x) + C$, identify $f(x)$ by inspection – a "reverse engineering" question that tests recognition of the pattern rather than computing an integral from scratch.

Formula Used: $f(x)$ is whichever term is NOT the derivative of the other

Source: Exercise 7.12, Q6(i) (ISC 2024)

Given: If $\int (\cot x - \operatorname{cosec}^2 x)\, e^x\, dx = e^x f(x) + C$, find $f(x)$

Solution:

Check whether one term is the derivative of the other: $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$ – this exactly matches the second term (with its given negative sign already present in the integrand). Since $\cot x + \frac{d}{dx}(\cot x) = \cot x - \operatorname{cosec}^2 x$ matches the given integrand exactly, we identify $f(x) = \cot x$.

Final Answer

$f(x) = \cot x$

Common Mistake: Trying to actually perform the integration (e.g. via substitution or by parts) instead of simply checking which term is the other's derivative – this type of question is answered by inspection alone, not by computing the integral.

Quick Recall: For a "find $f(x)$" question of this form, just check: does one of the two given terms differentiate to (a sign-adjusted version of) the other? Whichever term is the "original" (not the derivative) is $f(x)$.

Type 101

The base is $e^{g(x)}$ for a non-linear $g(x)$ (such as $\cot^{-1} x$) rather than plain e^x – the same recognition pattern applies, but now the accompanying terms must combine into $f(g(x)) \cdot g'(x)$ plus the derivative of that whole composite, requiring an extra layer of chain-rule awareness.

Formula Used: $\int e^{g(x)} [f(x)g'(x) + f'(x)]\, dx = e^{g(x)} f(x) + C$

Source: Exercise 7.12, Q13(iii)

Given: $e^{\cot^{-1} x} \cdot \dfrac{1 - x + x^2}{1 + x^2}$

Solution:

Split the fraction into two pieces matching the general pattern: $\dfrac{1 - x + x^2}{1 + x^2} = \dfrac{1 + x^2}{1 + x^2} - \dfrac{x}{1 + x^2} = 1 - \dfrac{x}{1 + x^2}$.

Recognise $g(x) = \cot^{-1} x$, so $g'(x) = -\dfrac{1}{1 + x^2}$. We need the integrand to match $f(x) \cdot g'(x) + f'(x)$ for some $f(x)$: testing $f(x) = x$ (so $f'(x) = 1$), we get $f(x)g'(x) + f'(x) = x \cdot \left(-\dfrac{1}{1 + x^2}\right) + 1 = 1 - \dfrac{x}{1 + x^2}$ – exactly matching our split fraction.

Applying the general pattern with $g(x) = \cot^{-1} x$ and $f(x) = x$: $\int e^{\cot^{-1} x} \left[1 - \frac{x}{1+x^2}\right] dx = e^{\cot^{-1} x} \cdot x + C$.

Final Answer

$x\, e^{\cot^{-1} x} + C$

Common Mistake: Trying to force this into the plain $e^x[f + f']$ template without accounting for the extra $g'(x)$ factor that the chain rule introduces when the exponent isn't simply x – always identify $g(x)$ first and check the pattern $f(x)g'(x) + f'(x)$, not just $f(x) + f'(x)$.

Quick Recall: When the base is $e^{g(x)}$ for some non-linear $g(x)$, the pattern generalises to $f(x)g'(x) + f'(x)$ – identify $g(x)$ from the exponent first, then hunt for an $f(x)$ that makes the rest of the integrand match this generalised template.

V. Square-Root Quadratic Forms & Definite Integrals (Ex 7.13–7.17)

Type 102

Direct application of the standard formula $\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$.

Formula Used: $\int \sqrt{a^2 - x^2}\, dx = \frac{x}{2}\sqrt{a^2 - x^2} + \frac{a^2}{2}\sin^{-1}\frac{x}{a} + C$

Source: Exercise 7.13, Q1(i)

Given: $\sqrt{4 - x^2}$

Solution:

This matches the standard formula directly with $a = 2$.
Applying it: $\int \sqrt{4 - x^2}\, dx = \frac{x}{2}\sqrt{4 - x^2} + \frac{4}{2}\sin^{-1}\frac{x}{2} + C$.

Final Answer

$\frac{x}{2}\sqrt{4 - x^2} + 2\sin^{-1}\frac{x}{2} + C$

Common Mistake: Trying to derive this formula from scratch via by parts every time instead of simply memorising and applying it – while it's useful to know the formula *can* be derived by parts (as shown in the theory section), on a timed exam it should be applied directly.

Quick Recall: Memorise all three standard square-root formulas exactly (for $\sqrt{a^2 - x^2}$, $\sqrt{x^2 + a^2}$, $\sqrt{x^2 - a^2}$) – each has the same "half the product plus half of a^2 times the matching inverse/log function" shape, differing only in which one applies.

Type 103

Direct application of the two companion standard formulas $\int \sqrt{x^2 + a^2}\, dx$ and $\int \sqrt{x^2 - a^2}\, dx$, which use a log term instead of $\sin^{-1}$ since the expression under the root doesn't have the bounded $a^2 - x^2$ shape.

Formula Used: $\int \sqrt{x^2 - a^2}\, dx = \frac{x}{2}\sqrt{x^2 - a^2} - \frac{a^2}{2} \log\left|x + \sqrt{x^2 - a^2}\right| + C$

Source: Exercise 7.13, Q1(ii)

Given: $\sqrt{x^2 - 9}$

Solution:

This matches the standard formula with $a = 3$.
Applying it: $\displaystyle\int \sqrt{x^2 - 9}\, dx = \frac{x}{2}\sqrt{x^2 - 9} - \frac{9}{2} \log\left|x + \sqrt{x^2 - 9}\right| + C.$

Final Answer

$\frac{x}{2}\sqrt{x^2 - 9} - \frac{9}{2} \log\left|x + \sqrt{x^2 - 9}\right| + C$

Common Mistake: Confusing the sign pattern between the $\sqrt{a^2 - x^2}$ (arcsin) formula and the $\sqrt{x^2 \pm a^2}$ (log) formulas – the arcsin version has a $+\sin^{-1}$ term with no log anywhere, while the other two always end in a log term with a minus sign in front.

Quick Recall: $\sqrt{a^2 - x^2}$ (bounded region) gives $\sin^{-1}$; $\sqrt{x^2 \pm a^2}$ (unbounded) gives log – match the exact shape under the root to the exact formula before integrating.

Type 104

The same standard square-root formulas as Types 102–103, but with a non-unit coefficient on x^2 – factor that coefficient out from under the root first, exactly as in Type 54, picking up an extra constant multiplier.

Formula Used: Factor the x^2 coefficient out before matching a standard formula

Source: Exercise 7.13, Q3(i)

Given: $\sqrt{4x^2 - 9}$

Solution:

Factor 4 out from under the root: $4x^2 - 9 = 4\left(x^2 - \frac{9}{4}\right)$, so $\sqrt{4x^2 - 9} = 2\sqrt{x^2 - \left(\frac{3}{2}\right)^2}$.

This matches the standard $\sqrt{x^2 - a^2}$ formula with $a = \frac{3}{2}$, scaled by the constant 2 pulled out front.

Applying the formula: $2\left[\frac{x}{2}\sqrt{x^2 - \frac{9}{4}} - \frac{9/4}{2} \log\left|x + \sqrt{x^2 - \frac{9}{4}}\right|\right] + C$, then simplify $\sqrt{x^2 - 9/4} = \frac{1}{2}\sqrt{4x^2 - 9}$ to express everything back in terms of the original expression.

Final Answer

$\frac{x}{2}\sqrt{4x^2-9} - \frac{9}{4}\log\left|2x+\sqrt{4x^2-9}\right| + C$

Common Mistake: Forgetting to re-simplify $\sqrt{x^2-9/4}$ back into $\frac{1}{2}\sqrt{4x^2-9}$ (matching the form of the original problem) at the end – board answers are expected in terms of the original expression under the root, not the factored intermediate form.

Quick Recall: Factor out the coefficient exactly as in Type 54, apply the matching standard formula in the simplified variable, then convert back to the original expression's form for the final answer.

Type 105

Prepare the integrand algebraically – rationalizing a denominator or splitting a numerator via the $A \cdot D + B \cdot D'$ decomposition from Type 66 – before a standard square-root formula becomes directly applicable.

Formula Used: Rationalize/decompose first (Types 22, 62/66), then use a sqrt formula

Source: Exercise 7.13, Q4(ii)

Given: $\dfrac{1}{x-\sqrt{x^2-1}}$

Solution:

Rationalize by multiplying numerator and denominator by the conjugate $x+\sqrt{x^2-1}$: $\dfrac{1}{x-\sqrt{x^2-1}} \cdot \dfrac{x+\sqrt{x^2-1}}{x+\sqrt{x^2-1}} = \dfrac{x+\sqrt{x^2-1}}{x^2-(x^2-1)} = x+\sqrt{x^2-1}$, since the denominator collapses to 1.

Integrating term by term: $\displaystyle\int x\,dx = \frac{x^2}{2}$, and $\displaystyle\int \sqrt{x^2-1}\,dx$ is the standard formula with $a=1$: $\dfrac{x}{2}\sqrt{x^2-1} - \dfrac{1}{2}\log\left|x+\sqrt{x^2-1}\right|$.

Final Answer

$\frac{x^2}{2} + \frac{x}{2}\sqrt{x^2-1} - \frac{1}{2}\log\left|x+\sqrt{x^2-1}\right| + C$

Common Mistake: Trying to integrate $\dfrac{1}{x-\sqrt{x^2-1}}$ directly (via substitution) instead of rationalizing first – the denominator's structure (a difference involving a square root) is a strong visual cue to rationalize before attempting anything else.

Alternate Board-Style Example

Source: Exercise 7.13, Q5(ii)

Given: $(x-3)\sqrt{\dfrac{x+2}{x-2}}$

Solution:

Rationalize by multiplying inside the root by $\dfrac{\sqrt{x+2}}{\sqrt{x+2}}$: $\sqrt{\dfrac{x+2}{x-2}} = \dfrac{x+2}{\sqrt{x^2-4}}$, so the integrand becomes $\dfrac{(x-3)(x+2)}{\sqrt{x^2-4}} = \dfrac{x^2-x-6}{\sqrt{x^2-4}}$.

Decompose the numerator: $x^2-x-6 = (x^2-4)-(x+2)$, splitting the integral into $\displaystyle\int \sqrt{x^2-4}\,dx - \int \frac{x+2}{\sqrt{x^2-4}}\,dx$.

The first piece is the standard formula with $a=2$. The second splits further into $\displaystyle\int \frac{x}{\sqrt{x^2-4}}dx + 2\int \frac{dx}{\sqrt{x^2-4}}$, both direct standard forms.

Combining all pieces and simplifying the resulting logs (which combine into a single term):

Final Answer

$$\left(\frac{x}{2}-1\right)\sqrt{x^2-4} - 4\log\left|x+\sqrt{x^2-4}\right| + C$$

Quick Recall: A square root of a fraction rationalizes the same way as a plain fraction with roots – multiply to clear the root from the denominator first, then decompose the resulting numerator using the $A \cdot D + B \cdot D'$ trick before applying standard formulas.

Type 106

Substitute for an inner expression (like x^2 or $\log x$) first when the square-root structure is secretly built from that inner variable, then apply a standard sqrt formula in the new variable.

Formula Used: Substitute for the inner variable first, as in Types 58 and 67

Source: Exercise 7.13, Q6(ii)

Given: $\dfrac{\sqrt{9-(\log x)^2}}{x}$

Solution:

Let $t = \log x$, so $dt = \dfrac{1}{x}dx$ – the accompanying $\dfrac{1}{x}$ factor supplies exactly dt.

Substituting, the integral becomes $\displaystyle\int \sqrt{9-t^2}\,dt$, a direct standard formula with $a=3$.

Applying it: $\dfrac{t}{2}\sqrt{9-t^2} + \dfrac{9}{2}\sin^{-1}\dfrac{t}{3} + C$.

Converting back: $t = \log x$.

Final Answer

$\frac{\log x}{2}\sqrt{9-(\log x)^2}+\frac{9}{2}\sin^{-1}\left(\frac{\log x}{3}\right)+C$

Common Mistake: Not noticing the $\frac{1}{x}$ factor as the signal for a $t = \log x$ substitution, and instead trying to apply a square-root formula directly in terms of x (which doesn't match any standard shape) – always check for an accompanying derivative factor first.

Alternate Board-Style Example

Source: Exercise 7.13, Q6(i)

Given: $x\sqrt{x^4-1}$

Solution:

Let $t = x^2$, so $dt = 2x\,dx$.
Substituting: $\frac{1}{2}\int\sqrt{t^2-1}\,dt$, a direct standard formula with $a = 1$.
Applying it: $\frac{1}{2}\left[\frac{t}{2}\sqrt{t^2-1}-\frac{1}{2}\log\left|t+\sqrt{t^2-1}\right|\right]+C$.
Converting back: $t = x^2$.

Final Answer

$\frac{x^2}{4}\sqrt{x^4-1}-\frac{1}{4}\log\left|x^2+\sqrt{x^4-1}\right|+C$

Quick Recall: If the expression under the root is a function of x^2 or $\log x$ (not x directly), substitute for that inner variable first – the standard sqrt formulas then apply directly in the new variable.

Type 107

Integrate x (or a related algebraic expression) times an inverse trig function using by parts, where the resulting remainder integral is a standard square-root form – combines Type 87's by-parts approach with the Type 102–103 formulas.

Formula Used: $\frac{d}{dx}\cos^{-1}x = -\frac{1}{\sqrt{1-x^2}}$

Source: Exercise 7.13, Q8(i)

Given: $x\cos^{-1}x$

Solution:

By ILATE, let $u = \cos^{-1} x$, $dv = x\,dx$, so $du = -\dfrac{1}{\sqrt{1-x^2}}\,dx$ and $v = \dfrac{x^2}{2}$.
Applying by parts: $\displaystyle\int x\cos^{-1} x\,dx = \frac{x^2}{2}\cos^{-1} x + \frac{1}{2}\int \frac{x^2}{\sqrt{1-x^2}}\,dx$.
Rewrite the remaining numerator: $x^2 = -(1-x^2)+1$, splitting the fraction into $-\sqrt{1-x^2}+\dfrac{1}{\sqrt{1-x^2}}$.
Integrating this remainder: $\displaystyle\int \left(-\sqrt{1-x^2}+\frac{1}{\sqrt{1-x^2}}\right)dx = -\left[\frac{x}{2}\sqrt{1-x^2}+\frac{1}{2}\sin^{-1} x\right]+\sin^{-1} x = -\frac{x}{2}\sqrt{1-x^2}+\frac{1}{2}\sin^{-1} x$.

Final Answer

$$\frac{x^2}{2}\cos^{-1} x + \frac{1}{2}\left[-\frac{x}{2}\sqrt{1-x^2}+\frac{1}{2}\sin^{-1} x\right]+C = \frac{x^2}{2}\cos^{-1} x - \frac{x}{4}\sqrt{1-x^2}+\frac{1}{4}\sin^{-1} x + C$$

Common Mistake: Not recognising that $\dfrac{x^2}{\sqrt{1-x^2}}$ needs its numerator rewritten as $-(1-x^2)+1$ before it can be split into standard forms – this numerator trick (writing x^2 in terms of $1-x^2$) is essential whenever x^2 appears over $\sqrt{1-x^2}$.

Quick Recall: x times an inverse trig function is a standard by-parts setup; when the remainder involves $x^2/\sqrt{1-x^2}$ (or similar), rewrite the numerator in terms of $(1-x^2)$ to split it into a standard sqrt-formula piece and a standard arcsin piece.

Type 108

Complete the square under a square root (exactly as in Types 64–65), then apply the matching standard $\sqrt{x^2 \pm a^2}$ or $\sqrt{a^2-x^2}$ formula from Types 102–103 to the shifted variable.

Formula Used: $x^2+bx+c = \left(x+\dfrac{b}{2}\right)^2 + \left(c-\dfrac{b^2}{4}\right)$, then match a standard sqrt formula

Source: Exercise 7.14, Q1(i)

Given: $\sqrt{x^2+4x+6}$

Solution:

Complete the square: $x^2+4x+6 = (x+2)^2+2$.
This matches the standard $\sqrt{x^2+a^2}$ formula with the shifted variable $x+2$ and $a^2 = 2$ (so $a = \sqrt{2}$).

Applying the formula:

$$\int \sqrt{(x+2)^2+2}\,dx = \frac{x+2}{2}\sqrt{(x+2)^2+2} + \frac{2}{2}\log\left|(x+2)+\sqrt{(x+2)^2+2}\right| + C.$$

Rewriting $(x+2)^2+2 = x^2+4x+6$ throughout to express the answer in terms of the original expression.

Final Answer

$\frac{x+2}{2}\sqrt{x^2+4x+6} + \log\left|(x+2)+\sqrt{x^2+4x+6}\right| + C$

Common Mistake: Forgetting to convert the completed-square expression $(x+2)^2+2$ back into the original x^2+4x+6 form inside the square root for the final answer – board markers expect the answer written in terms of the original quadratic, not the shifted intermediate form.

Quick Recall: Complete the square exactly as in Type 61/64, identify which of the three standard sqrt formulas matches (bounded → arcsin, unbounded → log), apply it to the shifted variable, then rewrite back in terms of the original quadratic.

Type 109

For (linear)$\times\sqrt{\text{quadratic}}$, combine the numerator-decomposition trick from Type 66 with completing the square: decompose the linear factor as $A \cdot$ (derivative of the quadratic) $+ B$, giving a direct power-rule piece from A and a completing-the-square standard-formula piece from B.

★ **Board Favorite** – This combination of numerator decomposition and completing the square is a standard high-yield pattern – it's often considered the single hardest "routine" integral type in the whole chapter, since it requires two full techniques executed correctly in sequence.

Formula Used: Linear factor $= A(2x+b)+B,$ matching the quadratic's derivative

Source: Exercise 7.14, Q3(i)

Given: $(x+3)\sqrt{3-4x-x^2}$

Solution:

The derivative of $3-4x-x^2$ is $-4-2x$. Write $x+3 = A(-4-2x)+B$: matching coefficients, $-2A = 1 \Rightarrow A = -\frac{1}{2}$, and $-4A+B = 3 \Rightarrow B = 3-2 = 1$.

Split the integral:

$$\int (x+3)\sqrt{3-4x-x^2}\,dx = -\frac{1}{2}\int(-4-2x)\sqrt{3-4x-x^2}\,dx + \int\sqrt{3-4x-x^2}\,dx.$$

The first piece is a direct power-rule (chain-rule-reverse) form: $-\frac{1}{2} \cdot \frac{2}{3}(3-4x-x^2)^{3/2} = -\frac{1}{3}(3-4x-x^2)^{3/2}$.
For the second piece, complete the square: $3-4x-x^2 = 7-(x+2)^2$, matching the standard $\sqrt{a^2-x^2}$ formula with $a=\sqrt{7}$:

$$\int \sqrt{7-(x+2)^2}\,dx = \frac{x+2}{2}\sqrt{3-4x-x^2} + \frac{7}{2}\sin^{-1}\left(\frac{x+2}{\sqrt{7}}\right).$$

Final Answer

$$-\frac{1}{3}(3-4x-x^2)^{3/2} + \frac{x+2}{2}\sqrt{3-4x-x^2} + \frac{7}{2}\sin^{-1}\left(\frac{x+2}{\sqrt{7}}\right) + C$$

Common Mistake: Trying to complete the square before doing the numerator decomposition – the decomposition must be based on the derivative of the *original* (un-completed) quadratic; do the decomposition first, then complete the square only for the leftover constant-times-sqrt piece.

Quick Recall: This type always splits into exactly two pieces: a direct power-rule term (from the A part of the decomposition) and a completing-the-square standard-formula term (from the B part) – do the decomposition first, then complete the square only for the second piece.

Type 110

Evaluate a definite integral by finding the antiderivative exactly as in indefinite integration, then applying the Fundamental Theorem of Calculus: substitute the upper limit, subtract the value at the lower limit, and drop the constant of integration entirely (it cancels).

Formula Used: $\int_a^b f(x)\,dx = F(b) - F(a)$, where F is any antiderivative of f

Source: Exercise 7.15, Q1(iii)

Given: $\int_2^3 \frac{1}{x}\,dx$

Solution:

The antiderivative of $\frac{1}{x}$ is $\log|x|$ – no constant of integration is needed since it will cancel in the subtraction.
Applying the limits: $\displaystyle\int_2^3 \frac{1}{x}\,dx = \Big[\log x\Big]_2^3 = \log 3 - \log 2.$
Combine using the log-quotient rule: $\log 3 - \log 2 = \log\frac{3}{2}$.

Final Answer

$\log\left(\dfrac{3}{2}\right)$

Common Mistake: Writing $+C$ in the antiderivative and then not knowing what to do with it – for a definite integral, the antiderivative is evaluated at both limits and subtracted, so any constant automatically cancels; never include C when evaluating a definite integral.

Quick Recall: Find the antiderivative exactly as usual (no C needed), then compute $F(\text{upper}) - F(\text{lower})$ – this is the entire process for a direct definite integral.

Type 111

Evaluate a definite integral of $|x-a|$ (or the greatest integer function $[x]$) by splitting the interval at the point(s) where the expression inside changes sign or jumps, then integrating each piece separately with its own sign or constant value.

★ **Board Favorite** – Definite integrals of $|x|$-type functions are a standard high-yield pattern – they specifically test whether students remember to split the interval at the critical point rather than integrating $|x-a|$ as if it were a smooth function throughout.

Formula Used: $|x-a| = \begin{cases} -(x-a) & x < a \\ x-a & x \geq a \end{cases}$

Source: Exercise 7.17, Q1(ii) (ISC 2022)

Given: $\int_1^4 |x-2|\,dx$

Solution:

The expression $x-2$ changes sign at $x=2$, which lies inside the interval $[1,4]$, so we must split the integral there: for $x \in [1,2]$, $|x-2| = 2-x$; for $x \in [2,4]$, $|x-2| = x-2$.

Splitting: $\displaystyle\int_1^4 |x-2|\,dx = \int_1^2 (2-x)\,dx + \int_2^4 (x-2)\,dx.$

First piece: $\displaystyle\left[2x - \frac{x^2}{2}\right]_1^2 = (4-2) - \left(2 - \frac{1}{2}\right) = 2 - \frac{3}{2} = \frac{1}{2}.$

Second piece: $\displaystyle\left[\frac{x^2}{2} - 2x\right]_2^4 = (8-8) - (2-4) = 0 - (-2) = 2.$

Adding both pieces: $\dfrac{1}{2} + 2 = \dfrac{5}{2}.$

Final Answer

$\frac{5}{2}$

Common Mistake: Integrating $|x-2|$ as if it were simply $x-2$ (or $2-x$) across the entire interval without splitting at the critical point $x=2$ – always identify where the expression inside the absolute value changes sign first, and split the integral there before doing anything else.

Quick Recall: For $\int |g(x)|\,dx$, first find where $g(x)=0$ within the interval – if that point lies inside the limits, split the integral there and flip the sign of $g(x)$ on whichever piece makes it negative.

Type 112

Evaluate a definite integral requiring a substitution – convert the limits of integration to the new variable at the same time as substituting the integrand, so the final numeric answer can be read off directly without ever converting back to x.

Formula Used: $\int_a^b f(g(x))g'(x)\,dx = \int_{g(a)}^{g(b)} f(t)\,dt$, where $t=g(x)$

Source: Exercise 7.16, Q2(ii)

Given: $\int_0^{\pi/3} \frac{\cos x}{3+4\sin x}\,dx$

Solution:

Let $t=3+4\sin x$, so $dt=4\cos x\,dx$, i.e. $\cos x\,dx=\frac{dt}{4}$ – this substitution is motivated by the numerator being (a constant times) the derivative of the denominator.

Convert the limits: when $x=0$, $t=3+4\sin 0=3$; when $x=\frac{\pi}{3}$, $t=3+4\sin\frac{\pi}{3}=3+4\cdot\frac{\sqrt{3}}{2}=3+2\sqrt{3}$ – both limits are converted to t-values, so there is no need to ever substitute back to x.

The integral becomes $\displaystyle\int_3^{3+2\sqrt{3}} \frac{1}{4t}\,dt = \frac{1}{4}\Big[\log t\Big]_3^{3+2\sqrt{3}} = \frac{1}{4}\Big[\log(3+2\sqrt{3})-\log 3\Big]$.

Combine using the log-quotient rule.

Final Answer

$\frac{1}{4}\log\left(\frac{3+2\sqrt{3}}{3}\right)$

Common Mistake: Substituting for the integrand but forgetting to convert the limits, then trying to convert the antiderivative back to x before applying the original limits – for definite integrals, converting the limits at substitution time is both correct and far less error-prone than converting back afterward.

Quick Recall: For a definite integral via substitution, always convert the limits to the new variable at the same time as the integrand – this lets you skip converting the antiderivative back to x entirely, evaluating directly in the new variable.

Type 113

Evaluate a definite integral of an algebraic rational function, applying whichever indefinite-integration technique (splitting, substitution, standard form) is needed, then evaluating at the converted or original limits.

Formula Used: Apply the matching indefinite technique, then evaluate at the limits

Source: Exercise 7.16, Q4(i)

Given: $\int_0^2 \frac{6x+3}{x^2+4}\,dx$

Solution:

Split the numerator to match the standard forms: $\frac{6x+3}{x^2+4} = \frac{6x}{x^2+4} + \frac{3}{x^2+4}$, where the first piece is a log form and the second is an arctan form.

Integrate each piece: $\int \frac{6x}{x^2+4}\,dx = 3\log(x^2+4)$, and $\int \frac{3}{x^2+4}\,dx = \frac{3}{2}\tan^{-1}\frac{x}{2}$.

Evaluating at the limits 0 to 2: $\left[3\log(x^2+4) + \frac{3}{2}\tan^{-1}\frac{x}{2}\right]_0^2 = \left[3\log 8 + \frac{3}{2}\tan^{-1} 1\right] - \left[3\log 4 + \frac{3}{2}\tan^{-1} 0\right]$.

Simplify: $3\log 8 - 3\log 4 = 3\log 2$ (since $8/4 = 2$), and $\frac{3}{2}\tan^{-1} 1 = \frac{3}{2}\cdot\frac{\pi}{4} = \frac{3\pi}{8}$.

Final Answer

$3\log 2 + \frac{3\pi}{8}$

Common Mistake: Evaluating each piece of the split integral at the limits *before* combining and simplifying the logs – it's cleaner (and less error-prone) to combine both antiderivative pieces into one bracket first, then substitute both limits once.

Quick Recall: Definite integrals of algebraic fractions use exactly the same splitting/substitution techniques as their indefinite counterparts (Types 26, 52–60) – the only new step is evaluating the completed antiderivative at the two limits.

Type 114

Evaluate a definite integral involving an inverse trig or log function, typically via by parts, being careful that the "boundary term" $[u \cdot v]$ from by parts is itself evaluated at the limits alongside the remaining integral.

Formula Used: $\int_a^b u\,dv = \Big[uv\Big]_a^b - \int_a^b v\,du$

Source: Exercise 7.16, Q10(ii)

Given: $\int_0^1 \tan^{-1} x\,dx$

Solution:

Using by parts (as in Type 86) with $u = \tan^{-1} x,\ dv = dx$: $du = \dfrac{1}{1+x^2}\,dx,\ v = x$.

The boundary term is $\Big[x\tan^{-1} x\Big]_0^1 = (1 \cdot \tan^{-1} 1) - (0 \cdot \tan^{-1} 0) = \dfrac{\pi}{4} - 0 = \dfrac{\pi}{4}$.

The remaining integral is $\displaystyle\int_0^1 \frac{x}{1+x^2}\,dx = \left[\frac{1}{2}\log(1+x^2)\right]_0^1 = \frac{1}{2}\log 2 - \frac{1}{2}\log 1 = \frac{1}{2}\log 2$.

Combining: $\dfrac{\pi}{4} - \dfrac{1}{2}\log 2$.

Final Answer

$\dfrac{\pi}{4} - \dfrac{1}{2}\log 2$

Common Mistake: Forgetting to evaluate the boundary term $[uv]$ at the limits (treating it as if it were part of the indefinite antiderivative to be dealt with later) – the boundary term must be evaluated at the limits immediately, just like the remaining integral.

Quick Recall: By parts for a definite integral produces two pieces that both need the limits applied: the boundary term $[uv]$ and the remaining integral – evaluate both completely at the limits before combining.

Type 115

Given a definite integral set equal to a numeric value, with the upper (or lower) limit as an unknown a, integrate first (leaving a symbolic), then solve the resulting equation in a.

Formula Used: Integrate with the limit symbolic, then solve for the unknown limit

Source: Exercise 7.15, Q21(i)

Given: If $\int_0^a 3x^2\,dx = 8$, find a

Solution:

Integrate treating a as the (symbolic) upper limit: $\displaystyle\int_0^a 3x^2\,dx = \left[x^3\right]_0^a = a^3 - 0 = a^3$.
Set this equal to the given value: $a^3 = 8$.
Solving: $a = \sqrt[3]{8} = 2$.

Final Answer

$a = 2$

Common Mistake: Trying to solve for a before actually completing the integration – always integrate first (treating a symbolically as the upper limit, exactly as if it were a number), and only then set up and solve the resulting algebraic equation.

Quick Recall: Treat the unknown limit exactly like any other upper limit during integration – the result will be an expression in a; setting that expression equal to the given value turns the problem into ordinary equation-solving.

Type 116

Evaluate a definite integral of a trig power or product (requiring the same techniques as Types 24, 41–43 for the indefinite case), then apply the limits – watch particularly for limits that are multiples of π, which often make boundary terms vanish or simplify dramatically.

Formula Used: Apply the matching trig technique first, then evaluate at the limits

Source: Exercise 7.16, Q8(i)

Given: $\int_0^{\pi/4} \sin^3 2x \cos 2x\,dx$

Solution:

Let $t = \sin 2x$, so $dt = 2\cos 2x\,dx$, i.e. $\cos 2x\,dx = \dfrac{dt}{2}$ – the accompanying $\cos 2x$ factor supplies exactly dt (up to the constant).
Convert the limits: when $x = 0$, $t = \sin 0 = 0$; when $x = \dfrac{\pi}{4}$, $t = \sin\dfrac{\pi}{2} = 1$.

The integral becomes $\displaystyle\frac{1}{2}\int_0^1 t^3\,dt = \frac{1}{2}\left[\frac{t^4}{4}\right]_0^1 = \frac{1}{2}\cdot\frac{1}{4} = \frac{1}{8}$.

Final Answer

$\dfrac{1}{8}$

Common Mistake: Converting the integrand via substitution but then evaluating the antiderivative at the *original* x-limits instead of the converted t-limits – once you commit to converting the limits (Type 112's approach), stay in the new variable all the way through.

Quick Recall: Trig-power definite integrals use the exact same substitution techniques as their indefinite counterparts – convert the limits along with the substitution, and the final evaluation is often a very clean number, especially at limits like $0, \pi/4, \pi/2$.

Type 117

Evaluate a definite integral requiring completing the square (or a standard sqrt/arctan form from Types 61–67), applying the converted or evaluated inverse trig values at the limits carefully.

Formula Used: Complete the square first (Types 61, 64), then evaluate at the limits

Source: Exercise 7.16, Q12(i)

Given: $\int_0^4 \frac{dx}{\sqrt{x^2+2x+3}}$

Solution:

Complete the square: $x^2 + 2x + 3 = (x+1)^2 + 2$.
This matches the standard $\sqrt{x^2+a^2}$ log form with $a = \sqrt{2}$ and shifted variable $x+1$: antiderivative is $\log\left|(x+1) + \sqrt{x^2+2x+3}\right|$.
Evaluating at the limits: $\left[\log\left|(x+1)+\sqrt{x^2+2x+3}\right|\right]_0^4 = \log\left(5+\sqrt{27}\right) - \log\left(1+\sqrt{3}\right)$.
Simplify $\sqrt{27} = 3\sqrt{3}$, and combine the logs using the quotient rule.

Final Answer

$$\log\left(\frac{5+3\sqrt{3}}{1+\sqrt{3}}\right)$$

Common Mistake: Trying to simplify the completed-square form back to the original quadratic before evaluating at the limits – it's fine (and often clearer) to substitute the numeric limits directly into the completed-square antiderivative, since $(x+1)^2+2$ and x^2+2x+3 give identical numeric values at any specific x.

Quick Recall: Complete the square exactly as in the indefinite case, then substitute the numeric limits directly into whichever form (completed-square or original) of the antiderivative is more convenient – both give the same numbers.

Type 118

A definite integral requiring both a substitution *and* an algebraic manipulation (such as partial fractions or a numerator split) combined – apply each technique in the correct order exactly as in the indefinite case, converting limits at the substitution step.

Formula Used: Combine substitution and algebra, same order as the indefinite case

Source: Exercise 7.16, Q17(i)

Given: $\int_0^1 \frac{5x}{(4+x^2)^2}\,dx$

Solution:

Let $t = 4 + x^2$, so $dt = 2x\,dx$, i.e. $x\,dx = \dfrac{dt}{2}$.
Convert the limits: when $x = 0$, $t = 4$; when $x = 1$, $t = 5$.
The integral becomes $\dfrac{5}{2}\displaystyle\int_4^5 t^{-2}\,dt = \dfrac{5}{2}\left[-\dfrac{1}{t}\right]_4^5 = \dfrac{5}{2}\left(-\dfrac{1}{5}+\dfrac{1}{4}\right) = \dfrac{5}{2}\cdot\dfrac{1}{20}$.

Final Answer

$\dfrac{1}{8}$

Common Mistake: Attempting partial fractions on $(4+x^2)^2$ (treating it as if it had linear factors) instead of recognising the much simpler direct substitution available here – always check for a straightforward u-substitution before reaching for partial fractions on a power of an irreducible quadratic.

Quick Recall: Before choosing a technique for a definite integral, scan for the simplest applicable method first (direct substitution, then splitting, then partial fractions only if genuinely needed) – the same technique-selection logic from the indefinite chapter applies unchanged here.

VI. Properties of Definite Integrals (Ex 7.17)

Type 119

"King's Rule": $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$. For a rational function of $\sin x$ and $\cos x$ integrated from 0 to $\pi/2$, applying this property and *adding* the original and transformed integrals often makes the trig functions cancel into a constant.

★ **Board Favorite –** King's Rule combined with $[0, \pi/2]$ trig integrands is one of the most heavily tested definite-integral patterns on every ISC board paper – the "add the original and the transformed version" trick is a must-know technique.

Formula Used: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Source: Exercise 7.17, Q8

Given: $\int_0^{\pi/2} \dfrac{\sin x}{\sin x + \cos x}\,dx$

Solution:

Let $I = \displaystyle\int_0^{\pi/2} \frac{\sin x}{\sin x + \cos x}\,dx$. Apply King's Rule with $a = \dfrac{\pi}{2}$, replacing x with $\dfrac{\pi}{2} - x$ throughout: since $\sin\left(\frac{\pi}{2} - x\right) = \cos x$ and $\cos\left(\frac{\pi}{2} - x\right) = \sin x$, we get $I = \displaystyle\int_0^{\pi/2} \frac{\cos x}{\cos x + \sin x}\,dx$.
Add this new expression for I to the original: $2I = \displaystyle\int_0^{\pi/2} \frac{\sin x + \cos x}{\sin x + \cos x}\,dx = \int_0^{\pi/2} 1\,dx = \frac{\pi}{2}$ – the numerator and denominator become identical, collapsing the integrand to 1.
Solving: $I = \dfrac{\pi}{4}$.

Final Answer

$\dfrac{\pi}{4}$

Common Mistake: Trying to evaluate the integral directly (via Weierstrass substitution or otherwise) instead of recognising the King's Rule "add and cancel" pattern – whenever a rational trig function is integrated over $[0, \pi/2]$ and looks resistant to direct methods, try King's Rule first.

Quick Recall: On $[0, \pi/2]$, replacing $x \to \frac{\pi}{2} - x$ swaps $\sin x \leftrightarrow \cos x$ – if adding the original and swapped integrals makes the denominators match the numerators (or otherwise simplifies dramatically), you've found the fast route; always try this before any direct integration technique.

Type 120

The general form of King's Rule, $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$, applied to an interval that doesn't start at 0 – otherwise identical in spirit to Type 119.

Formula Used: $\int_a^b f(x)\,dx = \int_a^b f(a+b-x)\,dx$

Source: Exercise 7.17, Q23 (King's Rule general form)

Given: $\int_{\pi/6}^{\pi/3} \dfrac{dx}{1+\sqrt{\tan x}}$

Solution:

Let $I = \displaystyle\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\tan x}}$. Here $a = \frac{\pi}{6}$, $b = \frac{\pi}{3}$, so $a+b = \frac{\pi}{2}$; apply the general King's Rule by replacing $x \to \frac{\pi}{2} - x$: since $\tan\left(\frac{\pi}{2} - x\right) = \cot x$, we get $I = \displaystyle\int_{\pi/6}^{\pi/3} \frac{dx}{1+\sqrt{\cot x}}$.

Rewrite $\sqrt{\cot x} = \dfrac{1}{\sqrt{\tan x}}$, so $\dfrac{1}{1+\sqrt{\cot x}} = \dfrac{1}{1+\frac{1}{\sqrt{\tan x}}} = \dfrac{\sqrt{\tan x}}{\sqrt{\tan x}+1}$.

Adding the two expressions for I: $2I = \displaystyle\int_{\pi/6}^{\pi/3} \left[\frac{1}{1+\sqrt{\tan x}} + \frac{\sqrt{\tan x}}{1+\sqrt{\tan x}}\right] dx = \int_{\pi/6}^{\pi/3} 1\,dx = \frac{\pi}{3} - \frac{\pi}{6} = \frac{\pi}{6}$.

Solving: $I = \dfrac{\pi}{12}$.

Final Answer

$\dfrac{\pi}{12}$

Common Mistake: Using $x \to a-x$ (the $[0, a]$ version) instead of the correct general substitution $x \to a+b-x$ when the interval doesn't start at 0 – always identify a and b from the actual limits and use $a+b-x$.

Quick Recall: For any interval $[a, b]$ (not just those starting at 0), King's Rule replaces x with $a+b-x$ – identify the sum of the limits first, then apply the same "add and simplify" strategy as Type 119.

Type 121

Use the even/odd symmetric-interval property: $\displaystyle\int_{-a}^{a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$ if f is even ($f(-x) = f(x)$), or $= 0$ if f is odd ($f(-x) = -f(x)$) – always check the parity of the integrand *first*, since it can eliminate the entire calculation.

★ **Board Favorite** – Recognising odd-function symmetry to instantly get 0 is one of the highest-value shortcuts in the whole definite-integral toolkit – ISC frequently constructs integrands that look intimidating specifically so students who check parity first save enormous time.

Formula Used: $\int_{-a}^{a} f(x)\,dx = \begin{cases} 2\displaystyle\int_0^a f(x)\,dx & f(-x) = f(x) \text{ (even)} \\ 0 & f(-x) = -f(x) \text{ (odd)} \end{cases}$

Source: Exercise 7.17, Q3(iii)

Given: $\int_{-\pi/3}^{\pi/3} (x^2 \sin x + \tan^3 x)\,dx$

Solution:

Check the parity of each term: $x^2 \sin x$ has $f(-x) = (-x)^2 \sin(-x) = x^2(-\sin x) = -x^2 \sin x$, so it's odd; similarly $\tan^3 x$ has $f(-x) = \tan^3(-x) = (-\tan x)^3 = -\tan^3 x$, also odd.
Since the entire integrand is a sum of two odd functions, it is itself odd.
By the odd-function property on the symmetric interval $[-\pi/3, \pi/3]$, the integral is immediately 0 – no actual integration is required.

Final Answer

0

Common Mistake: Attempting to integrate $x^2 \sin x$ by parts and $\tan^3 x$ via the Pythagorean identity before checking parity – this wastes significant time on a problem that resolves instantly once you notice the symmetric limits and odd integrand.

Quick Recall: Whenever the limits are symmetric about 0 (i.e. $[-a, a]$), check parity *before* attempting any integration technique – an odd integrand always gives 0 immediately, and an even integrand lets you double a $[0, a]$ integral instead.

Type 122

Use the reflection-about-midpoint property $\displaystyle\int_{0}^{2a} f(x)\,dx = 2\int_{0}^{a} f(x)\,dx$ if $f(2a-x) = f(x)$, or $= 0$ if $f(2a - x) = -f(x)$ – the analogue of Type 121's symmetric-interval property, but centred at a instead of 0.

Formula Used: $\int_0^{2a} f(x)\,dx = \begin{cases} 2\displaystyle\int_0^a f(x)\,dx & f(2a-x)=f(x) \\ 0 & f(2a-x)=-f(x) \end{cases}$

Source: Exercise 7.17, Q7 (property proof, applied)

Given: $\int_0^{\pi} \sin^7 x\,dx$ (using the $[0,2a]$ property with $a=\pi/2$)

Solution:

Here the interval is $[0,\pi]$, matching $[0,2a]$ with $a=\dfrac{\pi}{2}$. Check the reflection property: $\sin^7(\pi-x) = (\sin(\pi-x))^7 = (\sin x)^7 = \sin^7 x$ (using $\sin(\pi-x)=\sin x$), so $f(2a-x)=f(x)$ – the function satisfies the "matching" case.

By the property, $\displaystyle\int_0^{\pi} \sin^7 x\,dx = 2\int_0^{\pi/2} \sin^7 x\,dx$, reducing the problem to a $[0,\pi/2]$ odd-power integral, which is computed via the standard reduction technique (peel one factor, apply the Pythagorean identity, as in Type 42) to give $2\cdot\dfrac{16}{35}=\dfrac{32}{35}$.

Final Answer

$\dfrac{32}{35}$

Common Mistake: Trying to integrate $\sin^7 x$ directly over $[0,\pi]$ using the full expansion technique from Type 42, rather than first halving the interval using this reflection property – halving the interval first, when applicable, roughly halves the amount of algebra needed.

Quick Recall: For an interval $[0,2a]$, check whether $f(2a-x)$ equals $f(x)$ or $-f(x)$ before integrating directly – this can halve the interval (doubling a $[0,a]$ result) or eliminate the integral entirely, exactly like the symmetric-interval property but centred differently.

Type 123

Split $\displaystyle\int_a^b |g(x)|\,dx$ at every point within $[a,b]$ where $g(x)=0$, integrating $g(x)$ or $-g(x)$ on each sub-piece as appropriate – extends Type 111 to cases with multiple sign changes or combined with other terms in the integrand.

Formula Used: $|g(x)| = \begin{cases} -g(x) & g(x)<0 \\ g(x) & g(x)\geq 0 \end{cases}$, split at every zero of $g(x)$ within the interval

Source: Exercise 7.17, Q10(iii)

Given: $\int_1^3 |x^2 - 2x|\, dx$

Solution:

Factor to find where the expression changes sign: $x^2 - 2x = x(x-2)$, which is zero at $x = 0$ and $x = 2$; within $[1, 3]$, only $x = 2$ is relevant.
For $x \in [1, 2]$, test $x = 1.5$: $x^2 - 2x = 2.25 - 3 = -0.75 < 0$, so $|x^2 - 2x| = -(x^2 - 2x) = 2x - x^2$ here. For $x \in [2, 3]$, test $x = 2.5$: $6.25 - 5 = 1.25 > 0$, so $|x^2 - 2x| = x^2 - 2x$ here.
Split the integral: $\displaystyle\int_1^3 |x^2 - 2x|\, dx = \int_1^2 (2x - x^2)\, dx + \int_2^3 (x^2 - 2x)\, dx$.
First piece: $\displaystyle\left[x^2 - \frac{x^3}{3}\right]_1^2 = \left(4 - \frac{8}{3}\right) - \left(1 - \frac{1}{3}\right) = \frac{4}{3} - \frac{2}{3} = \frac{2}{3}$.
Second piece: $\displaystyle\left[\frac{x^3}{3} - x^2\right]_2^3 = (9 - 9) - \left(\frac{8}{3} - 4\right) = 0 - \left(-\frac{4}{3}\right) = \frac{4}{3}$.
Adding: $\dfrac{2}{3} + \dfrac{4}{3} = 2$.

Final Answer

2

Common Mistake: Only checking the sign at one point and assuming it holds across the whole interval, instead of testing on each side of every zero found within $[a, b]$ – always test a specific point within each sub-interval to confirm the sign before integrating.

Quick Recall: Factor the expression inside the absolute value to find all its zeros, keep only the ones inside $[a, b]$, split the integral at each one, and test a point in each resulting sub-interval to determine the correct sign.

Type 124

Integrate $\dfrac{g(x)}{|g(x)|}$ (a "sign function" in disguise, equal to $+1$ or -1 depending on the sign of $g(x)$) by splitting at $g(x) = 0$ exactly as in Type 123, but integrating the constant ± 1 on each piece instead of $g(x)$ itself.

Formula Used: $\dfrac{g(x)}{|g(x)|} = \begin{cases} -1 & g(x) < 0 \\ +1 & g(x) > 0 \end{cases}$

Source: Exercise 7.17, Q1(iv)

Given: $\int_{-1}^1 \dfrac{|x+2|}{x+2}\, dx$

Solution:

Since $x + 2 > 0$ for every x in $[-1, 1]$ (as $x + 2 \geq 1 > 0$ throughout), there is no sign change within this interval, so $\dfrac{|x+2|}{x+2} = 1$ for the entire interval – no splitting is actually needed here.

Integrating the constant: $\displaystyle\int_{-1}^{1} 1\,dx = \Big[x\Big]_{-1}^{1} = 1 - (-1) = 2.$

Final Answer

2

Common Mistake: Automatically assuming a sign function needs splitting without first checking whether the inner expression actually changes sign within the given interval – always check whether $g(x) = 0$ actually falls inside $[a, b]$ before doing any splitting work.

Quick Recall: $g(x)/|g(x)|$ is always ± 1 – check first whether $g(x)$ changes sign inside the interval; if it doesn't, the integrand is a single constant and the whole problem is just $\pm(b - a)$.

Type 125

Integrate the greatest integer function $[x]$ by splitting the interval at every integer it contains, since $[x]$ is constant (equal to that integer) on each unit sub-interval $[n, n+1)$.

Formula Used: $[x] = n$ for $x \in [n, n+1)$, where n is an integer

Source: Exercise 7.17, Q2(ii)

Given: $\int_{-1}^{1} [x]\,dx$

Solution:

Split at the integer contained in the interval: for $x \in [-1, 0)$, $[x] = -1$; for $x \in [0, 1)$, $[x] = 0$; the single point $x = 1$ contributes nothing to the integral.

Splitting: $\displaystyle\int_{-1}^{1} [x]\,dx = \int_{-1}^{0} (-1)\,dx + \int_{0}^{1} (0)\,dx = (-1)(0 - (-1)) + 0 = -1 + 0.$

Final Answer

-1

Common Mistake: Treating $[x]$ as if it were continuous or trying to find a single antiderivative for it across the whole interval – $[x]$ is a genuine step function; it must be split at every integer within the limits, integrating the appropriate constant on each piece.

Quick Recall: $[x]$ is piecewise constant, jumping at every integer – list every integer within the given interval, split there, and integrate the corresponding constant value on each resulting sub-interval.

Type 126

Integrate $|\sin x|$, $|\cos x|$, or $|\cos kx|$ over an interval spanning more than one sign-change of the trig function – split at every zero of the trig function within the interval, exactly as in Type 123 but for a periodic function with infinitely many zeros (only the ones inside the given limits matter).

Formula Used: $\cos\theta = 0$ at $\theta = \frac{\pi}{2} + n\pi$; split the integral at every such pointinside the limits

Source: Exercise 7.17, Q11(iii)

Given: $\int_0^{\pi/2} |\cos 2x|\, dx$

Solution:

Find where $\cos 2x = 0$ within $x \in \left[0, \frac{\pi}{2}\right]$: this happens when $2x = \frac{\pi}{2}$, i.e. $x = \frac{\pi}{4}$ – the only zero inside this interval.

For $x \in \left[0, \frac{\pi}{4}\right]$, $2x \in [0, \frac{\pi}{2}]$ where cosine is non-negative, so $|\cos 2x| = \cos 2x$. For $x \in \left[\frac{\pi}{4}, \frac{\pi}{2}\right]$, $2x \in [\frac{\pi}{2}, \pi]$ where cosine is non-positive, so $|\cos 2x| = -\cos 2x$.

Splitting: $\displaystyle\int_0^{\pi/2} |\cos 2x|\, dx = \int_0^{\pi/4} \cos 2x\, dx - \int_{\pi/4}^{\pi/2} \cos 2x\, dx$.

First piece: $\displaystyle\left[\frac{\sin 2x}{2}\right]_0^{\pi/4} = \frac{1}{2} - 0 = \frac{1}{2}$. Second piece: $\displaystyle -\left[\frac{\sin 2x}{2}\right]_{\pi/4}^{\pi/2} = -\left(0 - \frac{1}{2}\right) = \frac{1}{2}$.

Adding: $\displaystyle\frac{1}{2} + \frac{1}{2} = 1$.

Final Answer

1

Common Mistake: Forgetting that a compound argument like $\cos 2x$ has its zero at $x = \pi/4$, not $x = \pi/2$ (the zero of plain $\cos x$) – always solve for exactly where the compound argument itself vanishes, not where the ”outer” function alone would vanish.

Quick Recall: For $|\cos(kx)|$ or $|\sin(kx)|$, solve $kx =$ (the usual zero locations) to find the actual split points in x – the compound argument shifts and compresses where those zeros fall.

Type 127

An odd power of $\sin x$ or $\cos x$ integrated over a full period or half-period often integrates to a clean fraction via the reduction technique from Type 42, applied directly at the definite-integral level using the standard "Walker" pattern (each factor drops the power by 2 and multiplies by a running fraction).

Formula Used: $\int_0^{\pi/2} \sin^n x\, dx = \int_0^{\pi/2} \cos^n x\, dx = \dfrac{(n-1)(n-3)\cdots}{n(n-2)\cdots} \times \left(\dfrac{\pi}{2} \text{ or } 1\right)$ (Walker's formula)

Source: Exercise 7.16, Q18(ii)

Given: $\int_0^{\pi/2} \cos^5 x\, dx$

Solution:

For an odd power $n = 5$, Walker's formula gives $\displaystyle\int_0^{\pi/2} \cos^5 x\, dx = \frac{4}{5} \times \frac{2}{3} \times 1$, multiplying alternating descending factors down to 1 (not $\pi/2$, since the power is odd).

Computing the product: $\dfrac{4}{5} \times \dfrac{2}{3} \times 1 = \dfrac{8}{15}$.

Final Answer

$\dfrac{8}{15}$

Common Mistake: Using the formula's "$\times\pi/2$" ending (reserved for even powers) instead of "$\times 1$" for an odd power – always check the parity of n first: even powers end the product with $\pi/2$, odd powers end with 1.

Quick Recall: Walker's (Wallis') formula gives $\int_0^{\pi/2} \sin^n x\, dx$ or $\cos^n x\, dx$ directly as a descending product of fractions – multiply $(n-1)/n \times (n-3)/(n-2) \times \cdots$ down to either 1 (odd n) or $\pi/2$ (even n); this is much faster than the full reduction technique for a $[0, \pi/2]$ integral specifically.

Type 128

Prove or apply the classical result $\displaystyle\int_0^{\pi/2} \log(\sin x)\, dx = \int_0^{\pi/2} \log(\cos x)\, dx = -\frac{\pi}{2} \log 2$, derived by combining King's Rule with the double-angle identity $\sin 2x = 2 \sin x \cos x$ and a substitution.

★ **Board Favorite** – This log-sine integral result is one of the most celebrated and frequently tested "prove that" results in the ISC syllabus – the derivation itself (not just the final value) is regularly asked for directly.

Formula Used: $\sin 2x = 2\sin x \cos x$, combined with King's Rule and asubstitution$t = 2x$

Source: Exercise 7.17, Q25

Given: Prove: $\int_0^{\pi/2} \log(\sin x)\,dx = -\frac{\pi}{2}\log 2$

Solution:

Let $I = \displaystyle\int_0^{\pi/2} \log(\sin x)\,dx$. By King's Rule ($x \to \frac{\pi}{2} - x$), $I = \displaystyle\int_0^{\pi/2} \log(\cos x)\,dx$ as well, so I equals both forms.

Adding: $2I = \displaystyle\int_0^{\pi/2} [\log(\sin x) + \log(\cos x)]\,dx = \int_0^{\pi/2} \log(\sin x \cos x)\,dx$, using $\sin x \cos x = \frac{1}{2}\sin 2x$, so

$$2I = \int_0^{\pi/2} \log\left(\frac{\sin 2x}{2}\right)dx.$$

Split the log: $\log\left(\dfrac{\sin 2x}{2}\right) = \log(\sin 2x) - \log 2$, so $2I = \displaystyle\int_0^{\pi/2} \log(\sin 2x)\,dx - \frac{\pi}{2}\log 2$.

For the remaining integral, substitute $t = 2x$, $dt = 2\,dx$; the limits become 0 to π: $\displaystyle\int_0^{\pi/2} \log(\sin 2x)\,dx = \frac{1}{2}\int_0^{\pi} \log(\sin t)\,dt$. By the Type 122 reflection property (since $\sin(\pi - t) = \sin t$), $\displaystyle\int_0^{\pi} \log(\sin t)\,dt = 2\int_0^{\pi/2} \log(\sin t)\,dt = 2I$.

So $\displaystyle\int_0^{\pi/2} \log(\sin 2x)\,dx = \frac{1}{2}(2I) = I$. Substituting back: $2I = I - \dfrac{\pi}{2}\log 2$, giving $I = -\dfrac{\pi}{2}\log 2$.

Final Answer

$I = -\frac{\pi}{2}\log 2$ (proved)

Common Mistake: Losing track of which integral is which during the substitution step (confusing the original I with the new $\int_0^{\pi} \log(\sin t)\,dt$) – this proof involves the same integral I reappearing through two different routes (King's Rule and the reflection property); label each appearance carefully.

Quick Recall: This is a genuine "self-referencing" proof, structurally similar to the cyclic by-parts technique – combine King's Rule, a double-angle identity, and the $[0, 2a]$ reflection property until the original integral I reappears, then solve algebraically.

Type 129

Apply the general result $\displaystyle\int_0^a \frac{f(x)}{f(x)+f(a-x)}\,dx = \frac{a}{2}$, which follows directly from King's Rule – recognise the specific denominator pattern $f(x)+f(a-x)$ to apply this shortcut instantly instead of computing from scratch.

Formula Used: $\int_0^a \dfrac{f(x)}{f(x)+f(a-x)}\,dx = \dfrac{a}{2}$

Source: Exercise 7.17, Q24

Given: $\int_0^{\pi/2} \dfrac{\sin^4 x}{\sin^4 x+\cos^4 x}\,dx$

Solution:

Recognise this matches the general pattern $\dfrac{f(x)}{f(x)+f(a-x)}$ with $f(x)=\sin^4 x$ and $a=\dfrac{\pi}{2}$, since $f\left(\frac{\pi}{2}-x\right)=\sin^4\left(\frac{\pi}{2}-x\right)=\cos^4 x$ – exactly the second term in the denominator.

Applying the general result directly: $\displaystyle\int_0^{\pi/2} \frac{\sin^4 x}{\sin^4 x+\cos^4 x}\,dx = \frac{1}{2}\cdot\frac{\pi}{2}$.

Final Answer

$\dfrac{\pi}{4}$

Common Mistake: Not recognising the denominator's exact shape $(f(x)+f(a-x))$ and instead attempting a full Weierstrass substitution or partial fraction approach – always check whether the denominator is literally "the numerator function plus itself with $x\to a-x$" before doing any real work.

Quick Recall: Whenever the denominator is the sum of the numerator function and that same function with $x\to a-x$ substituted, the whole integral is instantly $a/2$ – this is King's Rule's most powerful shortcut and worth checking for first on any unfamiliar-looking rational integrand.

Type 130

A genuinely hard combined integral requiring King's Rule together with a nontrivial algebraic or trigonometric simplification – the "add and simplify" step doesn't immediately collapse to a constant, and needs an extra identity or substitution to finish.

★ **Board Favorite –** This level of combined difficulty (King's Rule plus a nontrivial follow-up simplification) represents the hardest routine pattern in the definite-integral toolkit and is a genuine board favourite for the long-answer section.

Formula Used: Apply King's Rule, add, then simplify with whatever identity is needed

Source: Exercise 7.17, Q21 (King's Rule, hard combination)

Given: $\int_0^\pi \dfrac{x \sin x}{1+\cos^2 x}\,dx$

Solution:
Let $I = \displaystyle\int_0^\pi \frac{x \sin x}{1+\cos^2 x}\,dx$. Apply King's Rule with $a = \pi$: replacing $x \to \pi - x$, and using $\sin(\pi - x) = \sin x$, $\cos(\pi - x) = -\cos x$ (so $\cos^2(\pi - x) = \cos^2 x$ is unchanged), we get $I = \displaystyle\int_0^\pi \frac{(\pi - x)\sin x}{1+\cos^2 x}\,dx$.
Adding the original and transformed forms: $2I = \displaystyle\int_0^\pi \frac{\pi \sin x}{1+\cos^2 x}\,dx$ – the x and $(\pi - x)$ terms combine to leave just π, since $x\sin x + (\pi - x)\sin x = \pi \sin x$.
So $I = \dfrac{\pi}{2}\displaystyle\int_0^\pi \frac{\sin x}{1+\cos^2 x}\,dx$. Let $t = \cos x$, $dt = -\sin x\,dx$; the limits become 1 to -1: $\displaystyle\int_0^\pi \frac{\sin x}{1+\cos^2 x}\,dx = \int_1^{-1} \frac{-dt}{1+t^2} = \int_{-1}^{1} \frac{dt}{1+t^2} = \Big[\tan^{-1} t\Big]_{-1}^{1} = \frac{\pi}{4} - \left(-\frac{\pi}{4}\right) = \frac{\pi}{2}$.
Substituting back: $I = \dfrac{\pi}{2} \cdot \dfrac{\pi}{2}$.

Final Answer

$\dfrac{\pi^2}{4}$

Common Mistake: Stopping after the King's Rule addition step and not recognising that the remaining integral (now free of x in the numerator) still needs its own substitution to finish – King's Rule is often only the *first* step in a longer problem, not the whole solution.

Quick Recall: For $\int_0^a f(x)g(x)\,dx$ where $g(\pi - x) = g(x)$ but the x (or $\pi - x$) factor is linear, King's Rule always turns the linear factor into a constant ($a/2$ times the remaining integral) – but that remaining integral still needs to be evaluated with whatever technique it requires.

Type 131

Prove one of the definite-integral properties itself from first principles (e.g. $\displaystyle\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$), using only the substitution rule for definite integrals – a conceptual proof question rather than an application question.

Formula Used: Substitute $t = a - x$ and show it reproduces the same integral

Source: Exercise 7.17, Q8 (property proof)

Given: Prove: $\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$

Solution:

Start from the right-hand side and substitute $t = a - x$, so $x = a - t$ and $dx = -dt$ – this substitution is the entire content of the proof.
Convert the limits: when $x = 0$, $t = a$; when $x = a$, $t = 0$.
Substituting: $\displaystyle\int_0^a f(a-x)\,dx = \int_a^0 f(t)\,(-dt) = \int_0^t f(t)\,dt$ after flipping the limits (which cancels the negative sign): $= \displaystyle\int_0^a f(t)\,dt$.
Since the variable of integration is a dummy variable (renaming t back to x doesn't change the value of a definite integral), this equals $\displaystyle\int_0^a f(x)\,dx$, exactly the left-hand side.

Final Answer

$\int_0^a f(x)\,dx = \int_0^a f(a-x)\,dx$ (proved)

Common Mistake: Forgetting that flipping the limits of integration (from $a \to 0$ back to $0 \to a$) requires flipping the sign of the integral as well – the two sign flips (from $dx = -dt$ and from reversing the limits) cancel each other, which is easy to lose track of.

Quick Recall: Every "prove this property" question in this section follows the same three-step pattern: substitute $t =$ (the appropriate expression), convert the limits, and show the sign flips cancel to reproduce the original integral (relabelling the dummy variable at the end).

VII. Additional Board-Critical Techniques (Frequently Tested, Not in ML Aggarwal Exercises Directly)

Type 132

For $\dfrac{x^2+1}{x^4+1}$ or $\dfrac{x^2-1}{x^4+1}$ (and similar "palindromic" quartics), divide numerator and denominator by x^2, then substitute $t = x - \dfrac{1}{x}$ or $t = x + \dfrac{1}{x}$ – this is one of the most important and most frequently missed substitution tricks in the entire chapter, since the substitution isn't visible until *after* dividing by x^2.

★ **Board Favorite** – The $x \mp \dfrac{1}{x}$ substitution is a guaranteed board favourite – it is the standard technique for any quartic-denominator integral built from $x^4 + 1$, $x^4 + kx^2 + 1$, or similar symmetric ("palindromic") polynomials, and appears on nearly every ISC paper's long-answer integration section.

Formula Used: $x^2 + \dfrac{1}{x^2} = \left(x - \dfrac{1}{x}\right)^2 + 2 = \left(x + \dfrac{1}{x}\right)^2 - 2$

Source: Exercise 7.7, Extended (classic $x^4 + 1$ family)

Given: $\dfrac{x^2+1}{x^4+1}$

Solution:

Divide numerator and denominator by x^2: $\dfrac{x^2+1}{x^4+1} = \dfrac{1+\dfrac{1}{x^2}}{x^2+\dfrac{1}{x^2}}$ – this division is the essential first move; the substitution is completely invisible before it.

Let $t = x - \dfrac{1}{x}$, so $dt = \left(1 + \dfrac{1}{x^2}\right) dx$ – notice this matches the new numerator exactly.

Rewrite the new denominator in terms of t: $x^2 + \dfrac{1}{x^2} = \left(x - \dfrac{1}{x}\right)^2 + 2 = t^2 + 2$, using the algebraic identity above (add 2 to complete the square, since $\left(x - \frac{1}{x}\right)^2 = x^2 - 2 + \frac{1}{x^2}$).

The integral becomes $\displaystyle\int \frac{dt}{t^2+2}$, a direct standard arctan form with $a = \sqrt{2}$: $\dfrac{1}{\sqrt{2}} \tan^{-1} \dfrac{t}{\sqrt{2}} + C$.

Converting back: $t = x - \dfrac{1}{x} = \dfrac{x^2-1}{x}$.

Final Answer

$$\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{x^2-1}{\sqrt{2}\,x}\right)+C$$

Common Mistake: Trying every other technique (partial fractions on the quartic, direct substitution $t = x^2$, etc.) without ever dividing by x^2 first – for x^2+1 over x^4+1, dividing by x^2 is the specific, non-obvious first step that makes the whole problem tractable; nothing else works cleanly.

Alternate Board-Style Example (the $x + 1/x$ version)

Source: Exercise 7.7, Extended (companion form)

Given: $\dfrac{x^2-1}{x^4+1}$

Solution:

Divide by x^2: $\dfrac{x^2-1}{x^4+1} = \dfrac{1-\dfrac{1}{x^2}}{x^2+\dfrac{1}{x^2}}$.

This time let $t = x+\dfrac{1}{x}$, so $dt = \left(1-\dfrac{1}{x^2}\right)dx$ – matching the new numerator (note the sign change in the numerator is what signals *this* substitution instead of the previous one).

Rewrite the denominator: $x^2+\dfrac{1}{x^2} = \left(x+\dfrac{1}{x}\right)^2 - 2 = t^2-2$.

The integral becomes $\displaystyle\int \frac{dt}{t^2-2}$, a direct standard log form with $a=\sqrt{2}$: $\dfrac{1}{2\sqrt{2}}\log\left|\dfrac{t-\sqrt{2}}{t+\sqrt{2}}\right| + C$.

Converting back: $t = x+\dfrac{1}{x} = \dfrac{x^2+1}{x}$, and simplifying the fraction by multiplying through by x.

Final Answer

$$\frac{1}{2\sqrt{2}}\log\left|\frac{x^2-\sqrt{2}\,x+1}{x^2+\sqrt{2}\,x+1}\right|+C$$

Quick Recall: Numerator x^2+1 (matches $1+1/x^2$ after dividing) $\Rightarrow$ use $t = x-\dfrac{1}{x}$ $\Rightarrow$ arctan answer. Numerator x^2-1 (matches $1-1/x^2$ after dividing) $\Rightarrow$ use $t = x+\dfrac{1}{x}$ $\Rightarrow$ log answer. The sign in the numerator after dividing by x^2 always tells you which of the two substitutions to use.

Type 133

Integrate a *sum* of two or more absolute-value terms, each with its own breakpoint (like $|x+2|+|x-3|$) – find every breakpoint within $[a,b]$ first, then split the whole interval into as many pieces as needed so that *every* absolute value has a fixed, known sign on each piece.

★ **Board Favorite –** Multi-breakpoint absolute-value sums are a standard high-yield pattern precisely because a single-breakpoint splitting instinct (from Type 111/123) isn't enough – ISC frequently tests whether students find *all* the breakpoints, not just the first one they notice.

Formula Used: List every absolute-value zero in $[a,b]$, split there

Source: Composite Board-Style Question

Given: $\int_{-4}^{4} \big(|x+2|+|x-3|\big)\,dx$

Solution:

Find every breakpoint: $|x+2|$ changes sign at $x=-2$, and $|x-3|$ changes sign at $x=3$ – both lie inside $[-4,4]$, so the interval must be split into *three* pieces, not two.
For $x<-2$: $|x+2|=-(x+2)$ and $|x-3|=-(x-3)=3-x$, so the sum is $(-(x+2))+(3-x)=1-2x$.
For $-2\le x<3$: $|x+2|=x+2$ and $|x-3|=3-x$, so the sum is $(x+2)+(3-x)=5$ – a constant on this middle piece.
For $x\ge 3$: $|x+2|=x+2$ and $|x-3|=x-3$, so the sum is $(x+2)+(x-3)=2x-1$.
Splitting the integral into three pieces: $\displaystyle\int_{-4}^{-2}(1-2x)\,dx+\int_{-2}^{3}5\,dx+\int_{3}^{4}(2x-1)\,dx.$
First piece: $\left[x-x^2\right]_{-4}^{-2}=(-2-4)-(-4-16)=-6+20=14$. Second piece: $5\times 5=25$.
Third piece: $\left[x^2-x\right]_{3}^{4}=(16-4)-(9-3)=12-6=6$.
Adding all three: $14+25+6=45$.

Final Answer

45

Common Mistake: Only finding one of the two breakpoints (usually the first term's) and splitting the interval into just two pieces – always identify the zero of *every* absolute-value expression present before splitting, since each one is an independent potential breakpoint.

Quick Recall: For a sum of several absolute values, list all their individual breakpoints first, sort them, then split the interval into one piece per gap between consecutive breakpoints – the integrand is linear (or constant) and has a fixed sign structure on each resulting piece.

Type 134

Integrate the greatest integer function of a *compound* expression, such as $[x^2]$ – find where the compound expression itself crosses each integer value (not where x itself is an integer), then split there.

★ **Board Favorite** – This exact question (with $[x^2]$) appeared on the ISC 2025 Specimen Paper – it specifically tests whether students realise the split points are where x^2 (not x) hits an integer, which requires solving $x^2 = n$ for each relevant integer n.

Formula Used: $[g(x)] = n$ for $g(x) \in [n, n+1)$; solve $g(x) = n$ for the actual split points in x

Source: ISC Specimen Paper 2025

Given: $\int_0^{\sqrt{2}} [x^2]\, dx$

Solution:

Find where x^2 crosses each integer within the range: as x goes from 0 to $\sqrt{2}$, x^2 goes from 0 to 2 – so x^2 crosses the integer value 1 exactly when $x = 1$ (solving $x^2 = 1$ for $x > 0$). For $x \in [0, 1)$: $x^2 \in [0, 1)$, so $[x^2] = 0$. For $x \in [1, \sqrt{2})$: $x^2 \in [1, 2)$, so $[x^2] = 1$.

Splitting: $\displaystyle\int_0^{\sqrt{2}} [x^2]\, dx = \int_0^1 0\, dx + \int_1^{\sqrt{2}} 1\, dx = 0 + \left(\sqrt{2} - 1\right)$.

Final Answer

$\sqrt{2} - 1$

Common Mistake: Trying to split at integer values of x (like $x = 1$ only by coincidence) rather than solving for where the *inside expression* x^2 actually equals each integer – for a compound greatest-integer argument, always solve $g(x) = n$ explicitly for each relevant integer n, don't assume the split points are integers themselves.

Quick Recall: For $[g(x)]$ with a compound $g(x)$, first find the range of $g(x)$ over the given interval, list every integer inside that range, solve $g(x) = n$ for the exact x-value of each split point, then integrate the corresponding constant on each resulting piece.

Type 135

Integrate an expression combining *both* an absolute value and a greatest-integer term in the same integrand – find the breakpoints from both sources separately, merge and sort the complete list, then split into one piece per gap exactly as in Type 133, now tracking two different piecewise rules simultaneously.

Formula Used: Merge all $|\cdot|$ and $[\cdot]$ breakpoints into one sorted list

Source: Composite Board-Style Question

Given: $\int_0^3 \big(|x-1| + [x]\big)\,dx$

Solution:

Find every breakpoint from both pieces: $|x-1|$ changes rule at $x = 1$; $[x]$ jumps at every integer, so within $[0,3]$ it jumps at $x = 1$ and $x = 2$. Merging and sorting: the complete list of breakpoints is $x = 1, 2$, splitting $[0,3]$ into three pieces.
For $x \in [0,1)$: $|x-1| = 1-x$ and $[x] = 0$, so the integrand is $(1-x)+0 = 1-x$.
For $x \in [1,2)$: $|x-1| = x-1$ and $[x] = 1$, so the integrand is $(x-1)+1 = x$.
For $x \in [2,3)$: $|x-1| = x-1$ and $[x] = 2$, so the integrand is $(x-1)+2 = x+1$.
Splitting: $\displaystyle\int_0^1 (1-x)\,dx + \int_1^2 x\,dx + \int_2^3 (x+1)\,dx.$
First piece: $\left[x - \frac{x^2}{2}\right]_0^1 = 1 - \frac{1}{2} = \frac{1}{2}$. Second piece: $\left[\frac{x^2}{2}\right]_1^2 = 2 - \frac{1}{2} = \frac{3}{2}$. Third piece: $\left[\frac{x^2}{2} + x\right]_2^3 = (4.5+3) - (2+2) = 7.5 - 4 = 3.5$.
Adding all three: $\frac{1}{2} + \frac{3}{2} + \frac{7}{2} = \frac{11}{2}$.

Final Answer

$$\frac{11}{2}$$

Common Mistake: Handling the absolute value's breakpoint and the greatest-integer function's jumps as if they were two separate integrals to be computed independently and then combined – they must be split *together*, on one merged list of breakpoints, since both rules apply simultaneously within the same integrand on any given piece.

Quick Recall: When an integrand mixes $|\cdot|$ and $[\cdot]$ terms, gather breakpoints from *both* sources into a single sorted list first – then, on each resulting piece, write out both pieces' fixed rule (which sign for the absolute value, which constant for the greatest integer) before integrating the combined expression.

Type 136

When x appears only as high negative powers wrapped in a square root (like $\dfrac{\sqrt{1+x^2}}{x^4}$), substitute the *reciprocal* $x = \dfrac{1}{t}$ – this flips every power of x upside down and often converts an intractable-looking integrand into a simple polynomial-times-square-root form.

★ **Board Favorite** – The reciprocal substitution $x = 1/t$ is a genuine board favourite specifically because it looks like nothing else in the chapter – there is no numerator clue or visible derivative pattern to spot; recognising "high negative powers of x under a root" as the trigger is the entire skill being tested.

Formula Used: $x = \dfrac{1}{t}, \quad dx = -\dfrac{1}{t^2}\,dt, \quad \sqrt{1+x^2} = \dfrac{\sqrt{t^2+1}}{|t|}$

Source: Exercise 7.16, Q23(ii) (Exemplar)

Given: $\dfrac{\sqrt{1+x^2}}{x^4}$

Solution:

Let $x = \dfrac{1}{t}$, so $dx = -\dfrac{1}{t^2}\,dt$ – this substitution is chosen because x^4 in the denominator is exactly the kind of high negative power that becomes a clean positive power of t once flipped.

Rewrite each piece in terms of t: $\sqrt{1+x^2} = \sqrt{1+\dfrac{1}{t^2}} = \dfrac{\sqrt{t^2+1}}{|t|}$ (for $t > 0$, this is $\dfrac{\sqrt{t^2+1}}{t}$), and $\dfrac{1}{x^4} = t^4$.

Substituting everything together: $\dfrac{\sqrt{1+x^2}}{x^4}\,dx = \dfrac{\sqrt{t^2+1}}{t} \cdot t^4 \cdot \left(-\dfrac{1}{t^2}\right) dt = -t\sqrt{t^2+1}\,dt$ – the messy original expression collapses to a simple power-rule-friendly form.

Integrating: $-\displaystyle\int t\sqrt{t^2+1}\,dt = -\dfrac{1}{3}(t^2+1)^{3/2} + C$, using the chain-rule-reverse power formula.

Converting back: $t = \dfrac{1}{x}$, so $t^2+1 = \dfrac{1}{x^2}+1 = \dfrac{1+x^2}{x^2}$, giving $(t^2+1)^{3/2} = \dfrac{(1+x^2)^{3/2}}{x^3}$ (for $x > 0$).

Final Answer

$$-\frac{(1+x^2)^{3/2}}{3x^3} + C$$

Common Mistake: Trying every standard sqrt-formula and substitution from Types 102–107 first, since the integrand *looks* like a square-root-of-quadratic problem – none of those apply here; the giveaway is the x^4 in the denominator (a high negative power once you think of it as x^{-4}), which is the specific trigger for $x = 1/t$, not any of the earlier techniques.

Quick Recall: If x appears under a square root together with a high negative power of x multiplying it (as x^{-n} for $n \geq 2$), try $x = 1/t$ – this flips every power and very often reduces the whole problem to a one-line power-rule integral in t.

Type 137

When the integrand mixes two or more different fractional powers of x (like $\sqrt{x}$ and $\sqrt[3]{x}$ together in a denominator), substitute $t = x^{1/n}$ where n is the LCM of all the roots present – this converts every fractional power of x into a whole-number power of t simultaneously.

★ **Board Favorite** – The LCM substitution is a standard high-yield technique whenever an integrand mixes different roots of x – it is the only way to make every term rational in the new variable at once, and ISC regularly tests exactly this recognition.

Formula Used: Let $t = x^{1/n},\ n = \text{LCM of root indices, so } x = t^n,\ dx = nt^{n-1}\,dt$

Source: Exercise 7.5, Q22(i)

Given: $\dfrac{1}{\sqrt{x}+\sqrt[3]{x}}$

Solution:

The two roots present are $\sqrt{x} = x^{1/2}$ and $\sqrt[3]{x} = x^{1/3}$; the LCM of the indices 2 and 3 is 6, so let $t = x^{1/6}$, giving $x = t^6$ and $dx = 6t^5\,dt$ – this single substitution makes *both* roots whole-number powers of t simultaneously ($\sqrt{x} = t^3$ and $\sqrt[3]{x} = t^2$).

Rewrite the integrand: $\dfrac{1}{\sqrt{x}+\sqrt[3]{x}} = \dfrac{1}{t^3+t^2} = \dfrac{1}{t^2(t+1)}$.

Substituting fully: $\displaystyle\int \frac{1}{t^2(t+1)}\cdot 6t^5\,dt = 6\int \frac{t^3}{t+1}\,dt$ – now an ordinary improper-fraction integral in t.

Divide first (improper fraction): $\dfrac{t^3}{t+1} = t^2 - t + 1 - \dfrac{1}{t+1}$.

Integrating: $\displaystyle 6\int \left(t^2 - t + 1 - \frac{1}{t+1}\right)dt = 6\left[\frac{t^3}{3} - \frac{t^2}{2} + t - \log|t+1|\right] = 2t^3 - 3t^2 + 6t - 6\log|t+1|$.

Converting back: $t = x^{1/6}$, so $t^3 = \sqrt{x}$, $t^2 = \sqrt[3]{x}$.

Final Answer

$2\sqrt{x} - 3\sqrt[3]{x} + 6x^{1/6} - 6\log\left|x^{1/6}+1\right| + C$

Common Mistake: Substituting only for one of the two roots present (e.g. $t = \sqrt{x}$ alone) – this leaves the other root ($\sqrt[3]{x}$) still irrational in the new variable, so the integral doesn't actually simplify; the LCM of *all* indices present must be used for the single substitution to clear every root at once.

Quick Recall: Whenever two or more different roots of x appear together, find the LCM of their indices and substitute $t = x^{1/\text{LCM}}$ – this is the only substitution that clears every root simultaneously, turning the whole problem into an ordinary rational-function integral in t.

Type 138

For $\dfrac{x^2}{\sqrt{x^2+a^2}}$ (numerator degree exactly matching what's under the root), rewrite the numerator as $x^2 = (x^2+a^2) - a^2$ – this splits the fraction directly into a plain $\sqrt{x^2+a^2}$ piece (integrable by the standard formula) and a $\dfrac{1}{\sqrt{x^2+a^2}}$ piece (also a standard formula), with no substitution needed at all.

Formula Used: $x^2 = (x^2+a^2) - a^2$, splits $\dfrac{x^2}{\sqrt{x^2+a^2}}$ into $\sqrt{x^2+a^2} - \dfrac{a^2}{\sqrt{x^2+a^2}}$

Source: Exercise 7.13, Q4(i)

Given: $\dfrac{x^2}{\sqrt{x^2+6}}$

Solution:

Rewrite the numerator by adding and subtracting the constant under the root: $x^2 = (x^2+6) - 6$ – this is a purely algebraic trick, not a substitution, and it works because the numerator's degree exactly matches the degree of the expression under the root.

Splitting the fraction: $\dfrac{x^2}{\sqrt{x^2+6}} = \dfrac{(x^2+6)-6}{\sqrt{x^2+6}} = \sqrt{x^2+6} - \dfrac{6}{\sqrt{x^2+6}}$ – both pieces are now direct standard forms.

Integrate each piece separately: $\displaystyle\int \sqrt{x^2+6}\,dx$ uses the Type 103 formula with $a = \sqrt{6}$: $\dfrac{x}{2}\sqrt{x^2+6} + 3\log\left|x+\sqrt{x^2+6}\right|$; and $\displaystyle\int \frac{6}{\sqrt{x^2+6}}\,dx = 6\log\left|x+\sqrt{x^2+6}\right|$, using the Type 56 log form.

Final Answer

$\dfrac{x}{2}\sqrt{x^2+6} + 3\log\left|x+\sqrt{x^2+6}\right| - 6\log\left|x+\sqrt{x^2+6}\right| + C = \dfrac{x}{2}\sqrt{x^2+6} - 3\log\left|x+\sqrt{x^2+6}\right| + C$

Common Mistake: Reaching for a substitution (like $x = \sqrt{6}\tan\theta$) or the by-parts approach from Type 107 instead of noticing the much faster "add and subtract the constant" split – this trick only needs the numerator's degree to match what's under the root, and it avoids by parts or trigonometric substitution entirely.

Quick Recall: If the numerator is x^2 (or x^2 plus a constant) sitting directly over $\sqrt{x^2 \pm a^2}$ with nothing else going on, don't reach for a technique – just rewrite x^2 as $(x^2 \pm a^2) \mp a^2$ and split into two standard-formula pieces immediately.

Type 139

Integrate $\cos^{-1}\sqrt{x}$ (or $\sin^{-1}\sqrt{x}$) by substituting $x = t^2$ first (exactly as in Type 92's $\tan^{-1}\sqrt{x}$), then applying by parts – the algebra differs slightly from Type 92 because the derivative of $\cos^{-1}$ carries a minus sign and pairs with $\sqrt{1-t^2}$ instead of $1+t^2$.

Formula Used: $\dfrac{d}{dt}\cos^{-1} t = -\dfrac{1}{\sqrt{1-t^2}}$

Source: Exercise 7.13, Q8(ii)

Given: $\cos^{-1}\sqrt{x}$

Solution:

Let $x = t^2$, so $\sqrt{x} = t$ and $dx = 2t\,dt$ – exactly the same opening move as Type 92, substituting to clear the square root from inside the inverse trig function.

Substituting: $\displaystyle\int \cos^{-1}\sqrt{x}\,dx = \int \cos^{-1} t \cdot 2t\,dt = 2\int t\cos^{-1} t\,dt$ – now a standard polynomial-times-inverse-trig by-parts problem (Type 87) in t.

Let $u = \cos^{-1} t$, $dv = t\,dt$, so $du = -\dfrac{1}{\sqrt{1-t^2}}\,dt$, $v = \dfrac{t^2}{2}$: $\displaystyle\int t\cos^{-1} t\,dt = \frac{t^2}{2}\cos^{-1} t + \frac{1}{2}\int \frac{t^2}{\sqrt{1-t^2}}\,dt$.

Rewrite the remaining numerator using the Type 138 trick: $t^2 = -(1-t^2) + 1$, splitting into $-\sqrt{1-t^2} + \dfrac{1}{\sqrt{1-t^2}}$.

Integrating this remainder: $\displaystyle\int \left(-\sqrt{1-t^2} + \frac{1}{\sqrt{1-t^2}}\right) dt = -\left[\frac{t}{2}\sqrt{1-t^2} + \frac{1}{2}\sin^{-1} t\right] + \sin^{-1} t = -\frac{t}{2}\sqrt{1-t^2} + \frac{1}{2}\sin^{-1} t$.

Combining: $2\left[\dfrac{t^2}{2}\cos^{-1} t + \dfrac{1}{2}\left(-\dfrac{t}{2}\sqrt{1-t^2} + \dfrac{1}{2}\sin^{-1} t\right)\right] = t^2\cos^{-1} t - \dfrac{t}{2}\sqrt{1-t^2} + \dfrac{1}{2}\sin^{-1} t$.

Converting back: $t = \sqrt{x}$.

Final Answer

$$x\cos^{-1}\sqrt{x} - \frac{1}{2}\sqrt{x}\sqrt{1-x} + \frac{1}{2}\sin^{-1}\sqrt{x} + C$$

Common Mistake: Copying Type 92's arctan-based algebra directly instead of redoing the by-parts remainder with the correct $\cos^{-1}$ derivative – the minus sign in $\dfrac{d}{dt}\cos^{-1} t$ and the $\sqrt{1-t^2}$ (not $1+t^2$) denominator change every subsequent step; don't assume the two problems finish identically just because they start with the same substitution.

Quick Recall: $\sin^{-1}\sqrt{x}$ and $\cos^{-1}\sqrt{x}$ both substitute $x = t^2$ first, exactly like $\tan^{-1}\sqrt{x}$ – but finish using the sqrt-based by-parts remainder (Type 138's numerator-split trick), not the rational-fraction remainder that $\tan^{-1}$ produces.